

Who Benefits from Capital Market Integration in the Euro Area?

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1 Introduction

Through financial integration, households can insure their consumption against domestic shocks by holding foreign assets. When a country experiences a negative shock, income from foreign investments remains largely unaffected, helping smooth consumption and mitigate the downturn. By weakening the correlation between national income and consumption, cross-border asset holdings act as a stabilizing mechanism that reduces aggregate volatility across countries.

This mechanism lies at the core of [Mundell \[1961\]](#) theory of Optimal Currency Area: in the absence of exchange rate adjustment, member states can only absorb asymmetric shocks through fiscal transfers, labor mobility, or financial integration.¹ Yet in the euro area, these adjustment channels remain incomplete—fiscal transfers are limited, labor mobility low, and capital markets fragmented—leaving the union vulnerable to recurrent asymmetric shocks, from the sovereign debt crisis to the pandemic and energy crises. The Capital Markets Union (CMU) initiative embodies this logic, aiming to deepen financial integration and enhance private risk-sharing across member states.²

This paper revisits the foundations of this adjustment mechanism by embedding household heterogeneity into the theory of capital-market integration. It introduces two interacting dimensions of integration—heterogeneity in financial participation within countries and portfolio diversification between them—and studies how their interaction shapes macroeconomic stabilization and welfare when asymmetric shocks hit a monetary union. Standard representative-agent frameworks capture the cross-country dimension but abstract from heterogeneity within countries, predicting that deeper integration unambiguously enhances risk-sharing and stabilizes the union. I show that this prediction does not survive once financial participation is incomplete and portfolios are asymmetric.

Combining newly harmonized empirical evidence with a two-country heterogeneous-agent model of a monetary union, I show that financial integration stabilizes the union in the aggregate but redistributes welfare unevenly—across both countries and wealth groups. Within countries, limited financial participation amplifies the impact of adverse shocks for constrained households, who remain exposed to fluctuations in labor income and real wages when capital reallocates across borders. Across countries, asymmetric portfolio structures and persistent home bias generate unequal exposure to shocks and

¹With a common nominal interest rate, countries cannot use exchange rate movements to restore competitiveness or stabilize demand after idiosyncratic shocks.

²The *Five Presidents' Report* (2015) and the *New Action Plan for a Capital Markets Union* (2020) frame the CMU as a pillar of Europe's broader strategy to strengthen resilience alongside fiscal and banking union. Yet they highlight that persistent fragmentation in supervision, insolvency, and taxation continues to hinder cross-border financial integration.

heterogeneity in their transmission.

This paper makes three main contributions. First, it provides new empirical and conceptual evidence on the unequal structure of capital-market integration in the euro area. Using microdata on financial participation and harmonized macrodata on cross-border portfolios, I document large cross-country and within-country gaps and introduce a distinction between two dimensions of integration: within countries, through households' access to financial assets, and between countries, through portfolio diversification. Second, it develops a two-country heterogeneous-agent model of a monetary union that microfound how these dimensions interact through two key channels: (i) a diversification channel, determined by the steady-state composition of portfolios, and (ii) a reallocation channel, driven by dynamic portfolio adjustments following shocks. Third, it quantifies how these mechanisms shape macroeconomic adjustment and welfare, showing that the conventional Mundellian prediction does not survive once heterogeneity and portfolio dynamics are accounted for: financial integration stabilizes the union in the aggregate but redistributes welfare unevenly across households and countries.

I start by documenting two new stylized facts for the post-sovereign crisis euro area. First, the distribution of wealth has diverged both within and between groups since 2010, driven primarily by the bottom of the wealth distribution in the Periphery, where households have failed to recover relative to their pre-crisis position. Second, the capital stock between groups has also diverged, as the Periphery decoupled from its pre-crisis trend while the Core remained broadly stable, and even slightly expanded, over the same period. This divergence in capital is partly driven by a reduction in home bias among Periphery investors, who have increasingly shifted from domestic assets toward those issued in Core countries, reinforcing cross-country asymmetries in ownership and exposure. These developments reflect the persistence of structural imbalances between groups of countries that were expected to converge but instead moved further apart.

To understand these patterns, I examine heterogeneity in financial participation and cross-border portfolio holdings across euro area countries between 2010 and 2021. Using the Household Finance and Consumption Survey (HFCS), I show that financial participation is substantially lower in the Periphery, where more than two-thirds of households hold no assets beyond deposits. Complementary evidence from the IMF's Coordinated Portfolio Investment Survey (CPIS) indicates that international portfolios remain asymmetric and strongly home-biased. In the Periphery, limited diversification provides only partial insurance against domestic shocks. By contrast, Core portfolios display an even stronger home bias, insulating them from foreign shocks and limiting their losses when downturns occur abroad. This unequal access to risk-sharing mechanisms—both within

and between country groups—constrains the extent to which capital market integration can smooth asymmetric shocks across the euro area.³

To analyze the consequences of these empirical asymmetries, I develop a two-country New Keynesian model with heterogeneous agents in a monetary union. Each country includes two household types—Savers and Non-Savers—reflecting the observed differences in financial participation. Savers hold capital and invest through a domestic mutual fund that allocates their portfolios between domestic and foreign assets, subject to home bias and adjustment costs.⁴

This structure isolates two mechanisms through which capital market integration affects inequality: (i) a diversification channel, determined by the steady-state composition of portfolios, and (ii) a reallocation channel, driven by dynamic portfolio adjustments following shocks.

These channels operate asymmetrically across households and countries. Diversification provides insurance by stabilizing the financial income of Savers exposed to domestic shocks, while the reallocation channel, through which capital flows toward regions with higher expected returns, amplifies downturns in the affected country by withdrawing investment. This reallocation depresses wages and exacerbates inequality, particularly for Non-Savers, who lack financial assets to buffer consumption.

I calibrate the model to euro area data for 2010, which serves as both the first year of observation and a pre-crisis benchmark, preceding the sovereign debt crisis that represented the most significant asymmetric shock of the period. The results show that these effects are quantitatively significant. The strong home bias of Core investors and the relatively small size of the Periphery limit cross-border risk-sharing. As a result, integration is redistributive: it narrows inequality in the Core but widens it in the Periphery, mirroring the post-crisis divergence observed in the data. Counterfactual exercises comparing the model to economies with either no capital market integration or a representative agent demonstrate that accounting for both inequality and portfolio heterogeneity is essential to understand and estimate the transmission of shocks across households and countries.

The model also serves as a laboratory to examine how the channels identified above operate under different environments. I first analyze a symmetric monetary union with identical countries but an asymmetric shock, confirming the working of both diversification and reallocation channels. It confirms that the resulting patterns are not driven

³Consistent with findings in, among others, [Lane and Milesi-Ferretti \[2018\]](#), [Vermeulen \[2018\]](#), [Fagereng et al. \[2020\]](#), and [Bach et al. \[2020\]](#).

⁴While this intermediary is not essential in the two-agent (TANK) structure, it provides a useful bridge to the heterogeneous-agent (HANK) extension and reflects the fact that, in practice, most household portfolios are held indirectly through financial institutions.

by calibration choices but by the endogenous interaction between integration and heterogeneity. I then introduce uncorrelated shocks across countries to study long-run dynamics: the variance of consumption declines for Savers, who benefit from diversification, and rises for Non-Savers, whose income depends on local labor markets. Finally, by varying the degree of financial participation and the extent of home bias across countries, I isolate the contribution of each dimension of integration and show that the main results are robust to structural asymmetries.

Extending the analysis to a fully heterogeneous-agent (HANK) framework confirms that endogenous precautionary savings and wealth heterogeneity further amplify these asymmetries, offering a microfounded explanation for the persistence of inequality within an integrated monetary union.

Literature review. This paper bridges two major strands of research: heterogeneous-agent macroeconomics and international risk-sharing through capital market integration.

The first strand of the literature highlights how household heterogeneity shapes aggregate dynamics and policy transmission. Models with incomplete markets and idiosyncratic income risk show that differences in wealth and portfolio composition affect consumption, saving, and welfare responses to shocks. The Heterogeneous-Agent New Keynesian (HANK) framework of [Kaplan et al. \[2018\]](#), extended to monetary unions by [Bayer et al. \[2024\]](#), has been central to this progress. Building on this foundation, studies such as [de Ferra et al. \[2020\]](#), [Auclert et al. \[2021\]](#), [Oskolkov \[2023\]](#), [Guo et al. \[2023\]](#), and [Acharya and Challe \[2025\]](#) examine how heterogeneity in asset holdings influences macroeconomic adjustment and policy effectiveness in open economies. Yet, this literature remains largely closed in terms of capital: households may borrow or lend abroad through bonds but cannot invest in foreign productive assets. My paper extends this work by introducing cross-border portfolio heterogeneity, allowing households to invest internationally and study how these choices shape macroeconomic adjustment.

A second strand of research focuses on how financial integration fosters international risk-sharing. Foundational contributions by [van Wincoop \[1999\]](#), [Obstfeld and Rogoff \[2001\]](#), [Heathcote and Perri \[2013\]](#), and [Coeurdacier and Rey \[2013\]](#) formalize how cross-border asset holdings allow consumption to be smoothed relative to domestic output. Within monetary unions, [Corsetti et al. \[2008\]](#), [Auray and Eyquem \[2014\]](#), [Farhi and Werning \[2017\]](#), [Senay and Sutherland \[2019\]](#), and [Cimadomo et al. \[2020\]](#) show that portfolio diversification can partly substitute for missing fiscal transfers in absorbing asymmetric shocks. More recent studies—[Blanchard et al. \[2017\]](#), [Almgren et al. \[2022\]](#), [Martin Fuentes et al. \[2023\]](#), and [Ferrari and Rogantini Picco \[2023\]](#)—highlight persistent Core–Periphery asymmetries in the euro area, where incomplete financial integration limits private risk-sharing. Yet this literature remains grounded in representative-agent frameworks that assume uniform market participation and therefore abstract from within-country heterogeneity. My paper departs from this approach by introducing household heterogeneity into the analysis of international risk-sharing. When access to capital markets is unequal, integration still stabilizes the union in aggregate but redistributes its benefits unevenly across households and countries. This result challenges the representative-agent intuition inherited from Mundell: in heterogeneous economies, financial integration does not necessarily equalize welfare or smooth consumption uniformly—it stabilizes some groups while exposing others to greater volatility.

A complementary strand of research examines capital flows and the global financial cycle, emphasizing that financial integration can amplify—rather than mitigate—cross-

country imbalances. [Lane \[2012\]](#) and [Rey \[2015\]](#) document the procyclicality of global capital flows, while [Broner et al. \[2014\]](#), [Caballero et al. \[2017\]](#), and [Maggiori \[2017\]](#) show how investors in safe-asset countries absorb shocks through portfolio reallocations, reinforcing asymmetries between creditor and debtor economies. Recent evidence by [Davis and van Wincoop \[2024\]](#) and [Beck et al. \[2024\]](#) indicates that these dynamics are particularly pronounced in the euro area, where flight-to-safety flows toward the Core have deepened intra-union divergence. My framework microfounds these mechanisms by endogenizing flight-to-safety behavior through heterogeneity in wealth and financial participation, thereby linking aggregate portfolio flows to the distribution of assets and inequality within countries.

Taken together, this paper provides the first unified framework that embeds household heterogeneity into models of international risk-sharing. It demonstrates that capital market integration—traditionally viewed as a stabilizing force—can at once mitigate aggregate volatility and amplify inequality, both within and across member states.

2 Empirical Evidence on Heterogeneous Effects of Capital Market Integration

This section documents heterogeneity in the effects of capital market integration across euro area countries. I begin by identifying two groups of euro area countries, defined as the Core and the Periphery, emerging from a clustering of structural characteristics observed in 2010. To analyze how integration has shaped wealth, income, and portfolio dynamics between 2010 and 2021, I adopt a dual empirical approach. At the household level, I use data from the Household Finance and Consumption Survey (HFCS) conducted by the European Central Bank, which provides detailed information on wealth, income, and asset composition. Since this dataset does not contain information on the geographical allocation of households' portfolios, I complement it with aggregate country-level data on cross-border equity and investment fund holdings from the Coordinated Portfolio Investment Survey (CPIS) of the International Monetary Fund. Combining these two data sources enables me to capture both within-country and between-country heterogeneity in exposure to capital market integration.

Two new stylized facts emerge. First, wealth distributions in the Core and the Periphery have diverged significantly between 2010 and 2021, driven primarily by increasing inequality at the lower end of the distribution. Second, capital stocks in the Core and the Periphery have also drifted apart from their pre-crisis trajectories, reflecting a sustained

shortfall in capital accumulation in the Periphery relative to the Core. To interpret these patterns, I distinguish two dimensions of integration. Within countries, poorer households are substantially less financialized, particularly in the Periphery. Between countries, portfolio diversification across euro area members remains limited: home bias is large and persistent, especially in the Core. Together, these findings highlight that inequality in exposure to capital market integration arises both within and between countries, underscoring the importance of accounting for these heterogeneous effects when analyzing the consequences of integration.

2.1 Core and Periphery: Classification Based on Pre-Crisis Imbalances

To capture the structural asymmetries that predated the euro area sovereign debt crisis, I identify two groups of countries, defined as the Core and the Periphery, emerging from a clustering of macroeconomic imbalances in 2010. The clustering relies on three key indicators: GDP per capita (Eurostat), Net International Investment Position (NIIP, Eurostat), and the current account balance as a share of GDP (ECB and Eurostat). These variables jointly summarize countries' relative income levels, external indebtedness, and financing needs on the eve of the crisis. I employ a two-step clustering approach. First, I apply Principal Component Analysis (PCA) to reduce dimensionality and capture the main sources of cross-country variation. Second, I perform k -means clustering (with two clusters and 50 random starts for robustness) on the first principal components to partition countries into two groups. The first cluster, labeled as the **Core**, includes Austria, Belgium, Finland, France, Germany, Luxembourg, and the Netherlands. The second cluster, labeled as the **Periphery**, includes Cyprus, Estonia, Greece, Ireland, Italy, Latvia, Lithuania, Malta, Portugal, Slovakia, Slovenia, and Spain.

Figure 1 demonstrates that Core countries are characterized by stronger external positions: higher current account balances, less negative (or positive) NIIPs, and higher GDP per capita. By contrast, Periphery countries exhibit persistent external deficits, large negative net positions vis-à-vis the rest of the world, and lower income levels. These patterns reflected expectations of real convergence toward Core countries, as Periphery members were receiving more investment inflows than their domestic savings could finance.

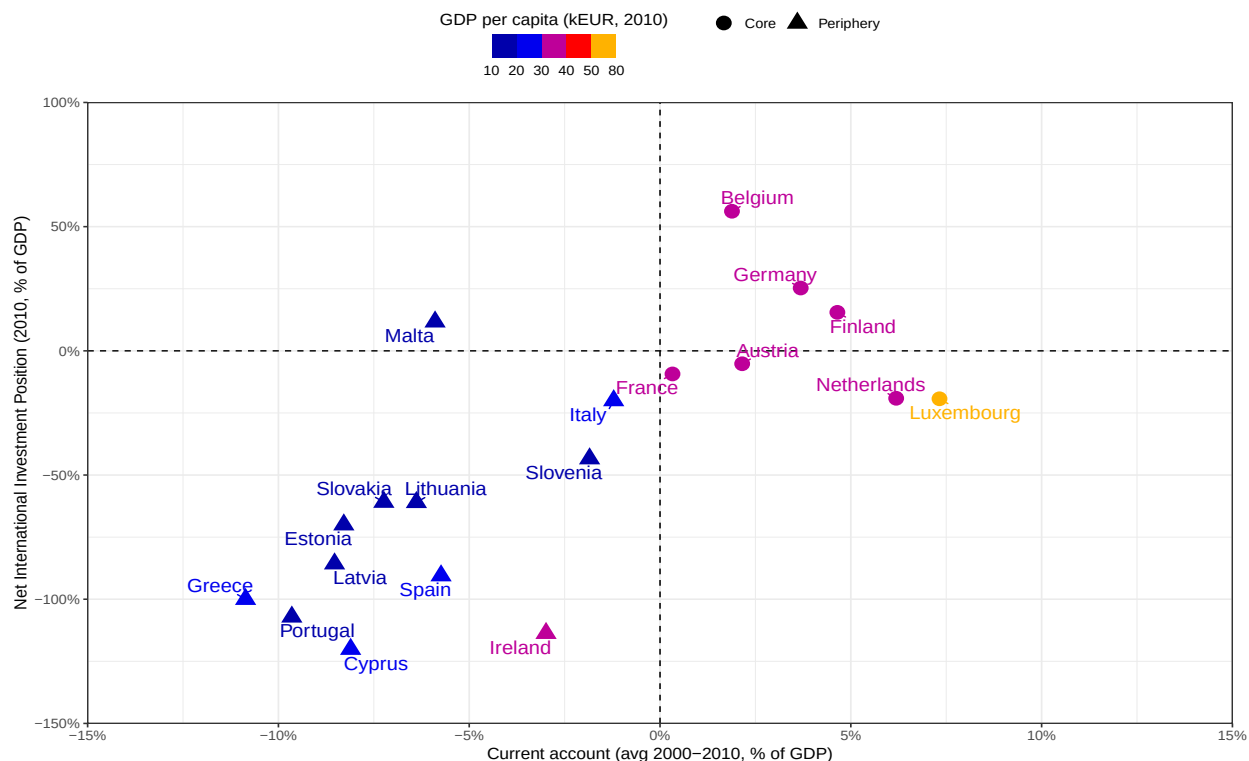


Figure 1: Clustering of Euro Area Countries in 2010. The figure plots euro area countries by their average current account balance (2000–2010, % of GDP, ECB & Eurostat) and their net international investment position (NIIP, 2010, % of GDP, Eurostat). Marker color reflects GDP per capita (in thousand EUR, 2010, Eurostat). Core countries (circles) and Periphery countries (triangles) are obtained from a PCA-*k*-means clustering based on these indicators.

Appendix A.1 provides complementary evidence and robustness checks. Using 2007 as a pre-crisis benchmark yields consistent cross-country patterns, confirming that the Core–Periphery division reflects pre-existing structural imbalances rather than crisis-driven adjustments. The classification remains stable across alternative specifications and aligns closely with Core–Periphery groupings commonly used in the literature⁵.

This Core–Periphery classification provides a natural framework to study how capital market integration has generated heterogeneous outcomes within the euro area. In the next subsection, I document how these structural differences translated into divergent wealth and capital accumulation dynamics over the 2010–2021 period.

⁵See, for example, Jaumotte and Sodsriwiboon [2010], Eggertsson et al. [2014], Alesina et al. [2017], Mazzocchi and Tamborini [2019], Cimadomo et al. [2020], Ferrari and Rogantini Picco [2023], Martin Fuentes et al. [2023], Kaufmann et al. [2023], and Gatti et al. [2024].

2.2 Two New Stylized Facts: Divergent Wealth and Capital Stocks

To assess heterogeneity in capital market integration both within and between countries, I combine household-level data from the ECB’s HFCS with aggregate cross-country data from the IMF’s CPIS.

2.2.1 Data

At the micro level, the HFCS provides household-level information on income, wealth, private pensions, and consumption across four waves (2010, 2014, 2017, 2021). I use these data to characterize wealth heterogeneity, financial participation, and portfolio diversification within each country. Because the HFCS lacks information on the geographic allocation of financial assets, I complement it with bilateral investment positions from the CPIS, which records cross-border holdings of equities and investment fund shares between countries. The focus is on the period 2010–2021 to align with HFCS survey waves.

I also use data from the European System of Central Banks (ESCB) on domestic holdings of listed shares and investment fund units, following [Beck et al. \[2024\]](#). All positions are converted into end-of-year U.S. dollars to ensure comparability. For countries with missing domestic data before 2012, I linearly retropolate values for 2010–2011 using country-specific trends. I then apply the correction proposed by [Beck et al. \[2024\]](#) to reallocate positions intermediated through Luxembourg, Ireland, and the Netherlands, using the Global Allocation database⁶. This adjustment introduces new bilateral links and significantly alters the distribution of holdings (Figure 25). Finally, I aggregate countries dynamically by group according to their euro area accession year: Estonia joins the Periphery from 2011, Latvia from 2014, and Lithuania from 2015.

⁶The correction proposed by [Beck et al. \[2024\]](#) addresses the misallocation of investment fund shares issued through onshore–offshore financial centers (OOFs) such as Luxembourg, Ireland, and the Netherlands. Using the European Central Bank’s *Securities Holdings Statistics* (SHS) combined with fund-level data from the Global Allocation Database and commercial sources, the procedure reallocates investment fund shares to their ultimate country of exposure. It does so by mapping each security to its ultimate parent entity and by distinguishing between holdings by euro area residents and those by the rest of the world. This correction mitigates the bias from financial intermediation through OOFs, yielding a more accurate measure of bilateral equity exposures within the euro area.

2.2.2 Diverging Wealth Distributions

Figure 2 plots the log change in total gross wealth⁷ between 2010 and 2021 across percentiles of the wealth distribution. In the Core, the entire distribution shifts upward, suggesting broad-based wealth gains and a tiny reduction of inequality. By contrast, in the Periphery, the top deciles experience moderate wealth gains while the bottom deciles lose significantly, leading to a widening of internal inequality. These dynamics mirror the divergent recovery trajectories following the sovereign debt crisis: between 2010 and 2017, the lower end of the distribution in the Periphery declined sharply, with only partial rebound by 2021.

The Theil decomposition⁸ (Figure 38) shows that within-country inequality accounts for nearly all euro area wealth inequality — 97.4% in 2010 and 97.1% in 2021 — indicating that disparities largely stem from heterogeneity within rather than between countries. Thus, understanding the impact of capital market integration requires accounting for household-level differences in financial exposure.

⁷Results are unchanged when using net wealth instead of gross wealth. As shown in Figure 37, the correlation between gross and net wealth by year and group is nearly one, indicating that the observed dynamics primarily reflect differences in asset accumulation rather than variations in household indebtedness.

⁸The Theil index measures the deviation of an observed income (or wealth) distribution from perfect equality. Higher values indicate greater inequality. Its main advantage is exact decomposability, allowing total inequality to be split into a within-group component (inequality within countries) and a between-group component (inequality across countries).

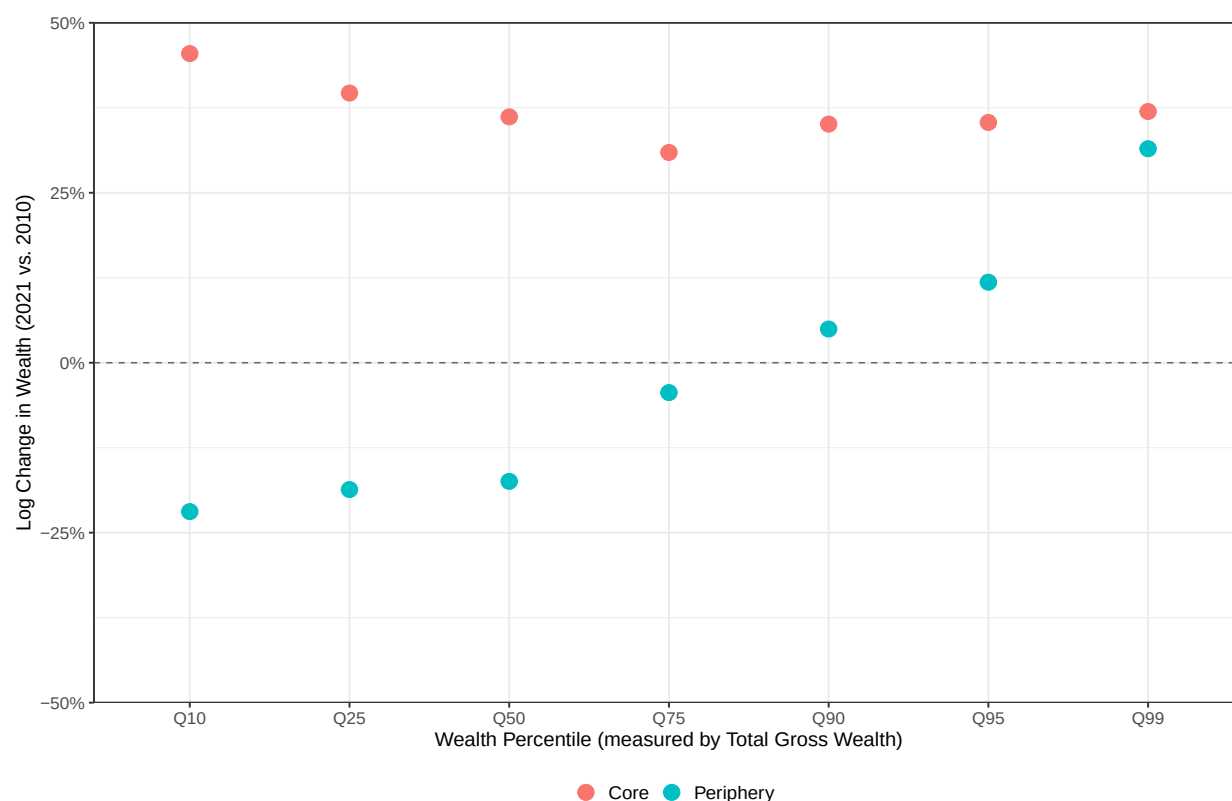


Figure 2: Change in Wealth Across the Wealth Distribution (2021 vs. 2010). The figure plots the log change in total gross wealth between 2010 and 2021 for households grouped by wealth percentile, comparing the Core (red) and the Periphery (blue).

Appendix [A.2.1](#) extends the analysis to financial wealth and income, showing that the widening Core–Periphery gap is driven primarily by lower-wealth households. Additional details on the distribution of wealth are reported in Appendix [A.2.2](#), and the relationship between income and wealth is further documented in Appendix [A.2.3](#).

2.2.3 Diverging Capital Stocks

Capital stock data⁹ (Eurostat) reveal a similar divergence at the macro level. Figure 3 plots the evolution of total net fixed assets (in current prices) relative to pre-crisis trends estimated over 2000–2007. Before the crisis, both groups exhibited broadly parallel trends,

⁹Capital stock data come from Eurostat’s dataset *Capital stocks by industry (NACE Rev.2) and detailed asset type*. The series measure net total fixed assets at current replacement costs, expressed in millions of euros and aggregated across all NACE activities (ESA 2010 definition). Figure 46 reports the same graph using gross total fixed assets.

with the Periphery growing slightly faster.¹⁰ After 2010, however, the Core’s capital stock recovered and outperformed its trend, while the Periphery’s stagnated below it. By 2021, the Periphery’s deviation from its pre-crisis trend reached 50 percentage points relative to the Core.

These patterns highlight a structural divergence in capital accumulation: the Core not only recovered but exceeded its pre-crisis trajectory, while the Periphery remained below. The asymmetry mirrors the wealth divergence documented above, suggesting that macro-level investment and micro-level wealth inequality are closely linked.

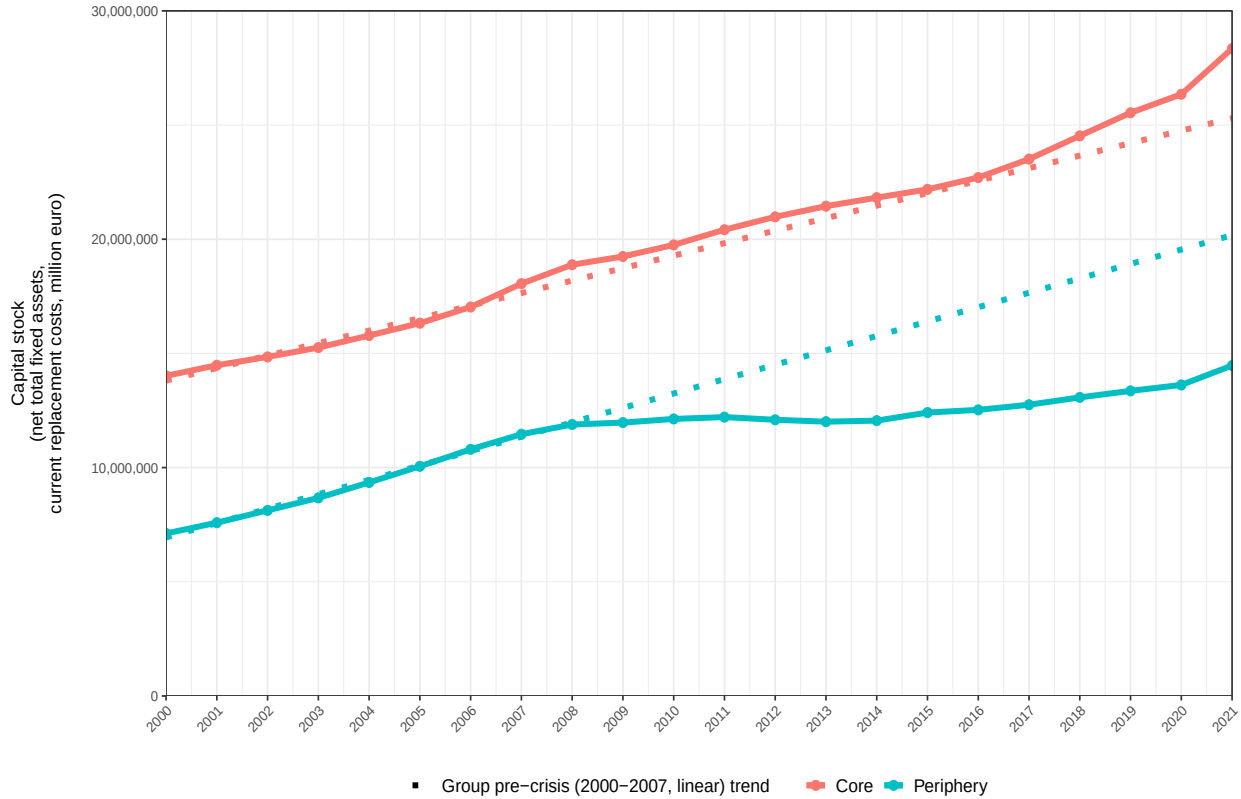


Figure 3: Capital Stock and Pre-Crisis Trends in the Core and Periphery (2000–2021). The figure plots total net fixed capital stock (million euros, current replacement costs) for the Core (red) and Periphery (blue). Dashed lines represent linear pre-crisis trends (2000–2007).

Appendix A.2.4 presents robustness checks using gross capital measures, sectoral decompositions, and country-level deviations, showing that the divergence in capital accu-

¹⁰The estimated pre-crisis slope for the Core is 547,271, while the interaction term ($Year \times Group_{Periphery}$) adds 83,112 (approximately 15% higher). These coefficients are obtained from an OLS regression of capital stock on year and its interaction with the group dummy over 2000–2007:

$$Capital_{it} = \alpha + \beta_1 Year_t + \beta_2 Group_{Periphery,i} + \beta_3 (Year_t \times Group_{Periphery,i}) + \varepsilon_{it}.$$

The interaction coefficient (β_3) measures the difference in pre-trend slopes; here $\beta_3/\beta_1 \approx 0.15$, with $p \approx 0.08$, indicating weak evidence that the Periphery’s capital was growing faster before 2007.

mulation is broad-based and not driven by any single country.

2.3 Two Dimensions of Integration

Having established divergence in both wealth and capital, I now examine the mechanisms driving these patterns through two complementary dimensions of integration — within and between countries. Within countries, poorer households are less financialized, particularly in the Periphery. Between countries, portfolio diversification remains limited, particularly among Core countries, whose portfolios exhibit a strong home bias. These two dimensions jointly explain the unequal transmission of shocks across the monetary union.

2.3.1 Within-Country: Financial Participation

Figure 4 illustrates the extensive and intensive margins of financial participation using HFCS data for 2010. The curve shows the share of households holding financial assets (extensive margin), while the bars show the share of financial assets in total wealth (intensive margin). Deposits are excluded, as they offer no cross-asset diversification and are insulated from international market conditions.

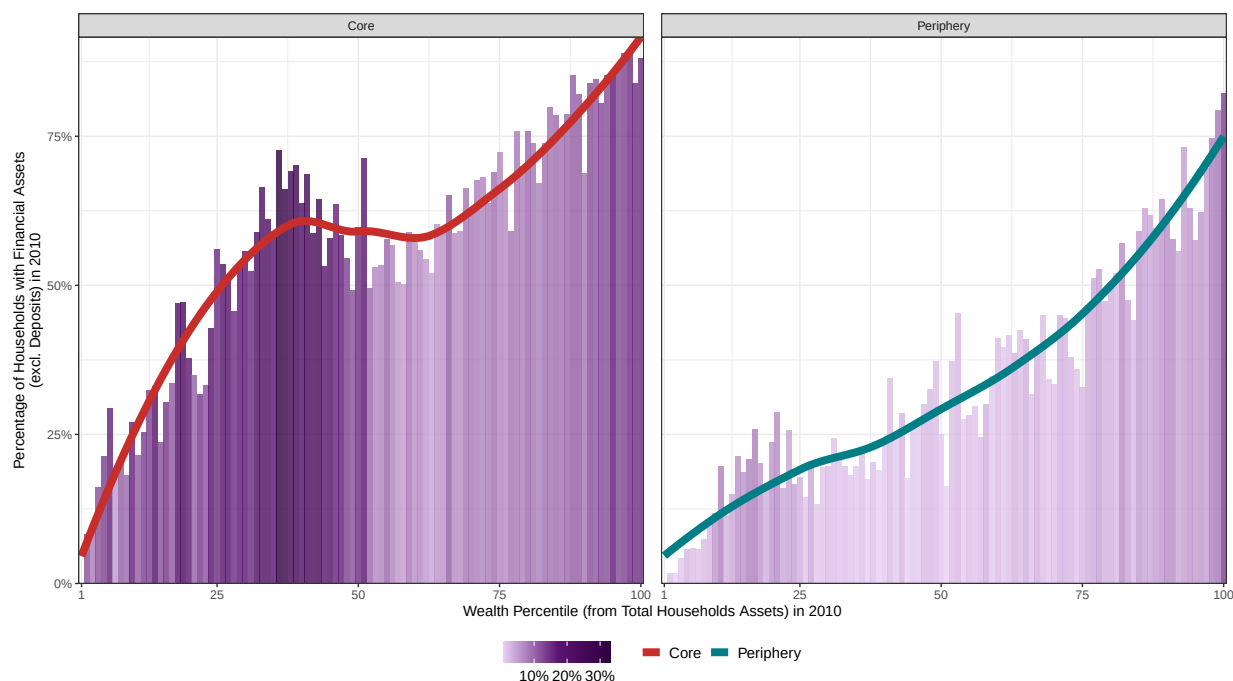


Figure 4: Financial Participation Along the Wealth Distribution (2010). The figure shows the share of households holding financial assets (curve, extensive margin) and the share of financial assets in total wealth (bars, intensive margin) across wealth percentiles.

Participation in financial markets rises steeply with wealth. In 2010, 43% of households in the Core and 67% in the Periphery held no financial assets other than deposits. The share of financial assets in total wealth also increases with wealth, reaching 13.6% in the Core and 3.8% in the Periphery. By 2021, participation improved slightly but remained highly unequal (Figure 56). Appendix A.3.1 replicates this figure for all survey waves. Overall, financialization is systematically deeper in the Core, where middle-wealth households tend to accumulate financial assets before purchasing housing, whereas portfolios in the Periphery remain dominated by real assets.

Appendix A.3.2 provides further detail on portfolio composition, confirming that real assets dominate at the bottom of the distribution, while voluntary pension funds and equities account for a growing share at the top, especially in the Core.

Sections 1–2 of the [Online Appendix](#) provide additional extensive-margin measures and further evidence of limited household diversification.

2.3.2 Between-Country: Diversification

A first proxy for household portfolio diversification is the aggregate investment portfolio, which captures both direct and indirect holdings through mutual funds and managed accounts. Using CPIS data, augmented with domestic holdings and adjusted for offshore-onshore centers, I compute the equity home bias following [Coeurdacier and Rey \[2013\]](#). The home bias equals one minus the share of foreign equities in country i 's portfolio relative to the share of foreign equities in the euro area market portfolio. The measure accounts for country size by comparing each country's foreign equity share to the euro area market portfolio, thereby controlling for the relative weight of domestic equities in the total market¹¹.

Figure 5 shows that diversification remains limited, particularly in the Core. In 2010, home bias stood at 86.96% in the Core and 84.68% in the Periphery. By 2021, the Core's bias was unchanged while the Periphery's fell to 75.69%. Hence, the Periphery became somewhat more diversified, but exposures remain asymmetric: the Periphery is more exposed to the Core than vice versa. This asymmetry constrains risk sharing and may amplify distributional consequences of asymmetric shocks within the monetary union.

¹¹Formally, from [Coeurdacier and Rey \[2013\]](#), $EHB_i = 1 - \frac{\text{Share of Foreign Equities in Country } i \text{ Equity Holdings}}{\text{Share of Foreign Equities in the Euro Area Market Portfolio}}$. A value of zero indicates full diversification consistent with the International CAPM; a value of one implies complete home bias.

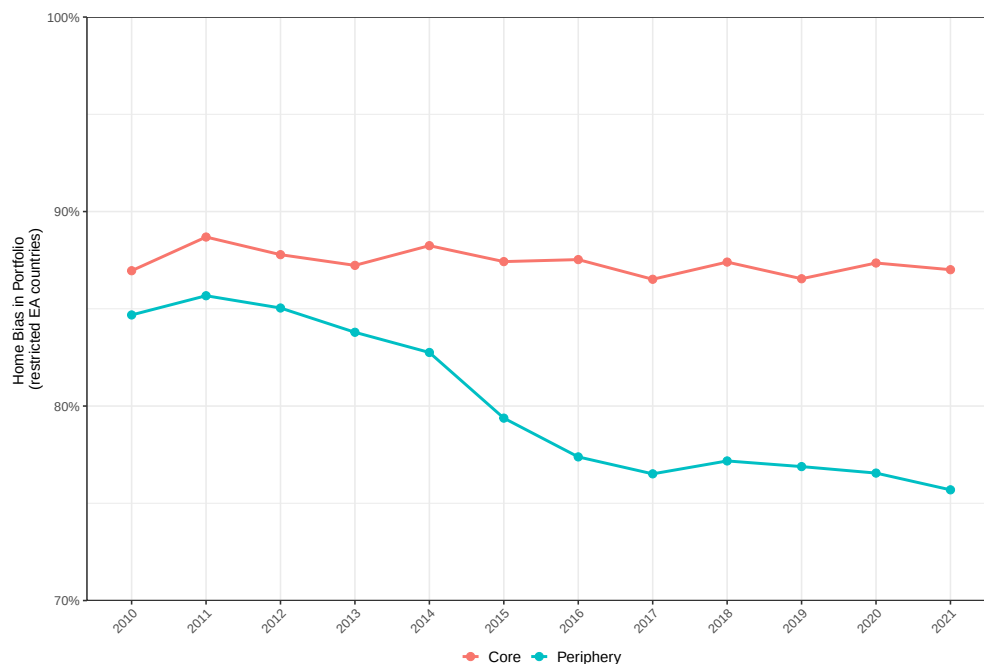


Figure 5: Home Bias in Portfolio Holdings (2010–2021). The figure shows the average share of domestic assets in households’ financial portfolios for Core and Periphery euro area countries, restricted to euro area assets only.

Additional results on portfolio composition, performance, and cross-country correlations—covering asset holdings, Sharpe ratio, returns, and consumption/GDP comovement—are reported in Sections 3–6 of the [Online Appendix](#).

Overall, the evidence reveals substantial heterogeneity in diversification both within and between countries. Within countries, limited financial participation restricts poorer households’ exposure to capital markets. Between countries, persistent home bias and asymmetric cross-border investment patterns constrain the scope for risk sharing across the monetary union. Core countries remain heavily invested in domestic and intra-Core assets, while Periphery countries have gradually diversified toward the Core but remain far less integrated overall.

These empirical findings point to a fundamental asymmetry in how capital market integration operates across the euro area. To interpret these patterns and assess their quantitative importance, the next section develops a unified modeling framework combining two complementary approaches: a Two-Agent New Keynesian (TANK) model enables to isolate the main mechanisms and quantify the role of capital market integration under controlled conditions, while a Heterogeneous-Agent New Keynesian (HANK) model extends the analysis to a richer environment with endogenous distributional dynamics.

3 Two-Agent New Keynesian (TANK) Model

3.1 Overview of the Model

The model features two countries, the Core (C) and the Periphery (P), forming a common monetary union calibrated to represent the corresponding euro area blocs. Each country produces a domestic good consumed both locally and abroad, with imperfect substitutability and country-specific import shares. Prices are flexible, while wages are sticky, ensuring zero profits each period so that the gains from diversification are not driven by countercyclical markups. A single central bank sets the nominal interest rate based on union-wide inflation. There is no migration between countries.

Each country $i \in C, P$ is populated by two household types $h \in N, S$: a share λ^i of Non-Savers (N) and $(1 - \lambda^i)$ of Savers (S). Setting $\lambda^i = 0$ collapses the model to a representative-agent (RA) framework. Non-Savers consume solely out of labor income and do not participate in asset markets, while Savers hold financial assets through country-specific, risk-neutral mutual funds. These funds pool domestic savings and allocate them between domestic and foreign capital according to relative returns. This structure departs from standard incomplete-markets frameworks, which typically restrict households to domestic (and occasionally foreign) bonds. Allowing funds to invest internationally is crucial to capture the mechanisms of capital market integration and cross-country portfolio reallocation.

The model features two distinct channels of international transmission. The first is the *diversification channel*, which captures average exposure to shocks and depends on the steady-state domestic portfolio share ζ^i . The second is the *reallocation channel*, which captures the ability to adjust portfolios in response to shocks and depends on the adjustment cost of deviating from the steady-state portfolio, denoted κ^i .

In this model, portfolio allocations are indeterminate at the steady state to a first-order approximation; therefore, the domestic share is set exogenously — although it could be rationalized through a zero-net portfolio condition — and is intended to capture structural determinants of home bias, such as preferences for domestic assets or portfolio frictions¹².

¹²Several mechanisms can rationalize home bias: (i) preferences and risk attitudes (CRRA vs. habit formation, recursive preferences; Merton [1969]; Campbell and Cochrane [1999]; Epstein and Zin [1989]; Coeurdacier and Rey [2013]); (ii) background risks from non-tradable income or housing (Heaton and Lucas [1997]; Heaton and Lucas [2000]; Baxter and Jermann [1997]; Curcuro et al. [2010]; Guiso et al. [2018]); (iii) market frictions such as fixed participation costs, transaction costs, taxes, or borrowing constraints (Constantinides [1986]; Vissing-Jørgensen [2002]; Martin and Rey [2004]; Du and Schreger [2022]); and (iv) institutional features such as restricted pension allocations, employer stock holdings, or informational asymmetries (Curcuro et al. [2010]; Sialm et al. [2015]; Lambert et al. [2024]).

In the dynamic setting, following shocks that alter relative returns, mutual funds adjust their portfolio composition endogenously, subject to costs of deviating from their initial portfolio shares. These adjustment costs can be interpreted as transaction or management fees typically incurred when rebalancing portfolios. This framework captures both the static and dynamic dimensions of portfolio allocation, allowing the model to quantify how capital market integration shapes macroeconomic adjustment and distributional outcomes across countries.

3.2 Risk-Neutral Mutual Fund

In each country i , a risk-neutral mutual fund intermediates all savings of domestic Savers. It collects the returns on its portfolio of interest-bearing assets and issues equity claims that households purchase. The fund allocates its portfolio between two assets: domestic capital, which produces the domestic good, and foreign capital, which produces the foreign good. Allowing mutual funds to invest across borders generates endogenous capital flows and net foreign asset positions, mechanisms that would otherwise be absent in a monetary union with a single common debt instrument.

The total equity value of the fund in country i , denoted E_t^i , equals the sum of its domestic and foreign capital holdings, $K_{i,t}^i$ and $K_{i,t}^j$:

$$E_t^i = K_{i,t}^i + S_t K_{i,t}^j,$$

where S_t denotes the relative price of country j 's consumption basket in terms of country i 's:

$$S_t \equiv \frac{CPI_t^j}{CPI_t^i}.$$

At each period, the fund chooses its portfolio allocation between the two assets based on their expected returns. The no-arbitrage (zero-profit) condition for the mutual fund in

country i is, under flexible prices,¹³:

$$r_t^i E_{t-1}^i = (r_t^{k,i} + 1 - \delta^i) K_{i,t-1}^i + (r_t^{k,j} + 1 - \delta^j) S_t K_{i,t-1}^j - \kappa^i \left(\frac{K_{i,t}^i}{E_t^i} - \zeta^i \right)^2 E_{t-1}^i,$$

where $\zeta^i \equiv \frac{\bar{K}_i^i}{\bar{E}^i}$ denotes the steady-state domestic portfolio share, and κ^i captures portfolio adjustment costs associated with deviations from ζ^i .

The fund's first-order condition implies a no-arbitrage relationship between expected returns on domestic and foreign capital:

$$\mathbb{E}_t[r_{t+1}^{k,i} + 1 - \delta^i] = \mathbb{E}_t \left[(r_{t+1}^{k,j} + 1 - \delta^j) \frac{S_{t+1}}{S_t} \right] + \kappa^i \left(\frac{K_{i,t}^i}{E_t^i} - \zeta^i \right).$$

A higher κ^i implies greater rigidity in portfolio adjustment, reflecting higher transaction costs or financial frictions. This parameter is country-specific, capturing structural differences in market flexibility.

Finally, capital in country i is jointly owned by domestic and foreign mutual funds:

$$K_t^i = K_{i,t}^i + K_{j,t}^i,$$

with investment dynamics given by

$$\begin{aligned} I_{i,t}^{F,i} &= K_{i,t}^i - (1 - \delta^i) K_{i,t-1}^i, \\ I_{j,t}^{F,i} &= K_{j,t}^i - (1 - \delta^j) K_{j,t-1}^i. \end{aligned}$$

Total investment in each country is thus the sum of domestic and foreign investments in its capital stock, linking portfolio decisions to aggregate capital accumulation.

¹³Under sticky prices, profits need not be zero, and the no-arbitrage condition can be written as:

$$r_t^i E_{t-1}^i = (r_t^{k,i} + 1 - \delta^i) K_{i,t-1}^i + (r_t^{k,j} + 1 - \delta^j) S_t K_{i,t-1}^j + \frac{K_{i,t-1}^i}{K_{i-1}^i} \Omega_t^i + \frac{K_{i,t-1}^j}{K_{j-1}^j} S_t \Omega_t^j - \frac{\kappa^{A,i}}{2} (K_{i,t-1}^i - \bar{K}_i^i)^2.$$

In this case, Ω_t^i denotes nominal profits in country i , and the last term represents portfolio adjustment costs relative to the steady-state allocation. Flexible prices imply $\Omega_t^i = 0$, leading to the simplified expression in the main text.

3.3 Households

3.3.1 Intratemporal Problem

In each country i , households consume a composite bundle of domestic and foreign goods:

$$C_t^i = \left[(1 - \omega_i)^{\frac{1}{\phi}} (C_{i,t}^i)^{\frac{\phi-1}{\phi}} + \omega_i^{\frac{1}{\phi}} (C_{j,t}^i)^{\frac{\phi-1}{\phi}} \right]^{\frac{\phi}{\phi-1}},$$

where $C_{i,t}^i$ and $C_{j,t}^i$ denote, respectively, domestic and imported goods consumed by households in country i . The parameter $\phi > 0$ is the elasticity of substitution between goods, and $0 \leq \omega_i < 1$ is the import share.

Under purchasing power parity (PPP), the composite price index simplifies to:

$$CPI_t^i = \left[(1 - \omega_i) P_{i,t}^{1-\phi} + \omega_i P_{j,t}^{1-\phi} \right]^{\frac{1}{1-\phi}},$$

where $P_{i,t}$ and $P_{j,t}$ are the prices of domestic and imported goods, respectively.

The intratemporal optimal allocation yields:

$$C_{i,t}^i = (1 - \omega_i) \left(\frac{P_{i,t}}{CPI_t^i} \right)^{-\phi} C_t^i, \quad C_{j,t}^i = \omega_i \left(\frac{P_{j,t}}{CPI_t^i} \right)^{-\phi} C_t^i.$$

Investment bundles are defined analogously, implying that the investment deflator equals the CPI.

Normalizing domestic prices by each country's CPI ($p_t^i \equiv P_{i,t}/CPI_t^i$), the CPI equations can be written as:

$$1 = (1 - \omega_i) (p_t^i)^{1-\phi} + \omega_i (p_t^j S_t)^{1-\phi}.$$

as S_t denotes the relative price of country j 's consumption basket in terms of country i 's.

3.3.2 Intertemporal Problem

Each household h in country i maximizes expected lifetime utility:

$$E_0 \sum_{t=0}^{\infty} \beta^t \left[\frac{(C_{h,t}^i)^{1-\sigma}}{1-\sigma} - \chi \frac{(L_t^{U,i})^{1+\eta}}{1+\eta} \right],$$

where $\beta \in (0, 1)$, $\sigma, \eta > 0$, and $\chi > 0$. Labor supply $L_t^{U,i}$ is set by unions and common across households within each country.

Savers. A share $(1 - \lambda^i)$ of households are Savers. They choose consumption $C_{S,t}^i$ and invest in (i) equity claims issued by the domestic mutual fund, $E_{S,t}^i$, and (ii) nominal risk-free bonds, $B_{S,t}^i$, which are in zero net supply at the union level. Real bond holdings are $b_{S,t}^i = B_{S,t}^i / CPI_t^i$. Savers earn a real return r_t^i on equity and a nominal return r_{t-1}^U on bonds, set by the common central bank. Because both countries share a monetary union, bonds yield identical returns across countries¹⁴.

The real budget constraint is:

$$C_{S,t}^i + E_{S,t}^i + b_{S,t}^i = w_t^i \Lambda L_t^{U,i} + r_t^i E_{S,t-1}^i + b_{S,t-1}^i \frac{r_{t-1}^U}{\pi_{CPI,i}^i} - \frac{\kappa_E^i}{2} p_t^i (E_{S,t}^i - \bar{E}_S^i)^2 - \frac{\kappa_D^i}{2} p_t^i (b_{S,t}^i)^2,$$

where $\Lambda \geq 1$ captures the higher productivity of Savers relative to Non-Savers, and κ_E^i and κ_D^i represent portfolio adjustment costs à la [Schmitt-Grohe and Uribe \[2003\]](#). The first-order conditions are:

$$\begin{aligned} (C_{S,t}^i)^{-\sigma} &= \lambda_t^i, \\ (C_{S,t}^i)^{-\sigma} \left(1 + \kappa_E^i p_t^i (E_{S,t}^i - \bar{E}_S^i) \right) &= \beta E_t \left[(C_{S,t+1}^i)^{-\sigma} r_{t+1}^i \right], \\ (C_{S,t}^i)^{-\sigma} \left(1 + \kappa_D^i p_t^i b_{S,t}^i \right) &= \beta E_t \left[(C_{S,t+1}^i)^{-\sigma} \frac{r_t^U}{\pi_{t+1}^{CPI,i}^i} \right]. \end{aligned}$$

Non-Savers. A share λ^i of households are Non-Savers. They consume their entire labor income each period:

$$C_t^{i,N} = w_t^i L_t^{U,i}.$$

As they have no access to financial markets, their consumption is determined by labor income alone.

3.4 Labor Supply and Wage Setting

In each country i , labor is supplied monopolistically by a continuum of unions indexed by $k \in (0, 1)$, similar to [Erceg et al. \[2000\]](#). A competitive labor aggregator combines union-specific inputs using a [Dixit and Stiglitz \[1977\]](#) technology with elasticity of substitution

¹⁴Since the common central bank sets a single nominal policy rate, domestic and foreign bonds are perfectly substitutable within the monetary union. Consequently, it is not meaningful to track country-specific bond holdings—only aggregate bond positions at the union level matter. This modeling choice follows [Bayer et al. \[2024\]](#), who make the same assumption in their analysis of a monetary union, noting that both government bonds pay the same interest rate and can therefore be treated as identical assets.

$\theta_W^i > 1$. The aggregate nominal wage index is then defined as:

$$W_t^i = \left(\int_0^1 [W_t^i(k)]^{1-\theta_W^i} dk \right)^{\frac{1}{1-\theta_W^i}},$$

and the corresponding labor demand faced by each union is:

$$L_t^{U,i}(k) = \left(\frac{W_t^i(k)}{W_t^i} \right)^{-\theta_W^i} L_t^{U,i}.$$

Each union sets its nominal wage $W_t^i(k)$ subject to quadratic adjustment costs à la Rotemberg [1982]:

$$UC_t^i(k) = \frac{\kappa_W^i}{2} \left(\frac{W_t^i(k)}{W_{t-1}^i(k)} - \bar{\pi}^W \right)^2,$$

where κ_W^i governs wage rigidity and $\bar{\pi}^W$ denotes steady-state wage inflation.

In a symmetric equilibrium, all unions set the same wage $W_t^i(k) = W_t^i$ and supply the same labor input $L_t^{U,i}(k) = L_t^{U,i}$. Gross wage inflation in country i satisfies:

$$\pi_t^{W,i} = \frac{W_t^i}{W_{t-1}^i} = \frac{w_t^i}{w_{t-1}^i} \pi_t^{CPI,i}.$$

The union's optimal wage-setting condition yields a New Keynesian Wage Phillips Curve of the form:

$$\begin{aligned} \pi_t^{W,i} (\pi_t^{W,i} - \bar{\pi}^W) &= \beta \mathbb{E}_t \left[(\pi_{t+1}^{W,i} - \bar{\pi}^W) \pi_{t+1}^{W,i} \right] \\ &+ \frac{\theta_W^i}{\kappa_W^i} \left[\chi (L_t^{U,i})^{1+\eta} - \frac{\theta_W^i - 1}{\theta_W^i} w_t^i L_t^{U,i} (\lambda^i (C_{H,t}^i)^{-\sigma} + (1 - \lambda^i) (C_{S,t}^i)^{-\sigma} \Lambda) \right]. \end{aligned}$$

This condition links nominal wage inflation to expected future inflation and the deviation of the marginal rate of substitution between consumption and leisure from the real wage. It captures the degree of real wage rigidity in each country, reflecting both structural labor market frictions and the heterogeneity of household consumption responses.

3.5 Production

Each country $i \in \{C, P\}$ produces a differentiated good under perfect competition with flexible prices. This assumption eliminates countercyclical profits that would otherwise

arise under price rigidities and allows the analysis to focus on risk-sharing and income heterogeneity rather than nominal distortions. Production follows a standard two-tier structure with monopolistically competitive intermediate firms and a representative final-good producer.

Final-good firms. Final output in country i is a Dixit–Stiglitz aggregate of a continuum of intermediate varieties:

$$Y_t^i = \left(\int_0^1 [Y_t^i(j)]^{\frac{\theta_p^i-1}{\theta_p^i}} dj \right)^{\frac{\theta_p^i}{\theta_p^i-1}},$$

where $\theta_p^i > 1$ is the elasticity of substitution across intermediate goods. Profit maximization yields the standard demand function $Y_t^i(j) = (P_{i,t}(j)/P_{i,t})^{-\theta_p^i} Y_t^i$, and the associated price index

$$P_{i,t} = \left(\int_0^1 [P_{i,t}(j)]^{1-\theta_p^i} dj \right)^{\frac{1}{1-\theta_p^i}}.$$

Intermediate firms. Each firm j in country i produces according to a Cobb–Douglas technology:

$$Y_t^i(j) = Z_t^i [K_{t-1}^i(j)]^\alpha [L_t^i(j)]^{1-\alpha},$$

where Z_t^i denotes total factor productivity (TFP), common across firms. Cost minimization implies the standard factor-price conditions:

$$w_t^i = (1-\alpha) \zeta_t^i \frac{Y_t^i(j)}{L_t^i(j)}, \quad r_t^{k,i} = \alpha \zeta_t^i \frac{Y_t^i(j)}{K_{t-1}^i(j)},$$

where ζ_t^i is the real marginal cost. Substituting these conditions gives

$$\zeta_t^i = \frac{1}{Z_t^i} \left(\frac{r_t^{k,i}}{\alpha} \right)^\alpha \left(\frac{w_t^i}{1-\alpha} \right)^{1-\alpha}.$$

Aggregating across firms yields the aggregate production function:

$$Y_t^i = Z_t^i (K_{t-1}^i)^\alpha (L_t^i)^{1-\alpha}, \quad \frac{K_{t-1}^i}{L_t^i} = \frac{\alpha}{1-\alpha} \frac{w_t^i}{r_t^{k,i}}.$$

Price setting and investment. Under flexible prices, firms set $P_{i,t}(j)$ as a constant markup over marginal cost:

$$P_{i,t} = \frac{\theta_P^i}{\theta_P^i - 1} (1 - \tau_t^i) CPI_t^i \zeta_t^i.$$

Setting $\tau_t^i = 1/\theta_P^i$ removes steady-state distortions, implying $P_{i,t} = CPI_t^i \zeta_t^i$ and zero profits in equilibrium. The domestic inflation rate is $\pi_t^i = P_{i,t}/P_{i,t-1}$, and capital evolves according to:

$$K_t^i = (1 - \delta^i) K_{t-1}^i + I_t^i, \quad I_t^i = I_{i,t}^{F,i} + I_{i,t}^{F,j}.$$

Investment is a CES bundle of domestic and imported goods, consistent with the structure of consumption.

Technology Shock. Each country i experiences an exogenous productivity process:

$$u_t^i = \rho^i u_{t-1}^i + \epsilon_t^i, \quad \epsilon_t^i \sim \mathcal{N}(0, \sigma_{z,i}^2),$$

where $\rho^i \in (0, 1)$ captures the persistence of technology shocks.

TFP evolves as:

$$Z_t^i = Z_0^i e^{-u_t^i}.$$

The history of aggregate shocks up to period t in country i is denoted $z^{i,t}$.

3.6 Monetary Policy

Monetary policy is conducted at the union level by a single central bank that sets a common nominal interest rate, r_t^U , applying to both the Core and the Periphery. The policy rate responds to union-wide inflation, measured as a weighted average of national inflation rates:

$$\pi_t^{HICP} = (\pi_t^{CPI,C})^\gamma (\pi_t^{CPI,P})^{1-\gamma},$$

where γ captures the relative weight of the Core in the union's consumption basket—reflecting its economic size or relative nominal rigidity. The central bank follows a standard Taylor-type rule:

$$r_t^U = \bar{r} + \Psi(\pi_t^{HICP} - 1),$$

where $\bar{r}$ denotes the steady-state nominal rate and Ψ the policy response to inflation deviations. For simplicity, I abstract from monetary shocks, though they could be readily incorporated.

3.7 Market Clearing Conditions

In each country $i \in \{C, P\}$, aggregate variables are obtained by summing over Savers and Non-Savers. Aggregate consumption and labor supply are given by:

$$C_t^i = \lambda^i C_{N,t}^i + (1 - \lambda^i) C_{S,t}^i, \quad L_t^i = [\lambda^i + (1 - \lambda^i)\Lambda] L_t^{U,i},$$

where $\Lambda \geq 1$ captures the productivity (and wage) premium of Savers.

Aggregate asset holdings reflect only Savers' portfolios:

$$E_t^i = (1 - \lambda^i) E_{S,t}^i, \quad b_t^i = (1 - \lambda^i) b_{S,t}^i.$$

The union-wide bond market clears as:

$$b_t^i + S_t b_t^j = 0,$$

where S_t is the real exchange rate.

The final good produced in country i is allocated to domestic and foreign consumption, investment, and portfolio adjustment costs:

$$Y_t^i = C_{i,t}^i + I_{i,t}^i + C_{i,t}^j + I_{i,t}^j + \frac{\kappa^{E,i}}{2(1 - \lambda^i)} (E_t^i - \bar{E}^i)^2 + \frac{\kappa^{D,i}}{2(1 - \lambda^i)} (b_t^i)^2.$$

The trade balance equals the value of exports minus imports:

$$TB_t^i = (C_{i,t}^j + I_{i,t}^j) p_t^i - (C_{j,t}^i + I_{j,t}^i) p_t^j S_t.$$

Linking real and financial accounts gives:

$$TB_t^i = b_t^i - b_{t-1}^i \frac{r_{t-1}^i}{\pi_t^{CPI,i}} + r_t^{k,i} K_{j,t-1}^i - r_t^{k,j} K_{i,t-1}^j S_t + S_t I_{j,t}^{F,i} - I_{i,t}^{F,j} + \frac{\kappa^i}{2} \left(\frac{K_{i,t-1}^i}{E_{t-1}^i} - \frac{\bar{K}_i^i}{\bar{E}^i} \right)^2 E_{t-1}^i.$$

Finally, the aggregate resource constraint in country i reads:

$$C_t^i + I_t^i + TB_t^i = p_t^i Y_t^i - \frac{\kappa^{E,i}}{2(1 - \lambda^i)} p_t^i (E_t^i - \bar{E}^i)^2 - \frac{\kappa^{D,i}}{2(1 - \lambda^i)} p_t^i (b_t^i)^2.$$

3.8 Equilibrium definition

Definition 1 (Monetary Union Equilibrium). A sequential competitive equilibrium is a collection of individual functions $\{C_{S,t}^i, C_{N,t}^i, E_{S,t}^i, b_{S,t}^i, L_t^{U,i}\}_{i \in \{C,P\}, t \geq 0}$, aggregate quantities $\{C_t^i, L_t^i, K_t^i, E_t^i, Y_t^i, I_t^i, TB_t^i\}_{i \in \{C,P\}, t \geq 0}$, portfolio positions $\{K_{i,t}^i, K_{i,t}^j\}_{i,j \in \{C,P\}, t \geq 0}$, and price processes $\{w_t^i, r_t^{k,i}, r_t^i, P_{i,t}, CPI_t^i, S_t, r_t^U\}_{i \in \{C,P\}, t \geq 0}$, such that, for given initial asset positions and exogenous processes $\{Z_t^i\}_{i \in \{C,P\}, t \geq 0}$, the following conditions hold for all $t \geq 0$:

1. **Households.** In each country i , Savers and Non-Savers solve their intertemporal optimization problems given prices and policy sequences. Savers choose consumption and asset holdings to satisfy the first-order conditions implied by their budget constraint, while Non-Savers consume their current labor income.
2. **Mutual funds.** The risk-neutral mutual fund in each country i allocates its portfolio between domestic and foreign capital to maximize expected returns net of portfolio adjustment costs, subject to the zero-profit and no-arbitrage conditions.
3. **Firms.** In each country i , firms hire labor and rent capital competitively to maximize profits, taking prices and productivity as given. Factor prices equal marginal products, and under flexible prices, firms earn zero profits.
4. **Wage setting.** Labor unions set nominal wages subject to Rotemberg adjustment costs, and the resulting wage inflation path satisfies the wage Phillips curve.
5. **Monetary policy.** The common central bank sets the nominal interest rate according to a Taylor-type rule responding to union-wide inflation, defined as a weighted average of national CPI inflations.
6. **Market clearing.** For each country i , goods, labor, and financial markets clear. Capital equals the sum of domestic and foreign holdings, and the union-wide bond market clears.

3.9 Channels

Capital market integration in the model operates through two distinct channels of international transmission: *diversification* and *reallocation*. Both channels are mediated by the structure of the risk-neutral mutual fund in each country, which determines how savings are allocated across domestic and foreign capital.

The first channel, **diversification**, captures average exposure to shocks and depends on the steady-state domestic portfolio share, ζ^i . Formally,

$$\zeta^i \equiv \frac{\bar{K}_i^i}{\bar{E}^i},$$

which represents the ratio of domestic capital to total equity holdings in the steady state. In a benchmark economy without foreign capital holdings, households can only invest domestically, corresponding to full home bias ($\zeta^i = 1$) and no diversification. In contrast, a lower ζ^i implies greater cross-border diversification and therefore lower exposure to domestic shocks. In this framework, ζ^i is treated as an exogenous, calibrated target, reflecting structural determinants of home bias such as preferences, market frictions, or institutional constraints¹⁵.

The second channel, **reallocation**, captures the dynamic adjustment of portfolios in response to shocks and depends on the adjustment cost parameter κ^i . This parameter determines how costly it is for investors to deviate from their steady-state portfolio composition. A higher κ^i implies greater rigidity, preventing reallocation and leading to fixed portfolio shares similar to models without international asset trade, whereas a lower κ^i reflects more flexible portfolios that can be rebalanced in response to changing relative returns.

The mutual fund's no-arbitrage condition illustrates this mechanism:

$$\mathbb{E}_t[r_{t+1}^{k,i} + 1 - \delta^i] = \mathbb{E}_t\left[(r_{t+1}^{k,j} + 1 - \delta^j) \frac{S_{t+1}}{S_t}\right] + \kappa^i \left(\frac{K_{i,t}^i}{E_t^i} - \zeta^i\right),$$

showing that deviations from the steady-state domestic share ζ^i are penalized proportionally to κ^i . The larger the adjustment cost, the slower the portfolio response to shocks.

Together, these two channels — diversification and reallocation — characterize how financial integration shapes macroeconomic and distributional outcomes. The parameter ζ^i determines average exposure to country-specific shocks, while κ^i governs the dynamic flexibility of capital reallocation. Varying these two parameters allows the model to trace how different degrees of integration affect the transmission of shocks and the heterogeneity in their effects across and within countries.

¹⁵See footnote 12.

4 Results - Two-Agent Model

In this section, I use the Two-Agent New Keynesian (TANK) model for two complementary purposes.

First, I calibrate the model to match the structural characteristics of the euro area in 2010. This calibration provides a realistic benchmark to evaluate how capital market integration and household heterogeneity shape macroeconomic adjustment and welfare. In particular, I contrast my model with two standard counterparts: (i) a setting where inequality is calibrated, comparing empirically consistent portfolio structures with an equity autarky alternative featuring full home bias and infinitely rigid portfolios; and (ii) a setting where portfolio structures are calibrated, comparing realistic inequality to a representative-agent benchmark that abstracts from unequal exposure to capital income. This calibrated environment allows me to quantify the relative importance of the diversification and reallocation channels and to assess how varying degrees of financial integration and inequality affect both aggregate stability and the distribution of welfare.

Second, I use the model as a controlled laboratory to isolate the underlying mechanisms at play. I consider a symmetric two-country monetary union, so that there is no ex-ante heterogeneity, and introduce an asymmetric technology shock affecting only one country. This experiment clarifies the transmission channels by disentangling the roles of within-country inequality (driven by the share of Non-Savers) and between-country inequality (determined by portfolio integration). The symmetric setup highlights how capital market integration shapes the propagation of shocks across countries and income groups in the absence of structural asymmetries.

4.1 Calibration

The calibration of parameters that remain constant across all cases is presented in the table below, while case-specific parameters are introduced at the beginning of each corresponding section. The overall calibration follows standard values commonly used in the literature.

Table 1: Calibration Parameters

Parameter	Value	Description	Source
β	0.99	Discount factor	Smets and Wouters [2003]
α	0.36	Capital share	Krueger et al. [2017]
σ	2.0	Relative risk aversion	CRRA
ϕ	2.0	Elasticity of substitution (home vs. foreign)	Pasch and Tervala [2024]
δ	0.025	Depreciation rate (quarterly)	Smets and Wouters [2003]
η	2.0	Inverse Frisch elasticity of labor supply	Chetty et al. [2011]
Ψ	1.5	Taylor rule coefficient on inflation	Darvas and Merler [2013]
κ_D	0.1	Debt adjustment cost	Schmitt-Grohe and Uribe [2003]
κ_E	0.1	Equity adjustment cost	Schmitt-Grohe and Uribe [2003]
ρ_z	0.95	TFP autocorrelation	Den Haan [2010]
Λ	1.0	Saver wage premium	Normalization
θ_W	21.0	Wage markup elasticity	Bilbiie and Ragot [2021]
κ_W	2100	Wage adjustment cost	Bilbiie and Ragot [2021]
Steady-State Normalization			
GDP^C	1.0	Gross nominal product in Core	Normalization
L^C	1/3	Hours worked in Core	Normalization

Additional technical details on the steady-state determination of portfolios are provided in Appendix C.1.

4.2 Calibrated Case for 2010 with Asymmetric Shock

This section calibrates the model to represent the euro area in 2010. The calibration targets two key empirical asymmetries: heterogeneity in financial participation within countries and heterogeneity in portfolio composition between countries.

Within countries, the share of Non-Savers is lower in the Core (43%) than in the Periphery (67%), consistent with cross-country differences in financial inclusion and market participation. Between countries, the Core exhibits a stronger home bias in equity holdings (97%) than the Periphery (89%). Import shares are computed as the ratio of imports to total absorption, aggregated by bloc and normalized by GDP shares, implying higher trade openness in the Periphery (39%) than in the Core (33%). Portfolio adjustment costs are set to 0.1 in both countries. The inflation weight, derived from Eurostat HICP data, assigns a higher policy weight to Core inflation (60%) than to Periphery inflation (40%).

The relative GDP ratio is calibrated to match the observed equity ratio between the Periphery and the Core (0.23) in the data. Country-specific negative technology shocks are calibrated to reproduce the observed deviations from capital linear pre-crisis trends (2000–2010) one year after the shock: a decline of 9.85% in the capital of the Periphery and

0.46% in the capital of the Core.

Table 2: Calibration of Parameters for Core and Periphery (2010)

Parameter	Core	Periphery	Source
Domestic portfolio share	$\zeta^C = 0.97$	$\zeta^P = 0.89$	CPIS (IMF), corrected with SHS, Beck et al. [2024]
Share of Non-Savers	$\lambda^C = 0.43$	$\lambda^P = 0.67$	HFCS (ECB, 2010 wave)
Import share	$\omega^C = 0.33$	$\omega^P = 0.39$	Eurostat, national accounts (imports/GDP)
Portfolio adjustment cost	$\kappa^C = 0.10$	$\kappa^P = 0.10$	Calibrated (see Section 4.1)
Inflation weight in union	$\gamma = 0.60$	$1 - \gamma = 0.40$	ECB (HICP weights, 2010)
TFP shock standard deviation	$\sigma_z^P = 0.456$	$\sigma_z^C = 0.0078$	Q4 decrease from pre-crisis trend in 2011
GDP ratio $\frac{GDP^P}{GDP^C}$	$\psi = 0.23$		Equity ratios $\frac{E^P}{E^C}$ from CPIS (IMF)

Capital market integration. To quantify the macroeconomic and distributional implications of capital market integration, I compare the calibrated 2010 configuration with a counterfactual equity-autarky benchmark that maintains the same inequality structure. The autarky case corresponds to a standard macroeconomic environment in which households can invest only in domestic capital, implying full home bias and no cross-border reallocation.

Table 3: Simulation Scenarios: Equity Autarky vs. 2010 Calibration

Case	ζ		κ	
	Core	Periphery	Core	Periphery
Equity autarky	1.0	1.0	∞	∞
Calibrated (2010)	0.97	0.89	0.1	0.1

Figure 6 compares the impulse responses of capital and consumption for Savers and Non-Savers under both configurations. The dotted lines correspond to equity autarky, while the solid lines represent the calibrated 2010 economy.

Relative to autarky, capital market integration amplifies cross-country divergence in both investment and consumption dynamics. Following a negative shock in both countries, larger in the Periphery, capital declines sharply in the Periphery (−132.5 percentage points) and rises in the Core (+32.2 percentage points), as savings are reallocated toward the higher-return region.

For Savers, the adjustment reflects the interplay between diversification and reallocation. In the Periphery, Savers experience a markedly smaller decline in consumption (+135.3 percentage points relative to autarky) owing to income insurance from Core asset

holdings. In contrast, Core Savers face a larger decline (−17.7 percentage points) due to their exposure to the Periphery downturn. Yet, because the Periphery is more diversified than the Core, the gains for Periphery Savers substantially outweigh the losses for Core Savers.

For Non-Savers, the transmission operates through labor income. The sharper contraction in Periphery capital depresses labor demand and real wages (−34.5 percentage points), reducing Non-Saver consumption. In the Core, higher capital inflows stimulate labor demand, moderating the wage decline and slightly increasing Non-Saver consumption (+9.5 percentage points).

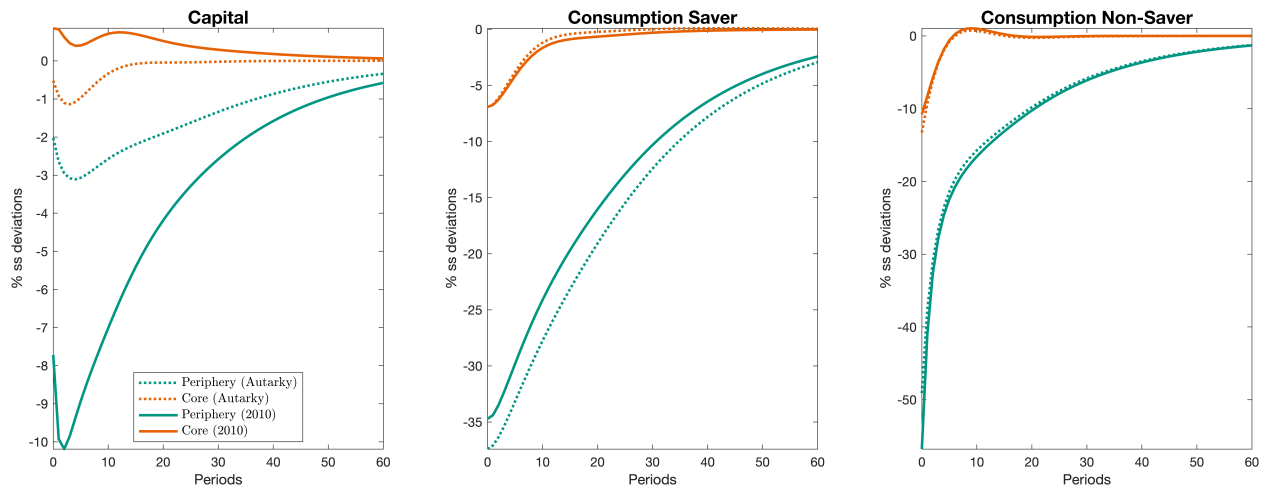


Figure 6: Impulse Responses of Capital and Consumption for Savers and Non-Savers in the Core and Periphery. Comparison between equity autarky (dotted) and calibrated 2010 (solid) configurations.

The welfare decomposition in Table 4 confirms these patterns. Capital market integration generates heterogeneous welfare effects across countries and household types. Savers in the Periphery experience substantial welfare gains, reflecting their ability to access higher-return assets and diversify away from local shocks once financial markets are integrated. Core Non-Savers also benefit, as cross-country risk sharing stabilizes consumption and mitigates the impact of domestic fluctuations. In contrast, Core Savers incur modest welfare losses due to lower returns on domestic assets following capital outflows, while Periphery Non-Savers experience welfare declines driven by adverse labor market adjustments and limited exposure to the capital income gains associated with integration. Overall, these results highlight that the distributional consequences of financial integration depend critically on households' portfolio composition and their position in the income distribution.

Table 4: Welfare Decomposition Relative to Autarky (Discounted Utilities)

Household	ΔU^C	ΔU^{Total}	CEV (%)
Capital market integration			
Core Saver	-4.470	-3.940	-3.278
Core Non-Saver	+8.236	+8.767	+5.082
Periphery Saver	+48.862	+47.079	+8.387
Periphery Non-Saver	-12.916	-14.700	-1.512

Consistent with the empirical evidence presented earlier, the calibrated model predicts that integration reduces cross-country inequality among Savers but widens within-country inequality, as the welfare gap between Savers and Non-Savers expands in the Periphery and narrows in the Core.

Appendix C.2 reports results under an alternative and more standard calibration, featuring a 1% decline in productivity in the Periphery only (a MIT shock).

Inequality. To isolate the role of household heterogeneity, I compare the calibrated 2010 economy with a counterfactual representative-agent benchmark, keeping the portfolio structure unchanged. The representative-agent case abstracts from financial participation heterogeneity by assuming universal market access ($\lambda^C = \lambda^P = 0$). This comparison identifies how the presence of liquidity-constrained households modifies the transmission of shocks through the capital allocation channel.

Table 5: Simulation Scenarios: Representative Agent vs. 2010 Calibration

Case	Core	Periphery
Representative agent	0.0	0.0
Calibrated (2010)	0.43	0.67

Figure 7 compares the impulse responses under the representative-agent and heterogeneous-household settings. Introducing household heterogeneity amplifies the asymmetry of macroeconomic adjustments across countries. In the Periphery, the decline in capital is significantly attenuated, by 57 percentage points, reflecting weaker aggregate demand when a large share of households is liquidity-constrained and unable to smooth consumption. In the Core, by contrast, capital rises by about 20 percentage points relative to the representative-agent benchmark, as excess savings from unconstrained households are reallocated toward the higher-return region.

The same mechanism shapes the consumption responses. In the Periphery, liquidity constraints magnify the fall in aggregate demand: Non-Savers cut consumption sharply following the adverse technology shock, while Savers experience a sizeable drop in consumption (−31 percentage points relative to the representative-agent case) due to lower returns on domestic assets. In the Core, unconstrained households increase savings and capital flows toward the Periphery, dampening the local recession but transmitting part of the adjustment through asset returns. Savers in the Core thus face a larger decline in consumption (−25 percentage points), while Non-Savers benefit from improved labor market conditions and higher real wages. Together, these dynamics imply that heterogeneity widens inequality within the Periphery—where constrained households bear most of the adjustment—but narrows it modestly within the Core, where Non-Savers benefit from stronger labor income.

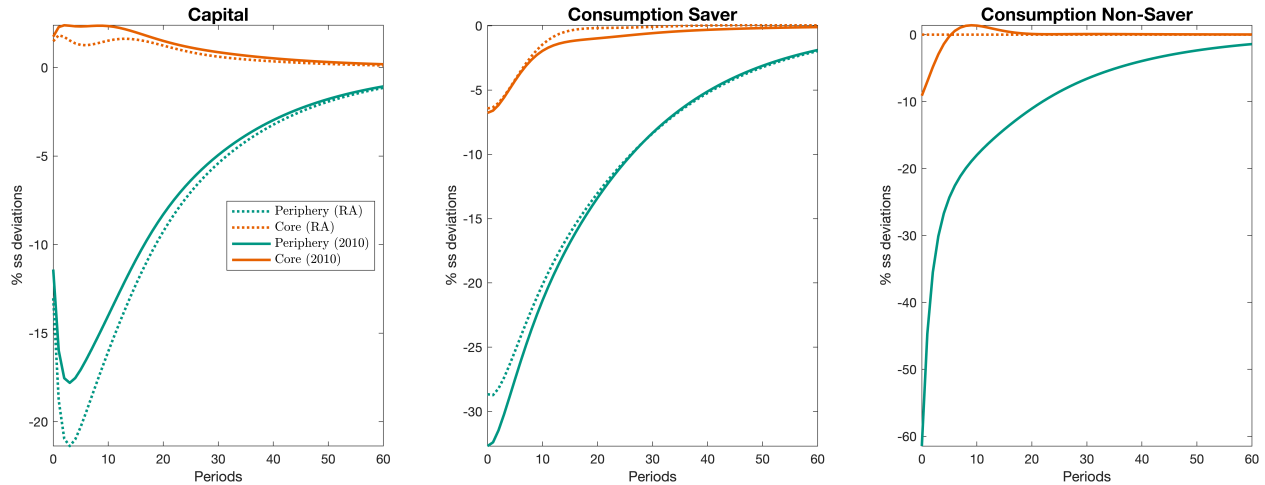


Figure 7: Impulse Responses of Capital and Consumption for Savers and Non-Savers in the Core and Periphery. Comparison between representative agent (dotted) and calibrated 2010 (solid) configurations.

The welfare decomposition in Table 6 corroborates these findings. The introduction of inequality generates asymmetric welfare effects across both countries and household types. Periphery Savers experience substantial gains (+25.6%), reflecting higher returns and improved diversification opportunities, while Core Non-Savers also benefit from stabilized consumption. In contrast, Core Savers face modest welfare losses due to reduced asset returns, and Periphery Non-Savers incur welfare declines driven by weaker labor markets and limited access to capital income. Overall, inequality strengthens cross-country asymmetries and redistributes welfare within countries, highlighting the central role of financial heterogeneity in shaping adjustment and redistribution in a monetary

union.

Table 6: Welfare Decomposition Relative to Autarky (Discounted Utilities)

Household	ΔU^C	ΔU^{Total}	CEV (%)
Inequality			
Core Saver	+10.201	+10.698	+7.480
Core Non-Saver	+4.069	+4.566	+2.511
Periphery Saver	+149.172	+137.788	+25.606
Periphery Non-Saver	+3.804	-7.580	+0.445

Overall, household heterogeneity amplifies both the cross-country and within-country asymmetries generated by capital reallocation. The presence of liquidity-constrained households deepens the downturn in the Periphery and reinforces capital accumulation in the Core, magnifying divergences in consumption, wages, and welfare. Compared with the representative-agent benchmark, the heterogeneous framework provides a microfoundation for the asymmetric adjustment observed during euro area crises: financially constrained households transmit local shocks more strongly through their limited consumption smoothing, while wealthier agents amplify cross-border spillovers through portfolio reallocation. This mechanism underscores that both inequality and financial integration must be modeled jointly to capture the full macroeconomic and distributional consequences of asymmetric shocks within a monetary union.

Appendix C.2 reports results under an alternative and more standard calibration, featuring a 1% decline in productivity in the Periphery only (a MIT shock).

To conclude, it appears that realistic calibration reveals that capital market integration mitigates aggregate volatility but redistributes its benefits unevenly. While integration improves insurance for financially connected agents, it amplifies inequality among households excluded from asset markets. Accounting for both within- and between-country heterogeneity is therefore crucial to understanding how financial integration shapes the distributional consequences of asymmetric shocks in a monetary union.

4.3 Understanding the Channels

In this section, I use my framework as a laboratory to illustrate the mechanisms through which capital market integration operates. I first analyze a symmetric monetary union, where countries are identical ex-ante and only one country experiences a negative technology shock. This setting isolates the transmission channels of diversification and reallocation in their pure form, abstracting from structural asymmetries. I then extend the

analysis to an asymmetric environment, introducing heterogeneity in financial participation and portfolio composition to assess how these dimensions shape the distributional effects of integration.

4.3.1 Symmetric Case with Asymmetric Shock

To isolate the mechanisms through which capital market integration shapes the transmission of asymmetric shocks and inequality, I begin with a symmetric two-country benchmark. Both countries share identical structural parameters: a 30% import share and a 50% share of Savers. This setup abstracts from ex-ante heterogeneity and captures a realistic degree of trade and wealth heterogeneity within a monetary union.

An asymmetric technology shock is then applied to the Periphery, while the Core remains unaffected. The shock is modeled as an unanticipated, temporary 1% decline in productivity (MIT shock). This specification provides a natural experiment that mimics the dynamics of the euro area crisis, combining a recession and a decline in the return to capital in the affected economy, while preserving comparability across countries.

I consider three configurations that successively introduce diversification and reallocation of portfolios across countries:

1. **Equity autarky:** households can only invest in domestic capital, implying full home bias ($\zeta^C = \zeta^P = 1$) and infinitely costly reallocation ($\kappa^C = \kappa^P = 10,000$). This case corresponds to the standard closed-economy benchmark.
2. **Diversification only:** portfolios are equally split between domestic and foreign capital ($\zeta^C = \zeta^P = 0.5$), but reallocation remains prohibited through infinitely costly adjustment ($\kappa^C = \kappa^P = 10,000$). This case isolates the insurance role of diversification under fixed portfolio weights.
3. **Capital market integration:** both **diversification** and **reallocation** are active. Portfolios are initially diversified ($\zeta^C = \zeta^P = 0.5$), and adjustment costs are low ($\kappa^C = \kappa^P = 0.1$), allowing mutual funds to rebalance across countries in response to shocks. This value aligns with other portfolio adjustment parameters in the model and prevents near-unit-root behavior in portfolio dynamics.

Table 7: Simulation Scenarios under Symmetry

Case	ζ (domestic portfolio share)	κ (portfolio adjustment cost)
Equity autarky	1.0 (full home bias)	10,000
Diversification only	0.5	10,000
Capital market integration	0.5	0.1

The following figures present the impulse responses of capital and consumption for the Core and the Periphery under each configuration. Additional model variables, shown both in absolute terms and relative to the equity autarky case, are reported in Appendix C.3.

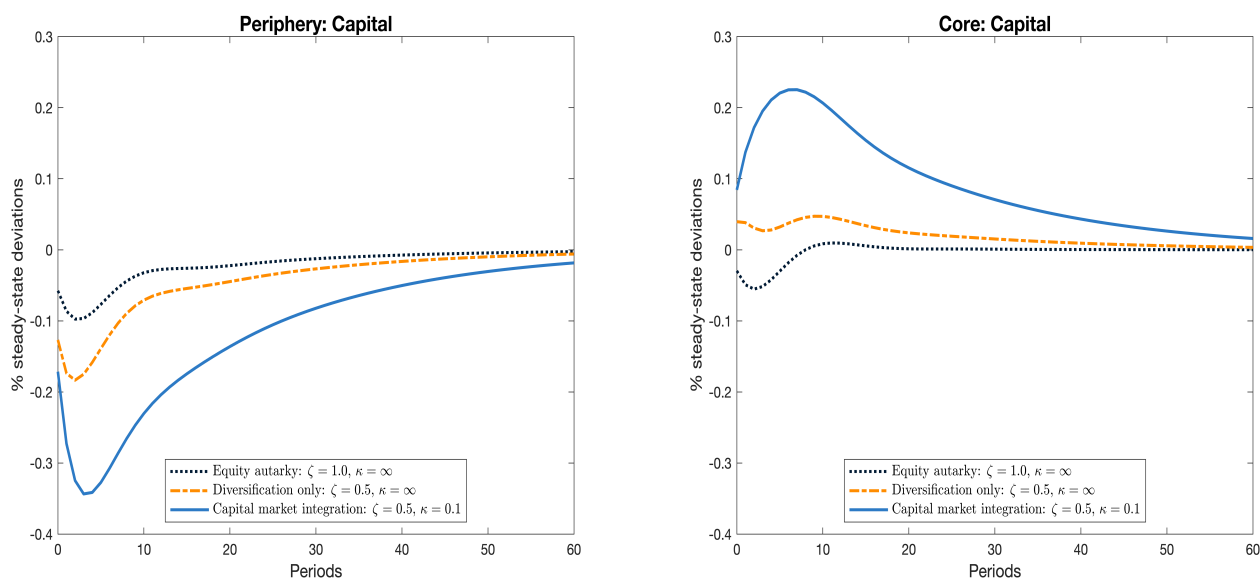


Figure 8: Impulse Response of Capital in the Core and the Periphery

In the equity autarky benchmark, a negative productivity shock in the Periphery reduces capital in both countries, with a larger contraction in the directly affected economy. Because households cannot diversify, the adjustment is purely domestic: investment falls, wages decline, and consumption drops across all households.

Introducing diversification, while keeping portfolio reallocation fixed, modifies this pattern through valuation effects. When Periphery productivity falls, domestic prices rise relative to the Core, reducing the real value of Periphery assets and increasing that of Core assets. Even though the portfolio composition remains constant, these valuation changes mechanically redistribute wealth across countries. This is the essence of the diversification channel: foreign asset holdings provide partial insurance by smoothing financial income across regions without altering the allocation of capital.

Allowing both diversification and portfolio reallocation introduces a second mechanism—the reallocation channel—which operates through active portfolio adjustment. As mutual funds respond to changing relative returns, capital flows toward the Core, where returns are temporarily higher, and away from the Periphery, where productivity and asset values have fallen. This endogenous reallocation amplifies cross-country divergence in capital accumulation: the Core attracts additional investment, while the Periphery experiences further capital losses.

Together, these two channels explain how integration transmits and magnifies shocks across borders. Diversification provides income insurance for asset holders by stabilizing returns, whereas reallocation amplifies real and distributional asymmetries by shifting productive resources toward stronger economies.

In the Periphery, these two channels operate asymmetrically across households. Diversification provides insurance to Savers, whose foreign asset holdings generate income unaffected by the domestic shock. However, reallocation amplifies the downturn by inducing capital outflows, which depress labor demand and wages. As a result, Non-Savers—who rely solely on labor income—face sharper consumption losses. Hence, while integration stabilizes aggregate financial income, it deepens the gap between Savers and Non-Savers, magnifying within-country inequality.

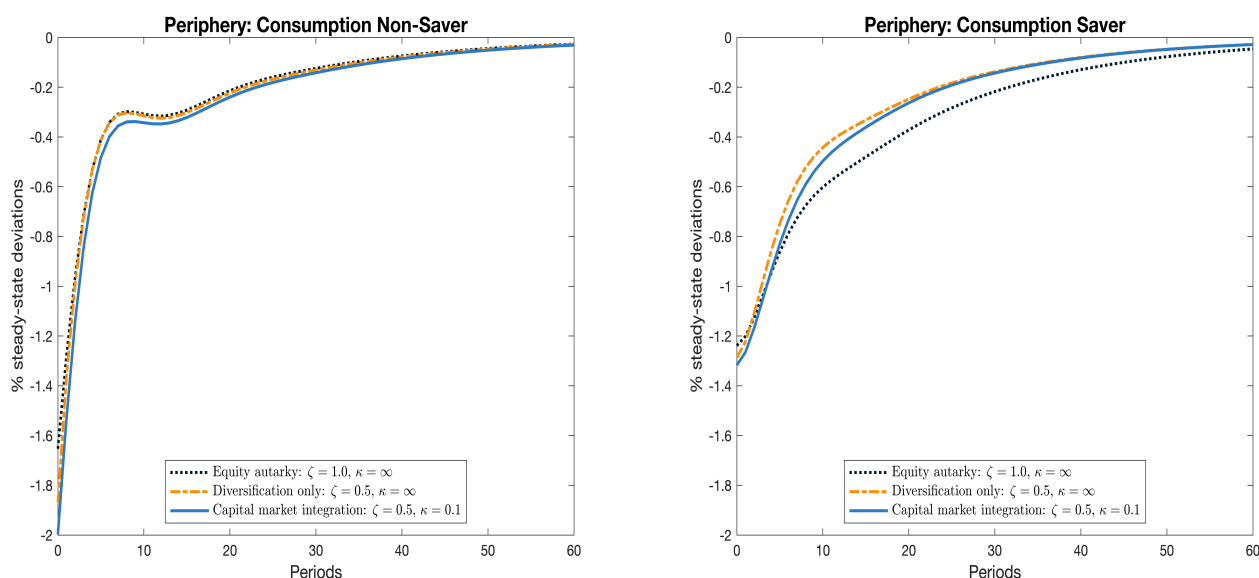


Figure 9: Impulse Response of Consumption for Non-Saver and Saver in the Periphery

In the Core, the same mechanisms operate in reverse. Diversification exposes Savers to foreign losses, as part of their portfolio is invested in the affected Periphery. Reallocation, however, attracts capital inflows that boost domestic investment and labor demand, raising wages and moderating the fall in Non-Savers' consumption. Thus, integration reduces inequality within the Core, as Non-Savers benefit from higher labor income while Savers face lower returns on capital.

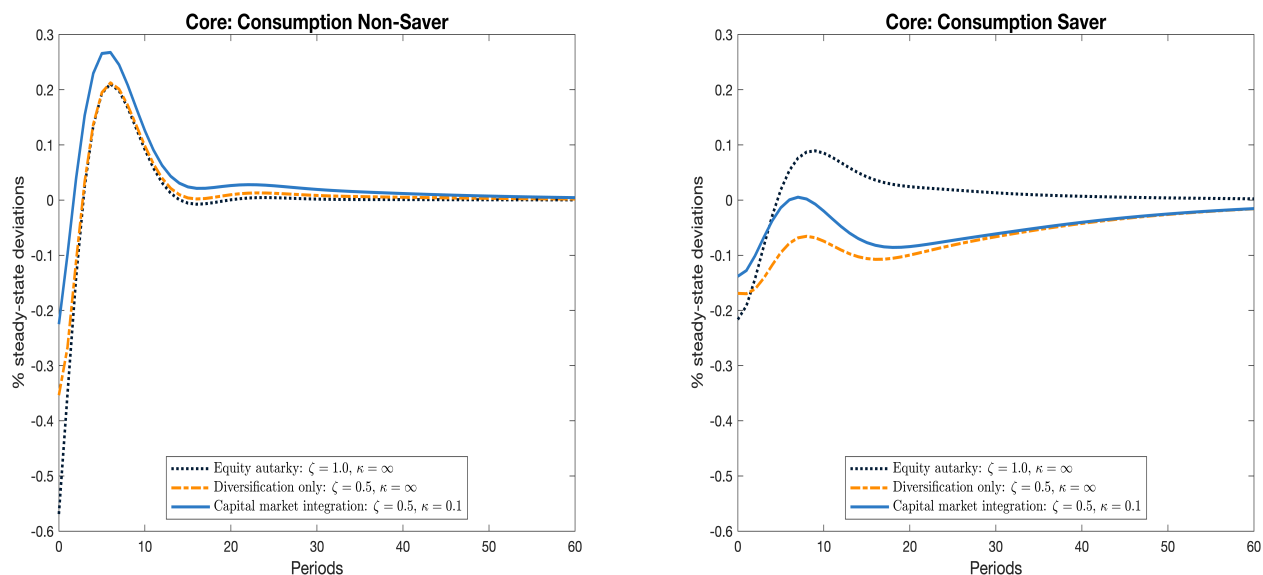


Figure 10: Impulse Response of Consumption for Non-Saver and Saver in the Core

Overall, diversification cushions aggregate fluctuations, while reallocation redistributes welfare both within and across countries. Capital market integration stabilizes the monetary union in the aggregate but amplifies inequality: Savers in the Periphery and Non-Savers in the Core gain, while liquidity-constrained households in debtor regions lose.

The novel insight of the model lies in the spillover to Non-Savers. Through capital reallocation, Non-Savers in the Periphery face a sharper decline in real wages as domestic capital falls, while Non-Savers in the Core benefit from higher capital inflows, which moderate the fall in their real wage relative to the autarky case. Consequently, capital market integration dampens aggregate volatility but amplifies the distributional wedge between households depending on their financial participation.

Computing welfare changes as consumption-equivalent variations relative to equity autarky demonstrates similar patterns. Under both diversification and full capital market integration, welfare improves for Savers in the Periphery and for Non-Savers in the Core, while it declines for Savers in the Core and Non-Savers in the Periphery. The welfare decomposition, reported in Table 8, confirms that the sign and magnitude of these effects align with the transmission channels described above.

Table 8: Welfare Decomposition Relative to Autarky (Discounted Utilities)

Household	ΔU^C	ΔU^{Total}	CEV (%)
Diversification only			
Core Saver	-0.411	-0.524	-0.444
Core Non-Saver	+0.084	-0.029	-0.018
Periphery Saver	+0.478	+0.584	+0.486
Periphery Non-Saver	-0.108	-0.001	-0.001
Capital market integration			
Core Saver	-0.300	-0.439	-0.372
Core Non-Saver	+0.263	+0.125	+0.080
Periphery Saver	+0.340	+0.459	+0.381
Periphery Non-Saver	-0.340	-0.221	-0.140

Additional figures regarding welfare decomposition are provided in Appendix C.4.

4.3.2 Symmetric Case with Uncorrelated Asymmetric Shocks

To assess the long-term stabilization properties of financial integration, I simulate uncorrelated technology shocks in both countries. This setting removes any systematic advantage or disadvantage associated with being the country directly hit by the shock and allows for an evaluation of the insurance properties of capital market integration. Given the structural symmetry of the two economies, the analysis focuses on the variance of consumption across household types, Savers and Non-Savers, relative to the equity autarky benchmark, which assumes full home bias and infinitely costly portfolio adjustment. This benchmark corresponds to a standard DSGE environment in which households can invest only in domestic capital.

Figure 11 reports the change in consumption variance relative to the autarky benchmark, with levels shown in Appendix C.5. The results indicate that capital market integration consistently reduces consumption volatility for Savers. The largest decline occurs under the diversification channel, where exposure to foreign financial income provides insurance without the wage-side trade-offs that arise under reallocation. In contrast, Non-Savers experience higher consumption volatility, with the largest increase when both channels are active. In this case, portfolio reallocation in response to idiosyncratic shocks amplifies wage fluctuations, destabilizing the income of Non-Savers, who rely entirely on labor earnings. Even under diversification alone, valuation effects associated with asymmetric capital accumulation across countries induce wage volatility and raise consumption variance for Non-Savers.

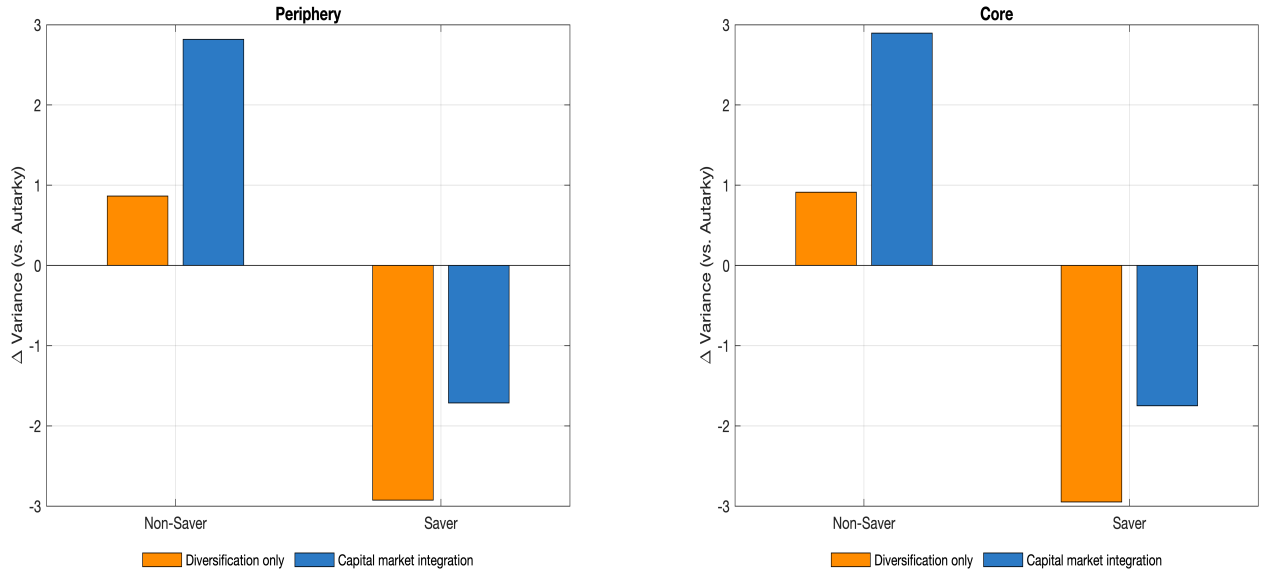


Figure 11: Variance of Consumption Deviations Relative to Equity Autarky

Overall, capital market integration has opposite effects across household types: it enhances risk-sharing and stabilizes consumption for Savers, while amplifying income risk for Non-Savers. This asymmetry highlights that the aggregate benefits of integration mask important distributional trade-offs, underscoring the need to account for within-country heterogeneity when assessing the welfare and policy implications of financial integration.

4.3.3 Asymmetric Case with Asymmetric Shock

To assess how structural heterogeneity shapes the distributional effects of capital market integration, I explore two dimensions: differences in the share of Non-Savers and differences in home bias across countries. The full results and figures are reported in Appendix C.6.

Varying the share of Non-Savers, which captures inequality in financial participation, shows that integration disproportionately benefits Savers while leaving Non-Savers largely unaffected. Because Savers alone hold financial assets, the aggregate gains from diversification become increasingly concentrated as their share declines. Non-Savers, relying solely on labor income, experience little direct change, as capital and wage dynamics remain broadly stable across configurations. Hence, the distributional effects of integration depend mainly on the overall prevalence of financial participation, rather than on its cross-country asymmetry.

Differences in home bias generate an additional layer of asymmetry through the portfolio structure. When the country hit by the shock is more diversified, its Savers benefit from foreign income that stabilizes domestic demand, moderating wage losses and cushioning Non-Saver consumption. Conversely, when the shocked country is more home-biased, the absence of foreign income insurance amplifies the contraction domestically and transmits it abroad through weaker demand, worsening outcomes for Non-Savers in both countries.

Together, these experiments highlight that heterogeneity in both financial participation and portfolio composition critically shapes the distributional consequences of capital market integration. The share of Non-Savers determines how aggregate gains are distributed across the population, while home bias governs how shocks are transmitted across borders through financial linkages. In combination, these dimensions explain why integration can stabilize the union as a whole while amplifying inequality within and between countries.

5 Heterogeneous-Agents New Keynesian (HANK) Model

The model extends the Two-Agent New Keynesian (TANK) framework by introducing incomplete markets and household heterogeneity, while preserving the aggregate structure described above. The Heterogeneous-Agents New Keynesian (HANK) version departs from the baseline along three dimensions.

First, households are subject to uninsurable idiosyncratic shocks to labor productivity, generating a non-degenerate stationary distribution of income and wealth.

Second, the presence of idiosyncratic risk gives rise to precautionary saving motives and endogenous market participation. Agents self-insure against income fluctuations by accumulating buffer-stock savings subject to borrowing constraints, which endogenously determine the fraction of households that are liquidity-constrained at any point in time. This mechanism generates a continuous distribution of marginal propensities to consume and savings behavior within each country, allowing the model to capture the contribution of within-country inequality to aggregate dynamics. In particular, the interaction between heterogeneous balance-sheet exposure and aggregate shocks amplifies the asymmetry of adjustment across households and shapes the redistributive consequences of capital market integration.

Finally, each country's risk-neutral mutual fund intermediates aggregate savings across all assets, domestic and foreign capital and a union-wide bond, issuing a composite claim to households. This formulation preserves the international portfolio structure of the

baseline model while embedding heterogeneity in wealth, consumption responses, and marginal propensities to consume.

5.1 Aggregate and Idiosyncratic Risk

Aggregate productivity follows an AR(1) process:

$$Z_t = e^{z_t}, \quad z_t = \rho_z z_{t-1} + \varepsilon_t, \quad \varepsilon_t \sim \mathcal{N}(0, \sigma_z^2),$$

and represents the sole aggregate source of uncertainty in the economy.

In addition, each household is subject to an idiosyncratic, uninsurable productivity shock $y_t \in \mathcal{Y} = \{y_1, \dots, y_n\}$, which follows a first-order Markov process with transition probabilities $\pi_{yy'}$. The invariant distribution S_y satisfies

$$S_y = \sum_{\tilde{y} \in \mathcal{Y}} \pi_{\tilde{y}y} S_{\tilde{y}}, \quad \text{for all } y \in \mathcal{Y}.$$

Idiosyncratic shocks generate heterogeneity in income and wealth, while the aggregate shock drives business-cycle fluctuations.

5.2 Risk-Neutral Mutual Fund

In each country i , a risk-neutral mutual fund issues a composite asset A_t^i to domestic Savers, combining equity and bond holdings:

$$A_t^i = E_t^i + b_t^i.$$

The fund reallocates internally between equity and bonds to maximize expected returns. The equity component aggregates domestic and foreign capital holdings:

$$E_t^i = K_{i,t}^i + S_t K_{i,t}^j,$$

where S_t denotes the relative price of country j 's consumption basket in terms of country i 's.

The gross return on the composite asset is given by:

$$r_t^A A_{t-1}^i = r_t^i E_{t-1}^i + \frac{r_{t-1}^U}{\pi_t^{CPI,i}} b_{t-1}^i,$$

where r_t^i is the real equity return and $\frac{r_{t-1}^U}{\pi_t^{CPI,i}}$ the real bond return.

The fund's no-arbitrage condition links expected returns on equity and bonds while incorporating portfolio adjustment costs:

$$\mathbb{E}_t[r_{t+1}^i] + \frac{\kappa_D}{1 - \lambda^i} p_t^i b_t^i = \frac{r_t^U}{\pi_{t+1}^{CPI,i}} + \frac{\kappa_E}{1 - \lambda^i} p_t^i (E_t^i - \bar{E}^i).$$

Higher κ_D or κ_E imply greater savings rigidity, reducing responsiveness to changes in relative returns. The internal allocation between domestic and foreign capital satisfies the same no-profit and no-arbitrage conditions as in the TANK model; the key difference is that households now interact with the fund through the single composite asset A_t^i .

5.3 Households

Each household h in country i faces an idiosyncratic productivity state $y_{h,t}$ and chooses consumption $c_{h,t}^i$ and asset holdings $a_{h,t}^i$ to maximize

$$\max_{\{c_{h,t}^i, a_{h,t}^i\}_{t=0}^{\infty}} E_0 \sum_{t=0}^{\infty} \beta^t \left(\frac{(c_{h,t}^i)^{1-\sigma}}{1-\sigma} - \chi \frac{(L_t^{U,i})^{1+\eta}}{1+\eta} \right)$$

subject to

$$a_{h,t}^i + c_{h,t}^i = r_t^i a_{h,t-1}^i + w_t^i y_{h,t} L_t^{U,i}, \quad a_{h,t}^i \geq -\bar{a}.$$

The Euler equation is

$$(c_{h,t}^i)^{-\sigma} = \beta E_t[r_{t+1}^i (c_{h,t+1}^i)^{-\sigma}].$$

Uninsurable idiosyncratic shocks induce precautionary savings, generating a stationary wealth distribution even in the absence of aggregate heterogeneity.

5.4 Labor Supply and Union Wage Setting

Labor supply and wage setting follow the same structure as in the two-agent model, now aggregated over the continuum of heterogeneous households. Each country $i \in \{C, P\}$ features monopolistic unions indexed by $k \in (0, 1)$, aggregated through a [Dixit and Stiglitz \[1977\]](#) technology with elasticity $\theta_W^i > 1$. Unions set nominal wages subject to quadratic adjustment costs à la [Rotemberg \[1982\]](#):

$$UC_t^i(k) = \frac{\kappa_W^i}{2} \left(\frac{W_t^i(k)}{W_{t-1}^i(k)} - \bar{\pi}^W \right)^2.$$

In the heterogeneous-agent environment, unions internalize the distribution of marginal utilities across households and choose wages to maximize aggregate welfare. In a symmetric equilibrium ($W_t^i(k) = W_t^i$), the first-order condition yields a New Keynesian Wage Phillips Curve:

$$\pi_t^{W,i}(\pi_t^{W,i} - \bar{\pi}^W) = \beta E_t[\pi_{t+1}^{W,i}(\pi_{t+1}^{W,i} - \bar{\pi}^W)] + \frac{\theta_W^i}{\kappa_W^i} \left[\chi(L_t^{U,i})^{1+\eta} - \frac{\theta_W^i - 1}{\theta_W^i} w_t^i L_t^{U,i} \overline{MU}_t^i \right],$$

where $w_t^i = W_t^i / CPI_t^i$ denotes the real wage and

$$\overline{MU}_t^i = \int_h (c_{h,t}^i)^{-\sigma} d\mu^i(h)$$

is the average marginal utility of consumption across households in country i . When inequality or liquidity constraints are high, the wage-setting process becomes more sensitive to the consumption fluctuations of constrained households, amplifying the real wage response to shocks.

5.5 Market Clearing Conditions

The aggregate equilibrium conditions in each country i mirror those in the TANK model, now integrated over the distribution of heterogeneous agents:

- Aggregate consumption: $C_t^i = \int_h c_{h,t}^i d\mu^i(h)$,
- Aggregate assets: $A_t^i = \int_h a_{h,t}^i d\mu^i(h) = E_t^i + b_t^i$.

Market clearing for capital, goods, the trade balance, and union-wide bonds follows the same structure as in the two-agent model.

6 Results - Heterogeneous-Agent Model

In this section, I extend the analysis to a fully Heterogeneous-Agent New Keynesian (HANK) framework, calibrated to represent the euro area in 2010. This extension allows me to validate the mechanisms highlighted in the two-agent (TANK) version and to quantify their magnitude in an economy with realistic wealth and income heterogeneity. In particular, I compare the responses to a similar asymmetric shock under both the TANK and HANK calibrations to assess the robustness of the results and the additional role played by within-country inequality in shaping aggregate and distributional dynamics.

The model is solved following a two-step procedure¹⁶. The stationary equilibrium is computed using the endogenous grid method (EGM) with interpolation over the joint asset–income space to obtain the individual policy and distribution functions. The aggregate transition dynamics around the steady state are then characterized using the [Reiter \[2009\]](#) method, which combines linearization of aggregate equations with numerical solution of the household block, ensuring full consistency between micro heterogeneity and macro adjustment.

6.1 Calibration

Table 9: Calibration Parameters

Parameter	Value	Description	Source
β	0.93	Discount factor	Smets and Wouters [2003]
α	0.36	Capital share	Krueger et al. [2017]
σ	2.0	Relative risk aversion	CRRA
ϕ	2.0	Elasticity of substitution (home vs. foreign)	Pasch and Tervala [2024]
δ	0.025	Depreciation rate (quarterly)	Smets and Wouters [2003]
η	2.0	Inverse Frisch elasticity of labor supply	Chetty et al. [2011]
Ψ	1.5	Taylor-rule coefficient on inflation	Darvas and Merler [2013]
κ_D	0.10	Debt adjustment cost	Schmitt-Grohe and Uribe [2003]
κ_E	0.10	Equity adjustment cost	Schmitt-Grohe and Uribe [2003]
ρ_Z	0.95	TFP autocorrelation	Den Haan [2010]
ρ_y	0.96	Persistence of idiosyncratic productivity	Author’s decision
σ_y	0.11	Std. dev. of idiosyncratic shock (annualized)	Author’s decision
N_a	300	Number of asset grid points	Discretization for asset holdings
N_y	2.0	Number of income states	Rouwenhorst discretization
$\bar{a}$	0.0	Borrowing limits	Mass at constraint
θ_W	21	Wage markup elasticity	Bilbiie and Ragot [2021]
κ_W	2100	Rotemberg wage adjustment cost	Bilbiie and Ragot [2021]
Steady-State Normalization			
r^U	1.065	Steady-state gross nominal rate	Asset clearing

¹⁶The stationary equilibrium is computed using the endogenous grid method (EGM) introduced by [Carroll \[2006\]](#), with interpolation over the joint asset–income space to obtain the individual policy and distribution functions. The aggregate transition dynamics are then characterized using the method of [Reiter \[2009\]](#), which combines linearization of the aggregate equations with numerical solution of the heterogeneous-agent block.

Table 10: Calibration of Parameters for Core and Periphery (2010)

Parameter	Core	Periphery	Source
Domestic portfolio share	$\zeta^C = 0.97$	$\zeta^P = 0.89$	CPIS (IMF), corrected with SHS, Beck et al. [2024]
Share of Non-Savers	$\lambda^C = 0.43$	$\lambda^P = 0.67$	HFCS (ECB, 2010 wave)
Import share	$\omega^C = 0.33$	$\omega^P = 0.39$	Eurostat, national accounts (imports/GDP)
Portfolio adjustment cost	$\kappa^C = 0.10$	$\kappa^P = 0.10$	Calibrated (see Section 4.1)
Inflation weight in union	$\gamma = 0.60$	$1 - \gamma = 0.40$	ECB (HICP weights, 2010)
TFP shock standard deviation	$\sigma_z^P = 0.456$	$\sigma_z^C = 0.0078$	Q4 decrease from pre-crisis trend in 2011
GDP ratio $\frac{GDP^P}{GDP^C}$	$\psi = 0.23$		Equity ratios $\frac{E^P}{E^C}$ from CPIS (IMF)

6.2 Calibrated Case for 2010 with Asymmetric Shocks

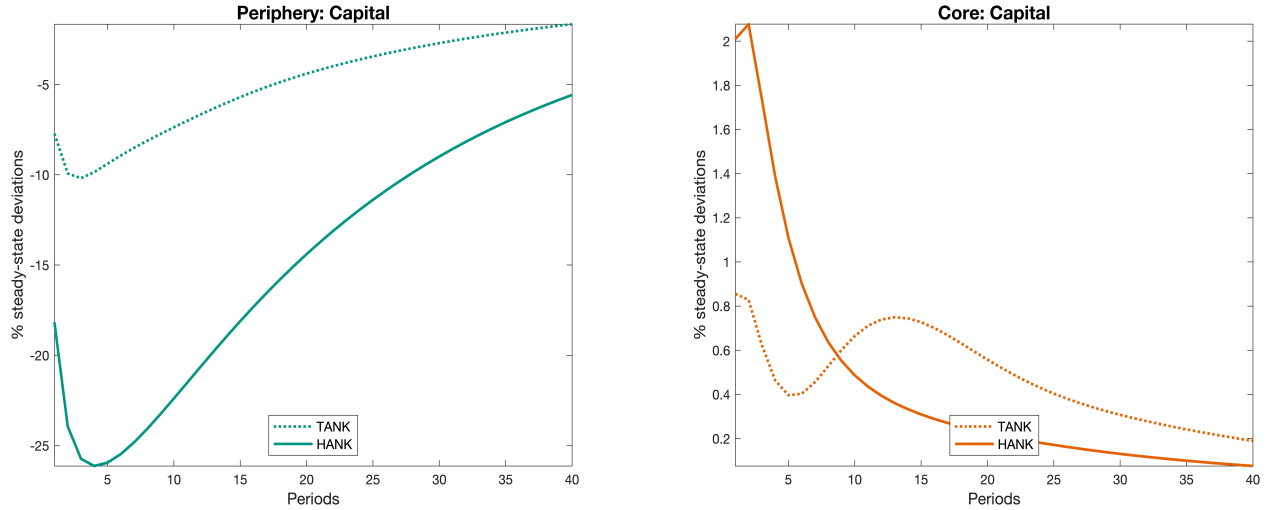


Figure 12: Impulse Response of Capital in the Core and the Periphery Across Models

Figure 12 compares the impulse responses of capital in the Periphery and the Core between the TANK and HANK models. The two frameworks generate qualitatively similar dynamics: following a negative productivity shock in the Periphery, capital falls domestically and rises in the Core as funds reallocate toward higher returns. Introducing household heterogeneity, however, amplifies these movements. In the HANK model, weaker aggregate savings by liquidity-constrained households and tighter balance-sheet conditions deepen and prolong the fall in Periphery capital. At the same time, unequal asset holdings and adjustment costs limit the immediate reallocation toward the Core, dampening cross-border spillovers. Hence, while the direction of adjustment remains the same, heterogeneity strengthens domestic amplification and smooths external responses.

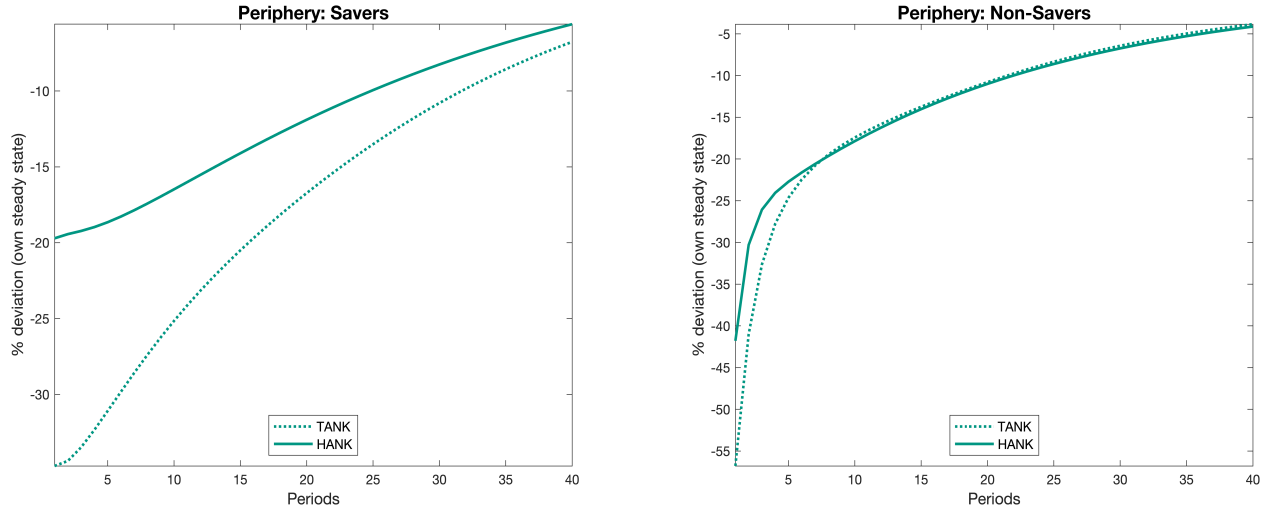


Figure 13: Impulse Response of Consumption in the Periphery Across Models

In the Periphery, both models display a sharp contraction in consumption following the shock, but the adjustment is more pronounced under HANK. Liquidity-constrained households bear the largest losses, as falling wages and asset prices reduce income and limit their ability to smooth consumption. Savers, in turn, increase precautionary savings, amplifying the drop in aggregate demand. As a result, the same mechanisms identified in the TANK model—declining labor income and lower returns—operate more forcefully once household heterogeneity is introduced, producing a deeper and more persistent downturn.

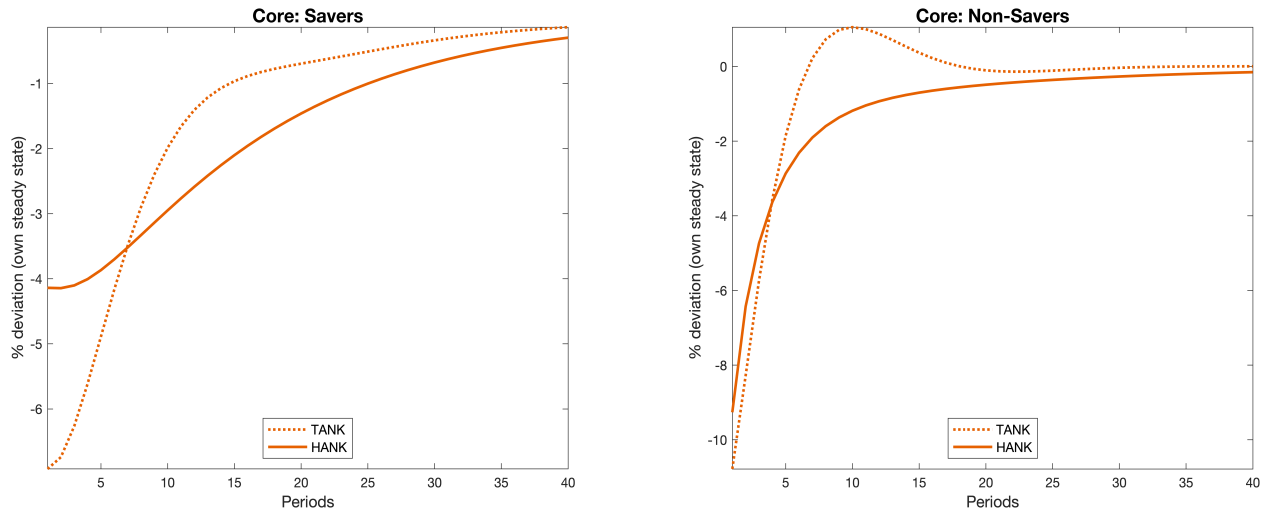


Figure 14: Impulse Response of Consumption in the Core Across Models

In the Core, the response is again qualitatively similar across models: consumption

falls moderately as the Periphery contracts, before gradually recovering. Under HANK, the adjustment becomes more persistent. Non-savers experience lower labor income and cannot smooth consumption, while savers reduce spending in response to weaker external demand and lower returns. The coexistence of high- and low-marginal-propensity-to-consume households introduces inertia into aggregate consumption, flattening the response relative to the representative-agent case.

Overall, introducing heterogeneity does not alter the direction of adjustment but amplifies its magnitude. The same channels identified in the two-agent framework, diversification, reallocation, and labor income effects, continue to operate, yet their strength depends on the distribution of financial wealth and liquidity constraints. In the Periphery, heterogeneity magnifies domestic amplification through balance-sheet and demand effects, while in the Core it moderates the transmission of capital inflows. Consequently, the HANK framework confirms the qualitative insights of the TANK model but reveals stronger and more persistent cross-country asymmetries driven by unequal household exposure.

7 Conclusion

The analysis shows that the answer to who benefits from capital market integration in the euro area depends critically on both where and who you are. Capital market integration operates through two channels, diversification and reallocation, that shape welfare and adjustment asymmetrically across households and countries. In the Periphery, integration stabilizes the consumption of wealthy savers through diversification but exposes non-savers to deeper labor income losses as capital outflows depress domestic activity. In the Core, by contrast, capital inflows cushion wages and benefit liquidity-constrained households, while financially integrated savers face lower returns through their exposure to the Periphery.

Taken together, these results imply that the main beneficiaries of capital market integration are asset-holding households in financially constrained regions and wage earners in capital-receiving economies, while liquidity-constrained households in debtor regions lose the most. At the aggregate level, integration stabilizes the monetary union but redistributes welfare both across and within countries, narrowing inequality in the Core while widening it in the Periphery.

The results also suggest that deep financial integration in a monetary union requires complementary mechanisms, such as labor mobility or fiscal transfers, to prevent the uneven distribution of gains and losses across regions. In their absence, the stabilizing

benefits of integration risk being offset by rising internal imbalances.

Finally, a natural next step is to extend this framework beyond the euro area to examine the global dimension of capital market integration, particularly in the context of United States–Euro Area linkages. The United States, as a major global investor with broader household participation in financial markets, offers an ideal benchmark for understanding how deeper participation and higher diversification capacity reshape the distributional consequences of integration.

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A Empirics

A.1 Cluster

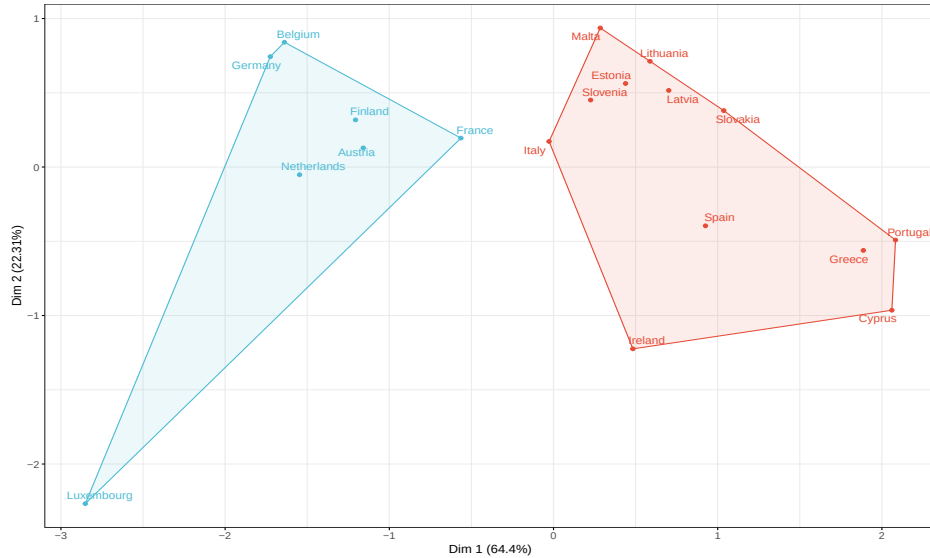


Figure 15: Clustering of Euro Area Countries in 2010. The figure shows the PCA- k -means classification of euro area countries into Core (red) and Periphery (blue) groups based on three indicators of pre-crisis imbalances: current account balance to GDP, net international investment position (NIIP, % of GDP), and GDP per capita. Dimension 1 mainly reflects external imbalances (CA/GDP: 38.7%; NIIP: 33.7%; GDP per capita: 27.6%), while Dimension 2 captures income differences (GDP per capita: 66.1%; NIIP: 31.2%; CA/GDP: 2.7%).

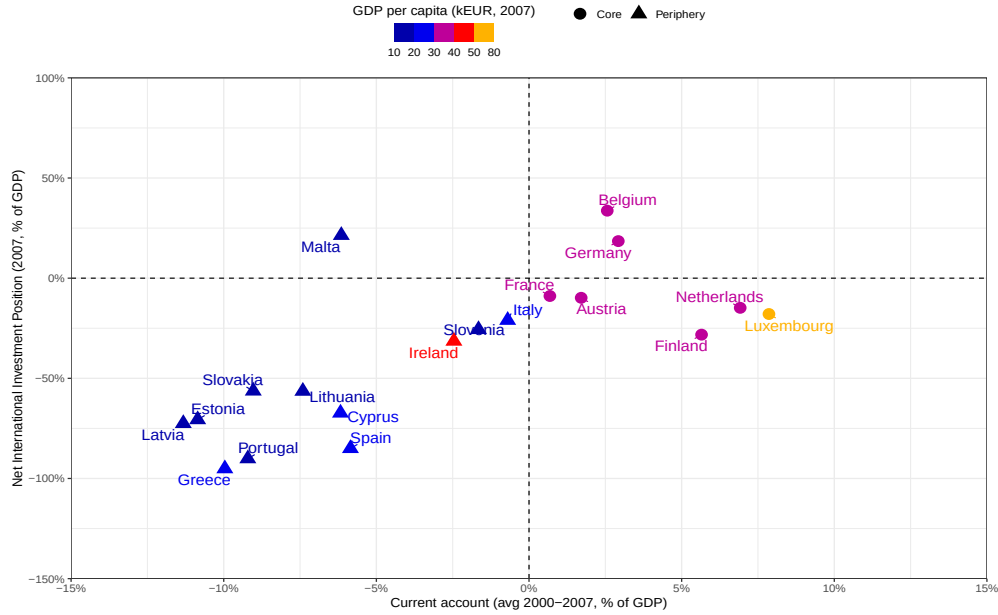


Figure 16: Clustering of Euro Area Countries in 2007. Euro area countries are plotted by their average current account balance (2000–2007, % of GDP) and net international investment position (NIIP, 2007, % of GDP). Marker color reflects GDP per capita (in thousand EUR, 2007). Core countries (circles) and Periphery countries (triangles) are derived from a PCA- k -means clustering based on these indicators.

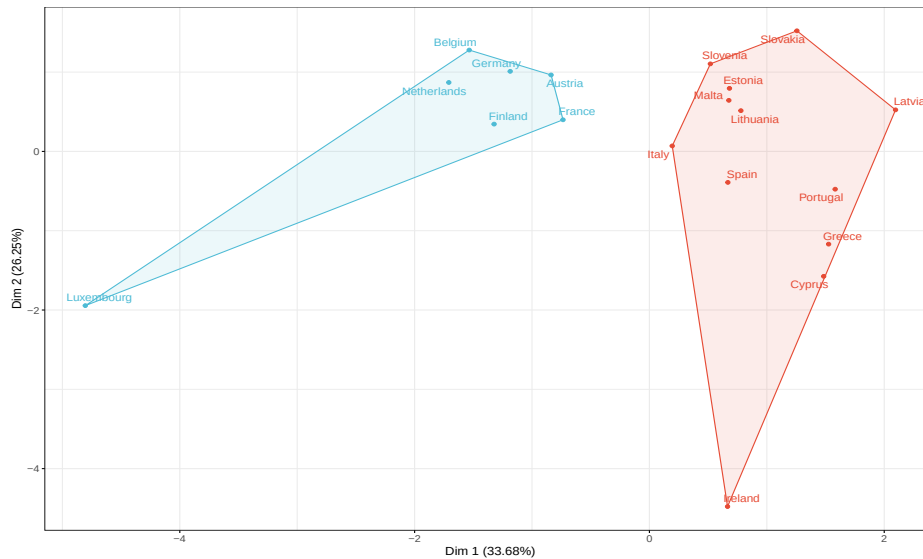


Figure 17: Robustness 1 - Clustering for 2010. The PCA- k -means classification is extended to include financial variables: market capitalization (% of GDP), equity return, portfolio domestic share, number of investment partners, and openness (% of GDP). The resulting Core–Periphery grouping remains identical to the baseline, confirming the stability of the classification. Main contributions to Dimension 1: GDP per capita (29.9%), market capitalization (26.9%), CA/GDP (17.9%), NIIP (15.9%); Dimension 2: domestic share (34.7%), equity return (24.2%), NIIP (14.5%), openness (13.7%).

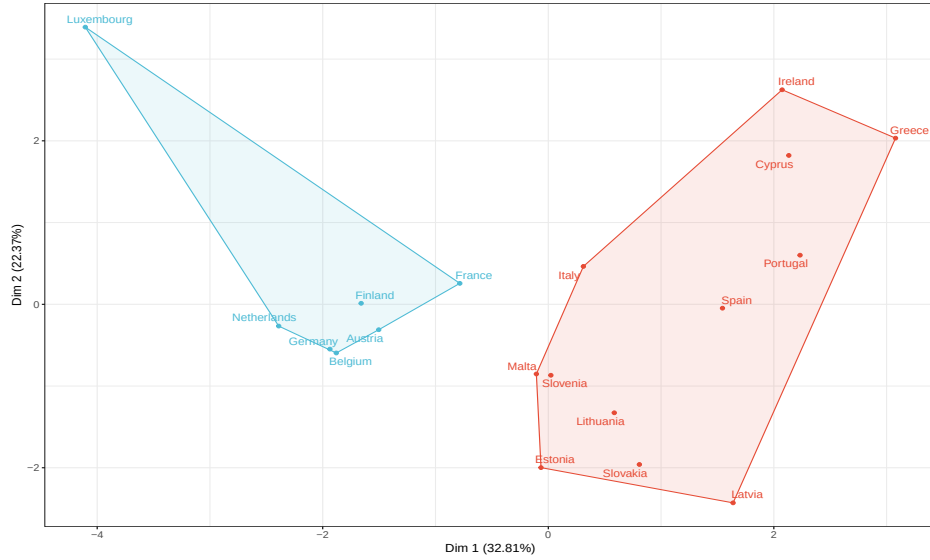


Figure 18: Robustness 2 - Clustering for 2010. Additional macro variables are included: unemployment rate, financial account (% of GDP), and real GDP growth (2009–2010). The Core–Periphery classification remains unchanged. Contributions to Dimension 1: NIIP (18.3%), CA/GDP (18.0%), FA/GDP (14.6%), GDP per capita (13.0%), market capitalization (12.4%), unemployment (10.7%). Contributions to Dimension 2: domestic share (24.2%), GDP per capita (19.1%), openness (13.1%), market capitalization (11.3%), FA/GDP (8.1%).

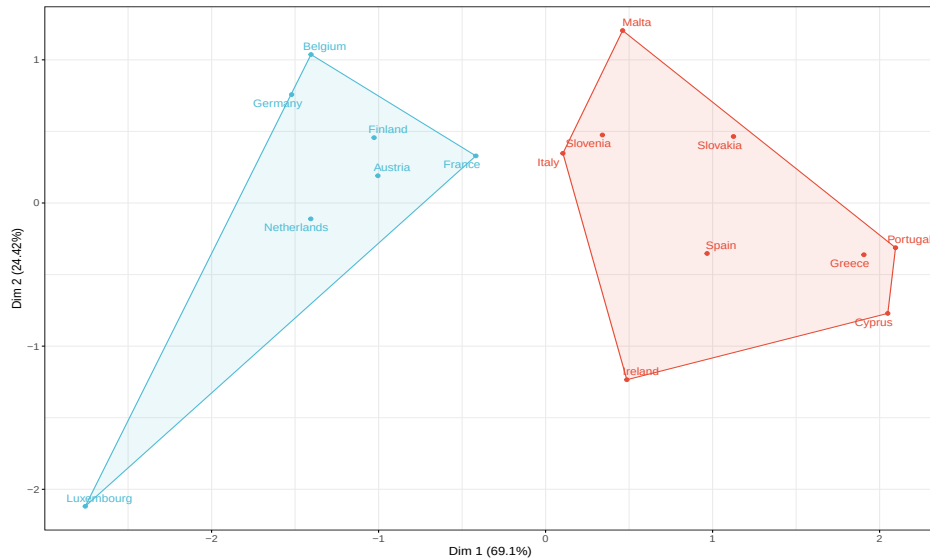


Figure 19: Clustering of Euro Area Countries in 2010 (restricted sample). The PCA- k -means clustering is restricted to euro area members as of 2010, excluding Estonia, Latvia, and Lithuania. Countries are classified into Core (red) and Periphery (blue) based on three indicators of pre-crisis imbalances: current account balance to GDP, net international investment position (NIIP, % of GDP), and GDP per capita. Dimension 1 mainly reflects external imbalances (CA/GDP: 42.9%, NIIP: 29.3%, GDP per capita: 27.8%), while Dimension 2 captures income differences (GDP per capita: 52.7%, NIIP: 47.3%, CA/GDP: 0.0%).

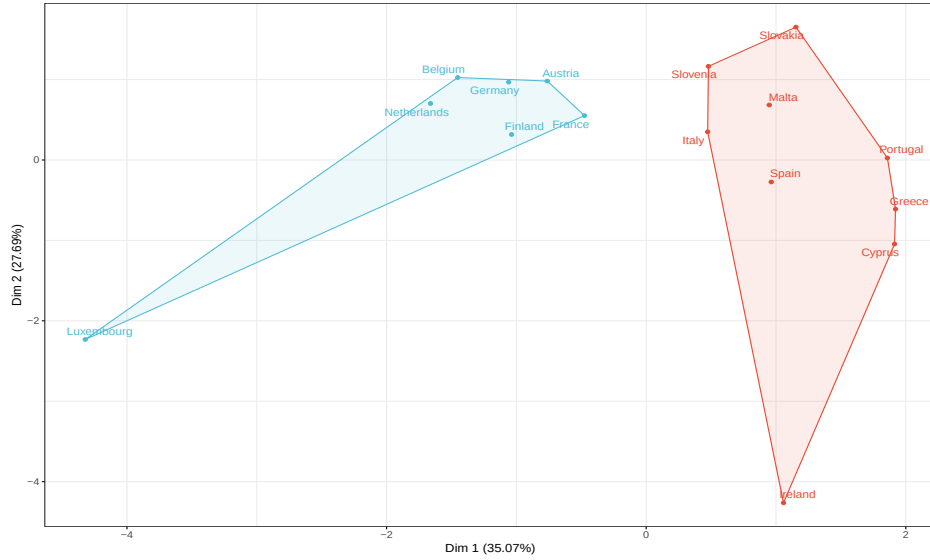


Figure 20: Robustness 1 - Clustering for 2010 (restricted sample). The PCA- k -means clustering is extended to include additional financial variables: market capitalization (% of GDP), equity return, portfolio domestic share, number of investment partners, and openness (% of GDP). Results remain consistent with the baseline classification. Main contributions to Dimension 1: CA/GDP (27.1%), GDP per capita (26.5%), market capitalization (22.7%), NIIP (15.7%), openness (3.2%). Main contributions to Dimension 2: portfolio domestic share (37.8%), equity return (26.3%), NIIP (12.6%), GDP per capita (9.6%), openness (8.8%).

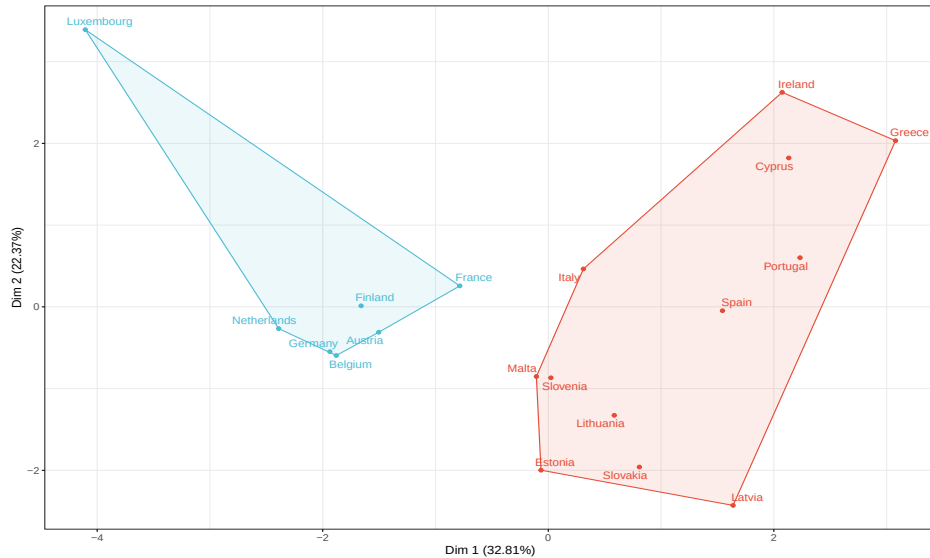


Figure 21: Robustness 2 - Clustering for 2010 (restricted sample). The classification includes additional macroeconomic variables—unemployment rate, financial account (% of GDP), and real GDP growth (2009–2010). The Core–Periphery structure remains unchanged. Main contributions to Dimension 1: NIIP (18.3%), CA/GDP (18.0%), FA/GDP (14.6%), GDP per capita (13.0%), market capitalization (12.4%), unemployment (10.7%). Main contributions to Dimension 2: domestic share (24.2%), GDP per capita (19.1%), openness (13.1%), market capitalization (11.3%), FA/GDP (8.1%).

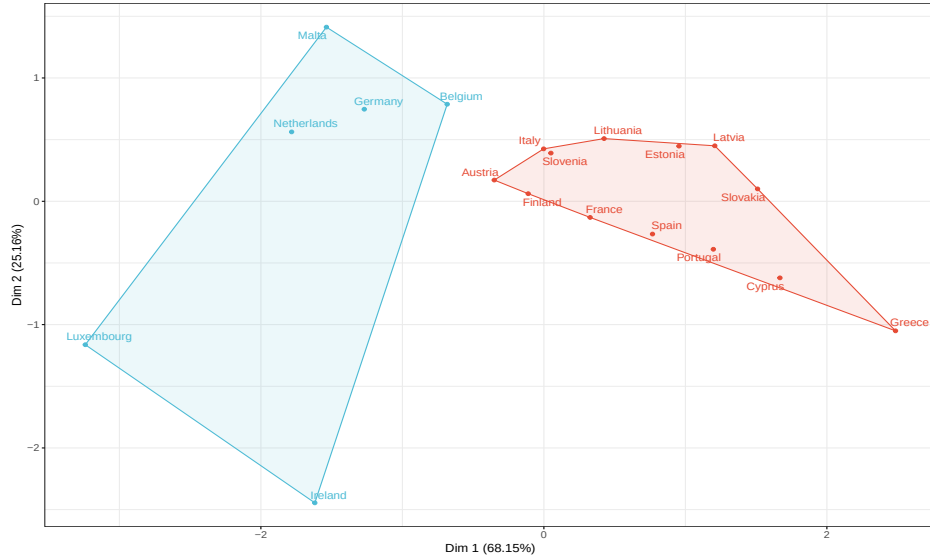


Figure 22: Clustering of Euro Area Countries in 2021. The PCA-*k*-means classification is replicated using 2021 data. France moves from the Core to the Periphery, indicating deeper heterogeneity since the crisis. Main contributions: Dimension 1 (CA/GDP: 43.1%, GDP per capita: 32.9%, NIIP: 23.9%) and Dimension 2 (NIIP: 64.6%, GDP per capita: 34.7%, CA/GDP: 0.7%).

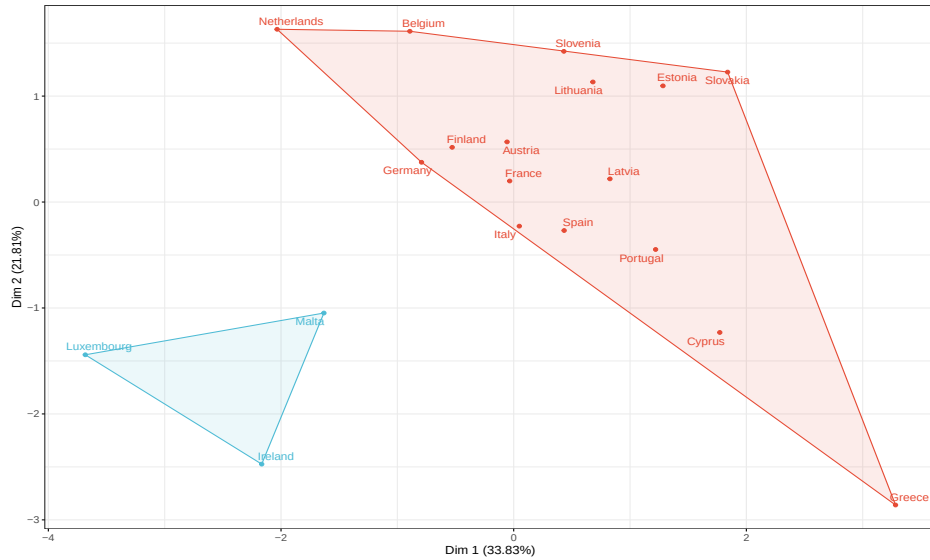


Figure 23: Robustness 1 - Clustering for 2021. The clustering includes additional financial indicators: market capitalization (% of GDP), equity return, portfolio domestic share, number of investment partners, and openness (% of GDP). The Core-Periphery classification remains broadly consistent. Main contributions to Dimension 1: CA/GDP (31.2%), GDP per capita (25.7%), NIIP (15.7%), portfolio domestic share (10.6%), market capitalization (9.1%). Main contributions to Dimension 2: openness (31.9%), equity return (28.3%), NIIP (16.0%), portfolio domestic share (10.7%), GDP per capita (5.4%).

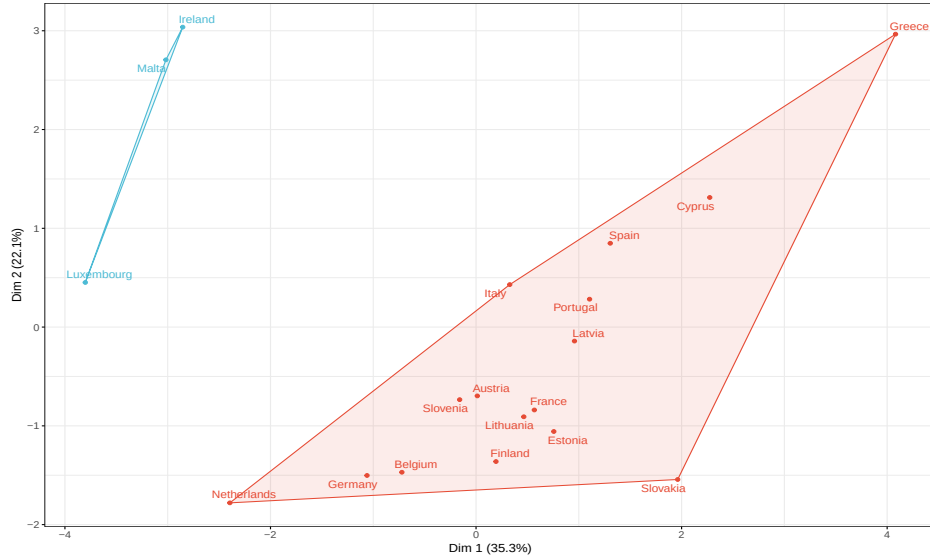


Figure 24: Robustness 2 - Clustering for 2021. The clustering further includes macroeconomic indicators—unemployment rate, financial account (% of GDP), real GDP growth, and portfolio structure variables. The Core–Periphery partition remains qualitatively stable. Main contributions to Dimension 1: CA/GDP (23.1%), FA/GDP (21.2%), GDP per capita (14.6%), NIIP (11.9%), unemployment (11.5%). Main contributions to Dimension 2: real GDP (28.0%), openness (16.0%), equity return (14.2%), number of investment partners (13.2%), NIIP (9.2%).

A.2 Two New Stylized Facts: Divergent Wealth and Capital Stocks

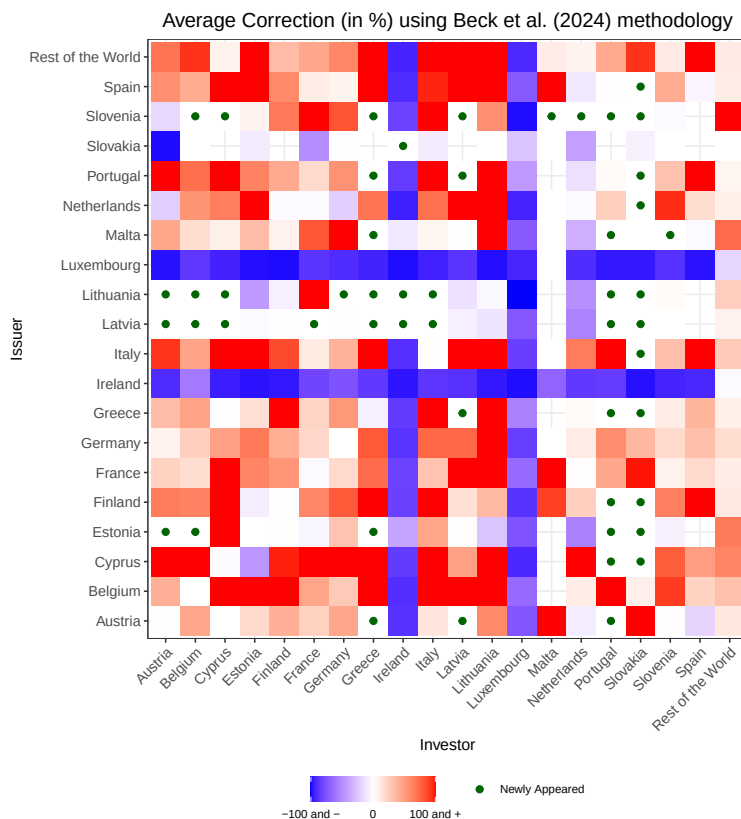


Figure 25: Average Correction (in %) Using the Beck et al. [2024] Methodology. The figure reports the average correction applied to bilateral investment positions (Issuer–Investor) between euro area countries and the Rest of the World, following the reallocation procedure of Beck et al. [2024]. Corrections are computed as the average percentage change between the original CPIS data and the adjusted dataset over 2014–2020. Red cells indicate downward corrections (reductions in reported holdings), blue cells indicate upward corrections (increases in holdings), and green dots mark pairs that newly appear in the corrected dataset. The adjustment reallocates portfolio positions routed through Luxembourg, Ireland, and the Netherlands to the Rest of the World, correcting for the fund-platform bias in euro area investment data.

A.2.1 Additional distributional evidence: financial wealth, income

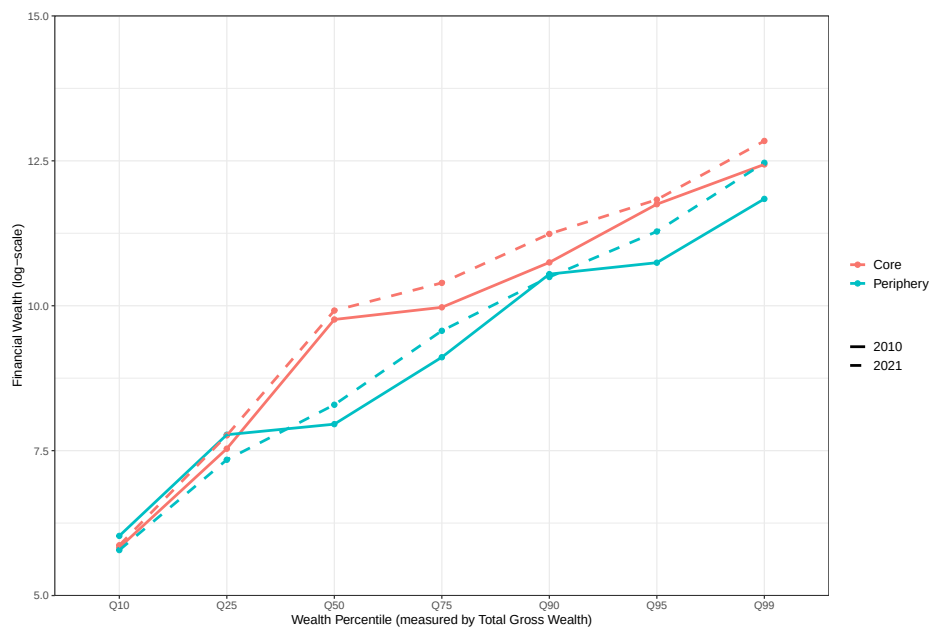


Figure 26: Change in Financial Wealth Across the Wealth Distribution (2021 vs. 2010). The figure plots the log deviation in financial wealth between 2021 and 2010 across percentiles of the wealth distribution for Core and Periphery households. Financial wealth includes bonds, equities, mutual funds, and private pensions, and excludes deposits.

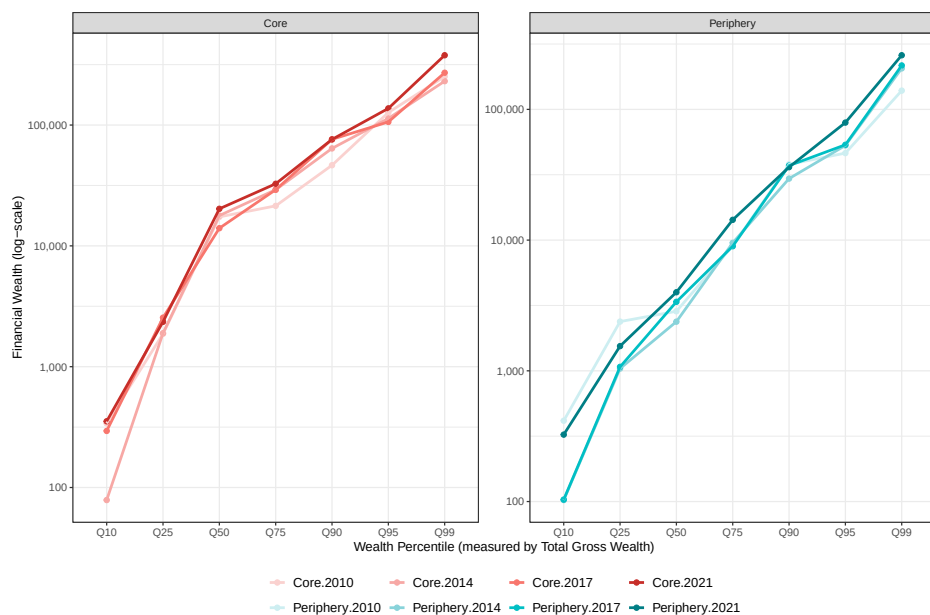


Figure 27: Financial Wealth Distribution (All Waves). The figure reports financial wealth (excluding deposits) by percentile for Core and Periphery households over the four HFCS waves (2010, 2014, 2017, and 2021).

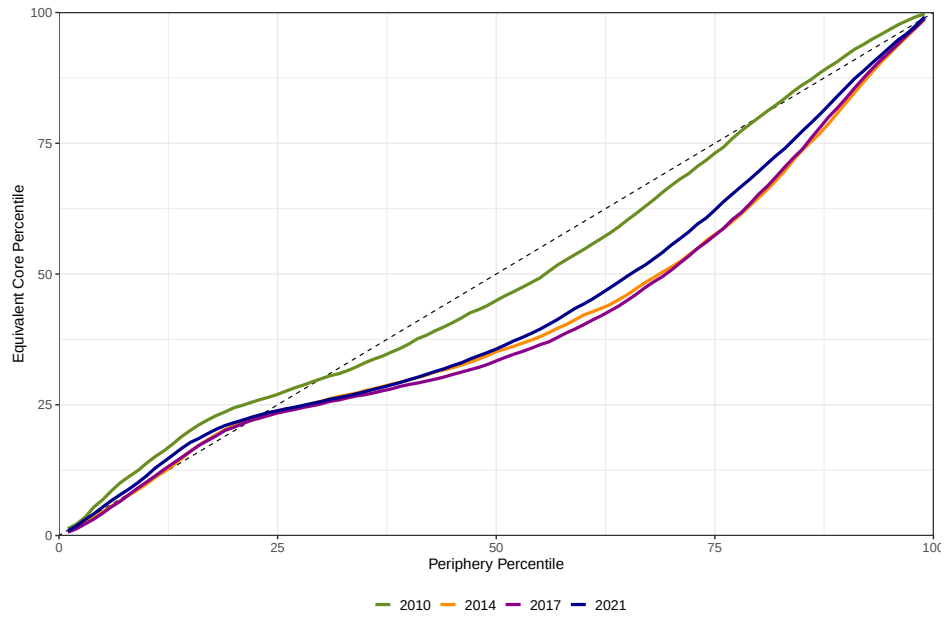


Figure 28: Equivalent Percentiles of Wealth Across Groups. This figure maps equivalent percentiles in the wealth distributions of the Core and Periphery.

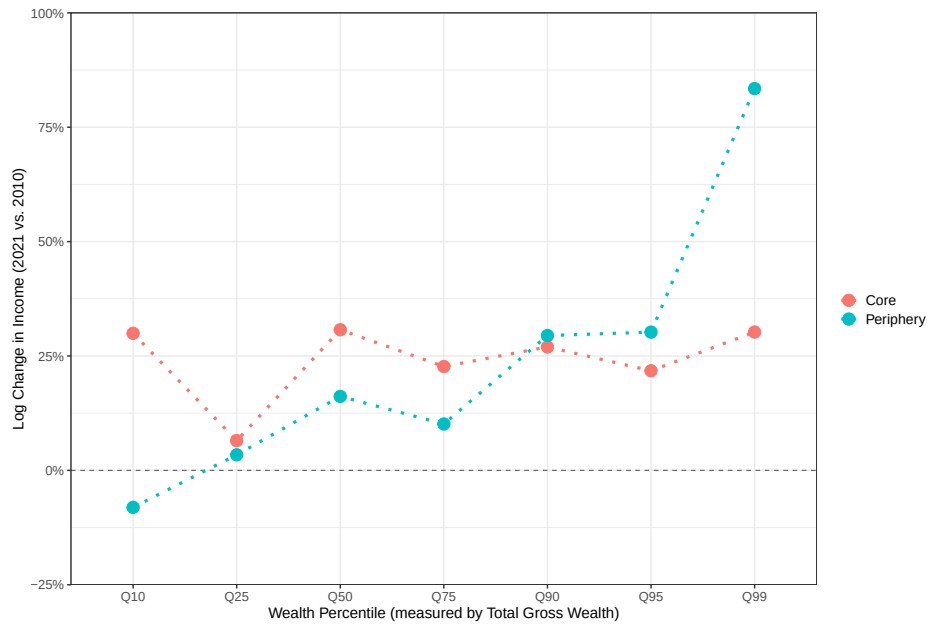


Figure 29: Change in Income Across the Wealth Distribution (2021 vs. 2010) The figure plots the log deviation in income between 2021 and 2010 across percentiles for Core and Periphery households.

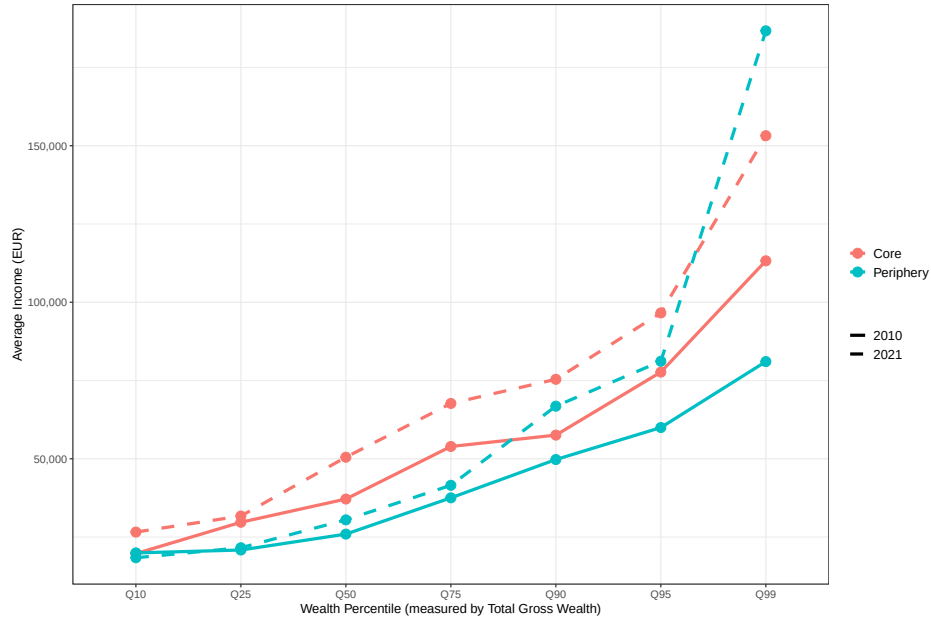


Figure 30: Income in Levels Across Wealth Distribution (2021 vs. 2010). The figure compares income in levels along the wealth distributions for Core and Periphery households in 2010 and 2021.

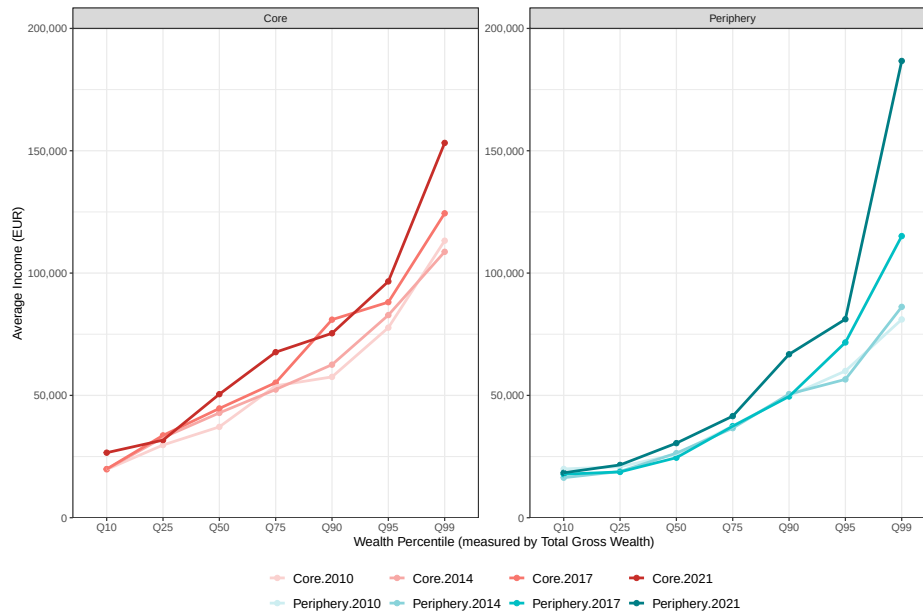


Figure 31: Income Across Wealth Distribution (All Waves) The figure reports household income by percentile for the Core and Periphery across the 2010, 2014, 2017, and 2021 HFCS waves.

A.2.2 Wealth Distributions

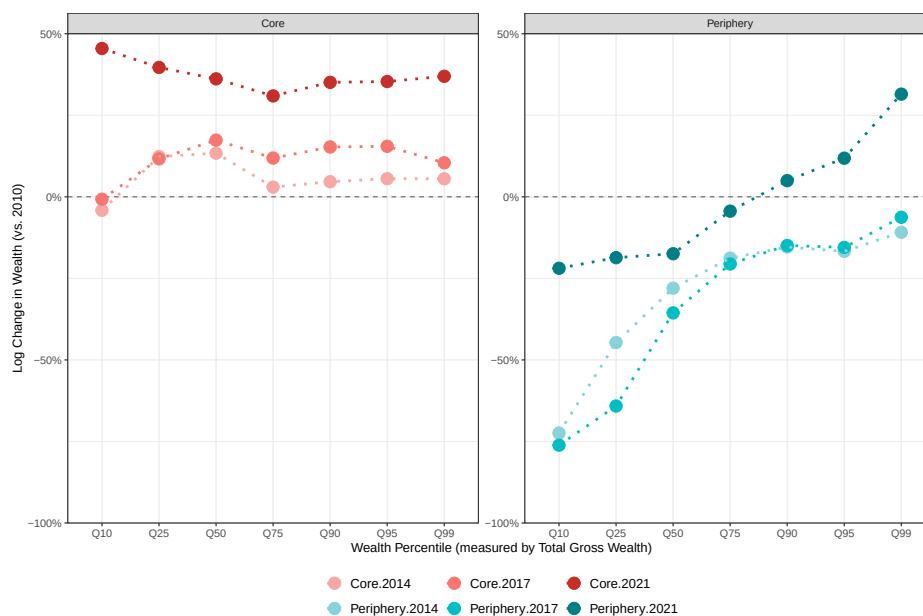


Figure 32: Evolution of Wealth Distributions (All Waves). The figure plots the log change in total gross wealth across percentiles relative to 2010 for each wave of the HFCS (2014, 2017, and 2021), comparing the Core and Periphery groups. The 2021 wave shows an upward shift in both distributions across all percentiles, likely reflecting post-COVID fiscal support.

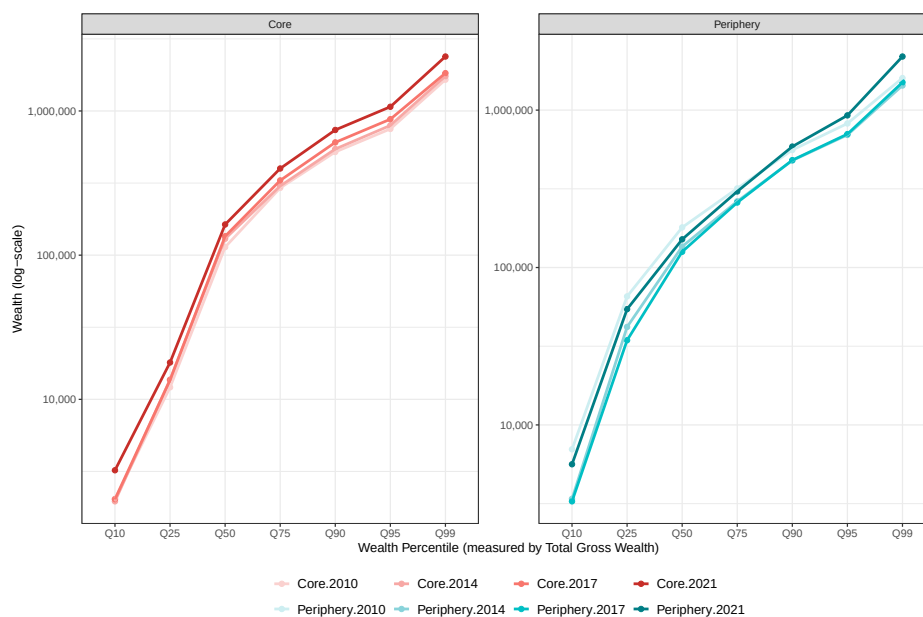


Figure 33: Wealth Distribution in Levels (All Waves). The figure presents the distribution of total gross wealth across percentiles for the Core and Periphery, shown in values (euros).

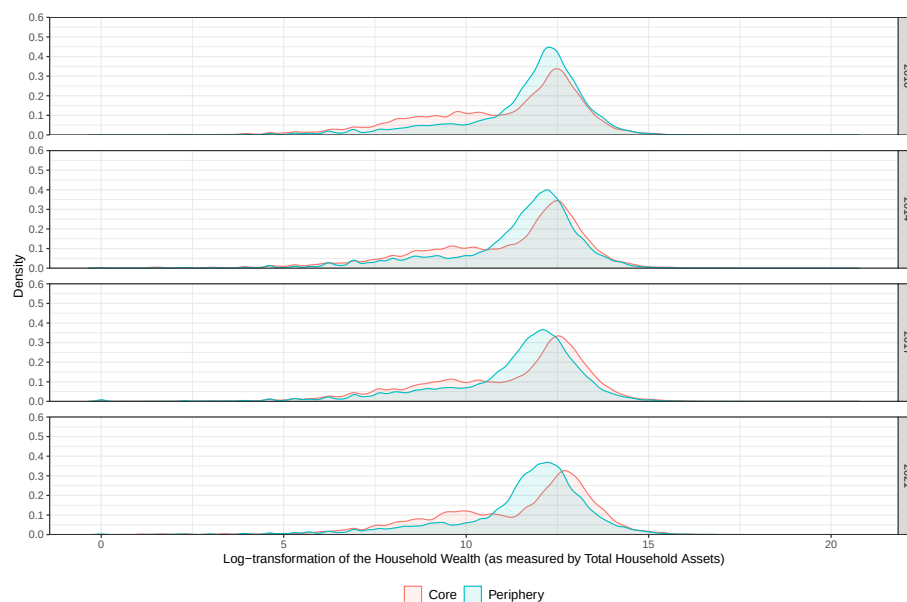


Figure 34: Wealth Distribution for Core and Periphery (All Waves). The figure compares the entire wealth distributions of Core and Periphery groups over all HFCS waves (2010–2021). The divergence between the two groups increases over time.

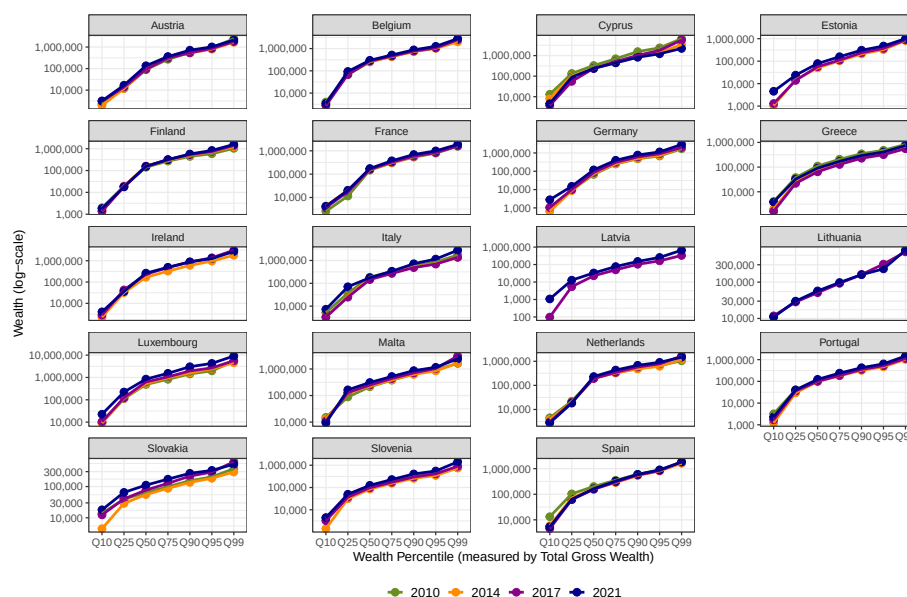


Figure 35: Wealth Distribution by Country and Quantile (All Waves). The figure plots the evolution of wealth across percentiles for each country between 2010 and 2021.

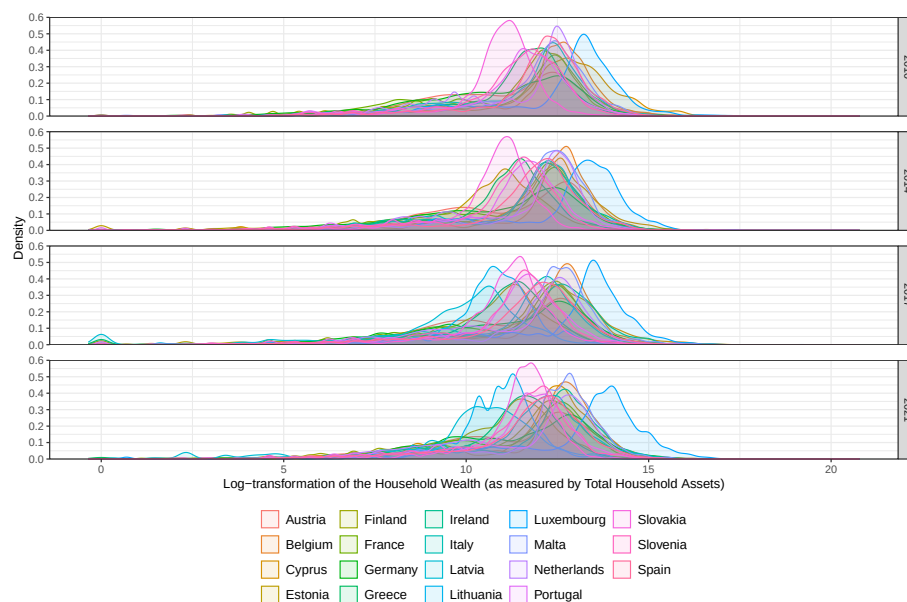


Figure 36: Wealth Distribution by Country (Full Sample). The figure shows the entire wealth distribution for each euro area country across all HFCS waves.

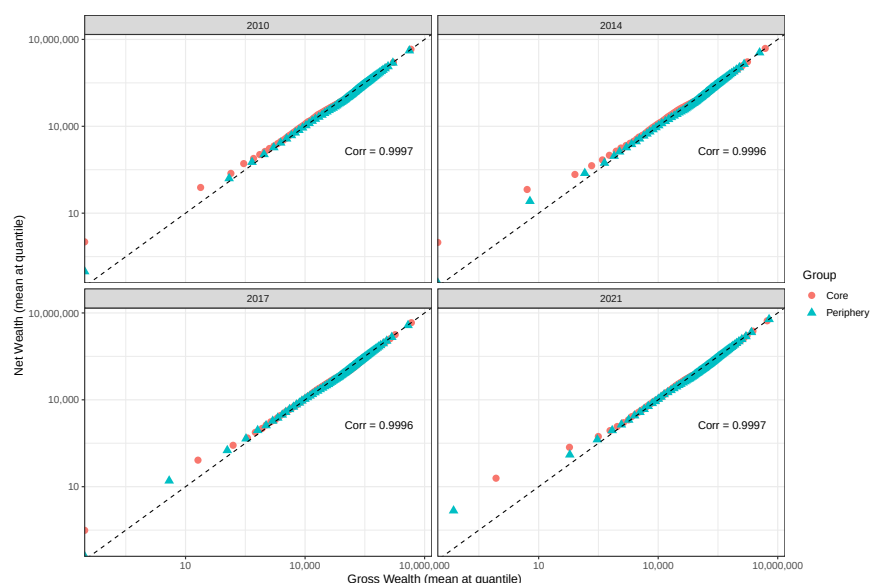


Figure 37: Correlation Between Net and Gross Wealth Across Waves. The figure reports the correlation between net and gross household wealth for each HFCS wave and group (Core and Periphery). The near-perfect correlation (close to one) indicates that the main patterns documented for gross wealth are not driven by household debt positions but by differences in asset accumulation.

A.2.3 Income/Wealth

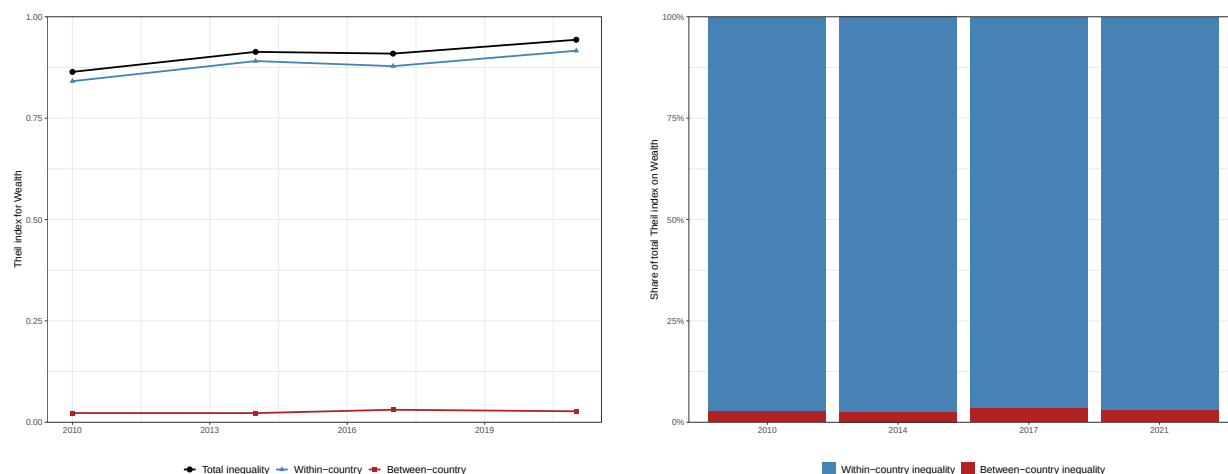


Figure 38: Wealth Inequality and Its Decomposition in the Euro Area (2010–2021). Panel (a) reports the Theil index of wealth inequality computed at the country level for each wave of the HFCS. Panel (b) decomposes total euro area wealth inequality into its *within-country* and *between-country* components.

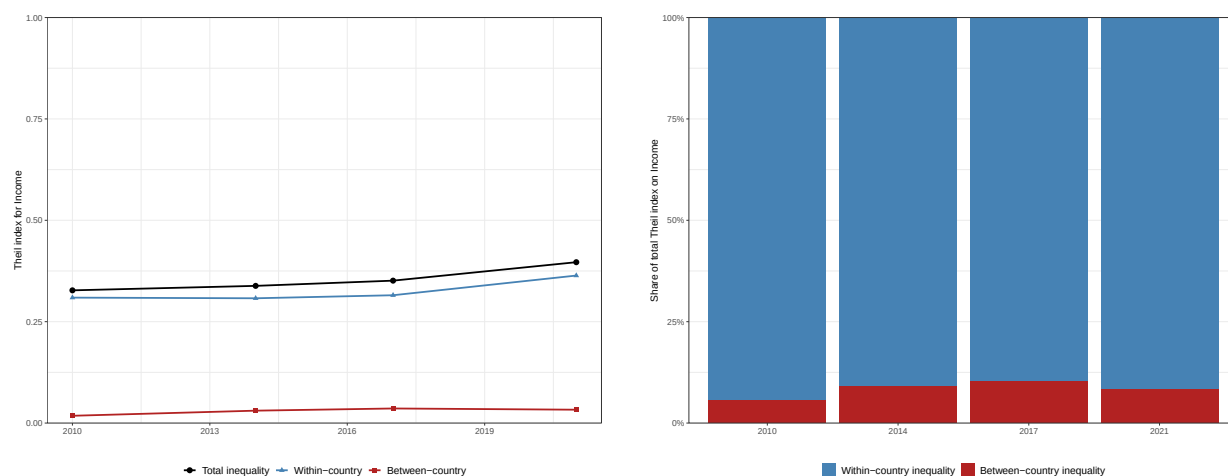


Figure 39: Income Inequality and Its Decomposition in the Euro Area (2010–2021). Panel (a) reports the Theil index of income inequality computed at the country level for each wave of the HFCS. Panel (b) decomposes total euro area income inequality into its *within-country* and *between-country* components.

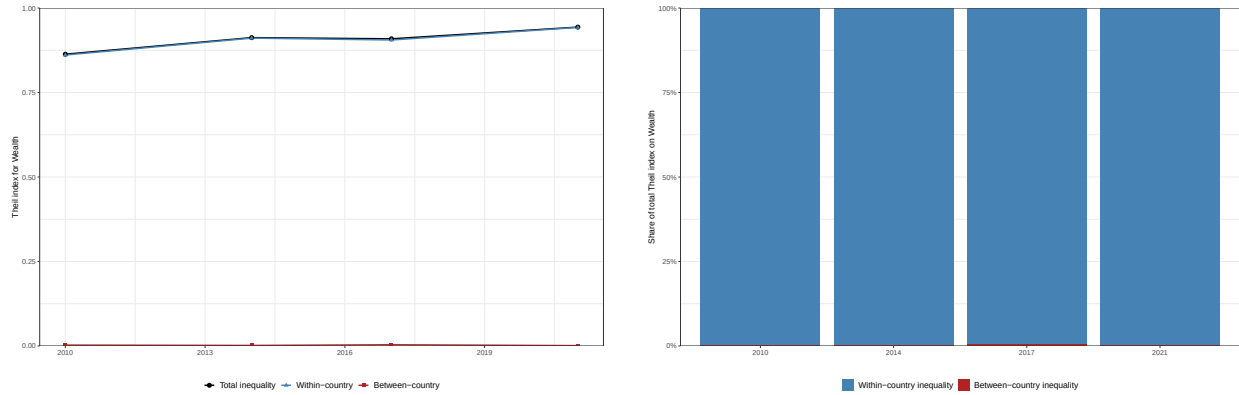


Figure 40: Wealth Inequality and Its Decomposition in the Euro Area (2010–2021). Panel (a) reports the Theil index of wealth inequality computed at the group level for each wave of the HFCS. Panel (b) decomposes total euro area wealth inequality into its *within-group* and *between-group* components.

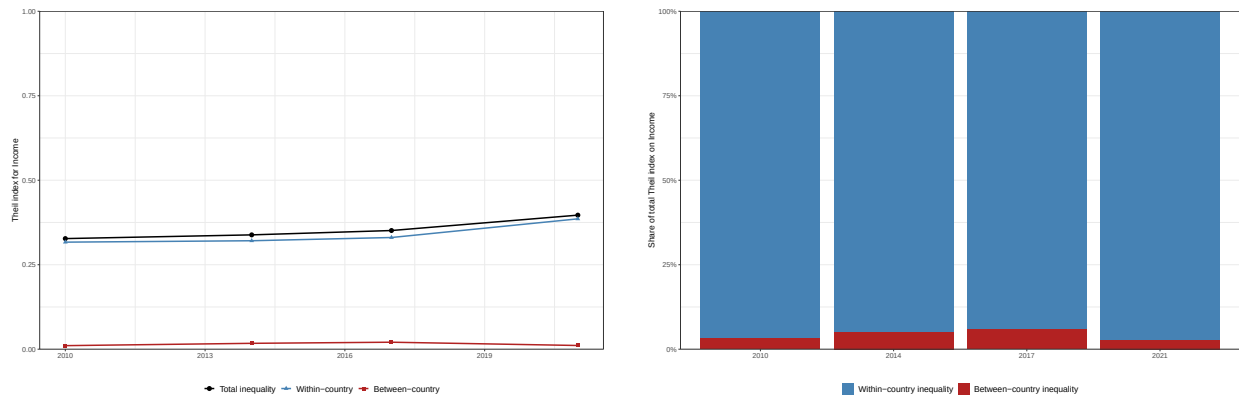


Figure 41: Income Inequality and Its Decomposition in the Euro Area (2010–2021). Panel (a) reports the Theil index of income inequality computed at the group level for each wave of the HFCS. Panel (b) decomposes total euro area income inequality into its *within-group* and *between-group* components.

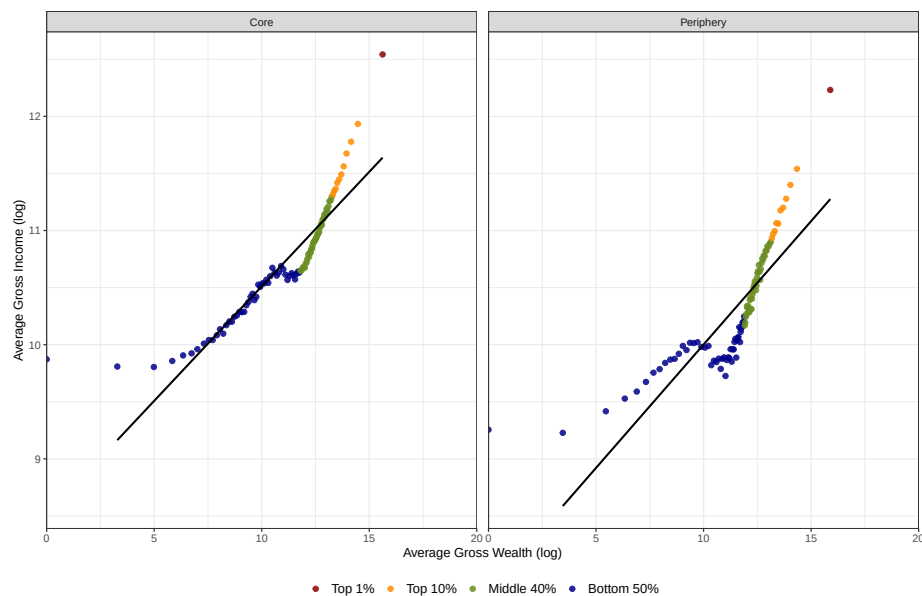


Figure 42: Wealth–Income Relationship in the Euro Area: Core vs. Periphery (All Years Combined). The figure plots the relationship between average wealth and income across percentiles for the Core and Periphery groups, pooling all HFCS waves (2010–2021). The positive slope reflects the strong association between income and wealth across the distribution, with higher wealth–income elasticity observed in the Core than in the Periphery.

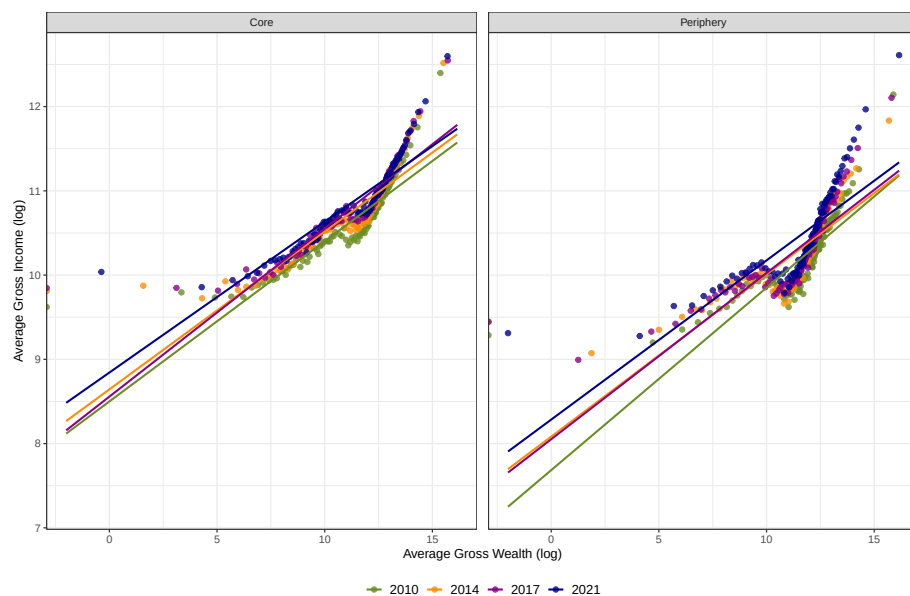


Figure 43: Wealth–Income Relationship by Survey Wave (Core vs. Periphery). The figure reports the wealth–income relationship separately for each HFCS wave (2010, 2014, 2017, and 2021). The Core exhibits a stable, upward-sloping relationship over time, while the Periphery shows a flattening after 2014, consistent with weaker post-crisis recovery among low- and middle-income households.

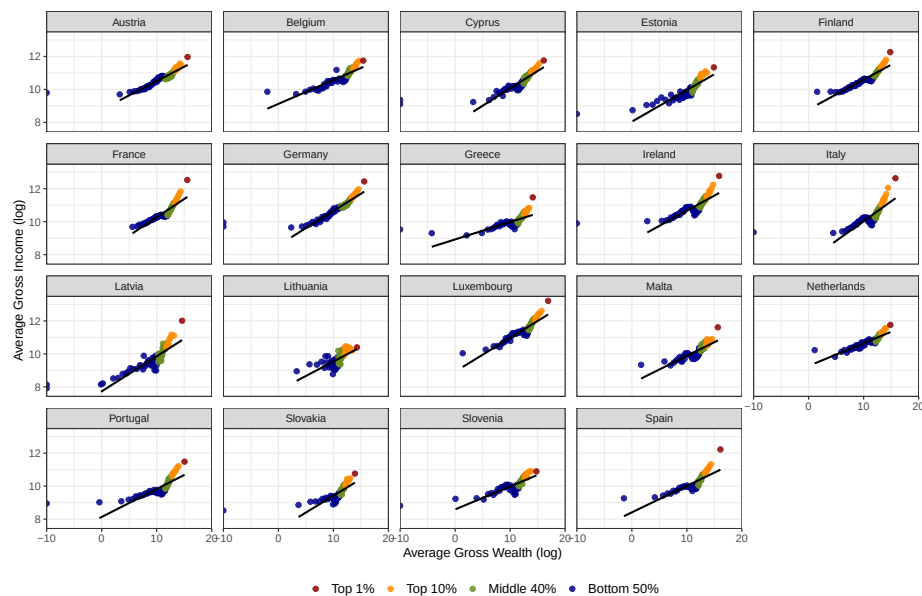


Figure 44: Wealth–Income Relationship by Country (All Years Combined). This figure displays the average relationship between wealth and income across countries, pooling all survey waves. Cross-country differences are substantial: Core countries cluster around higher wealth–income ratios, while Periphery countries display larger dispersion, especially in the middle of the distribution.

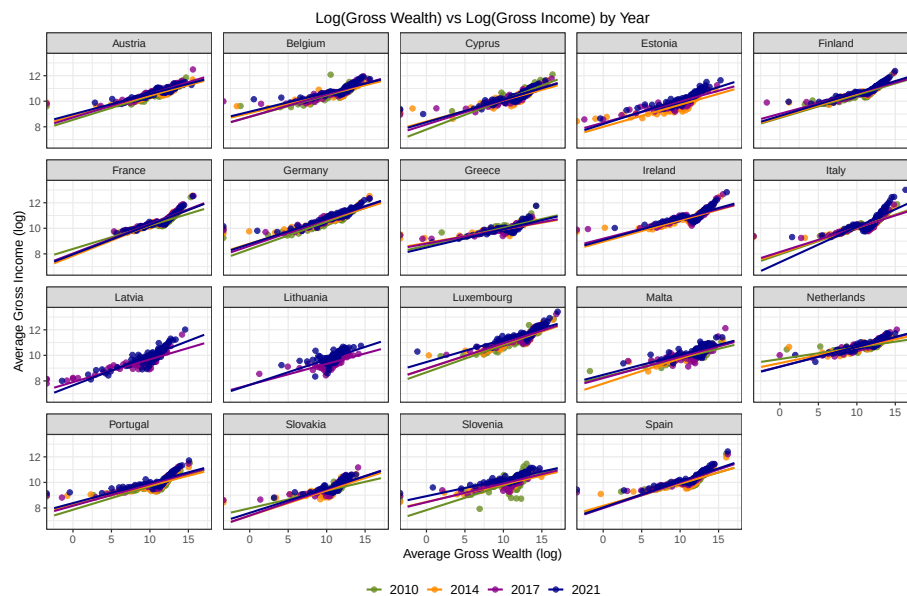


Figure 45: Wealth–Income Relationship by Country and Year. The figure illustrates the evolution of the wealth–income link across survey waves for each country. The slope of the relationship steepens in Core countries over time, whereas in several Periphery countries it flattens, reflecting persistently lower wealth accumulation for households with similar income levels.

A.2.4 Capital Stock

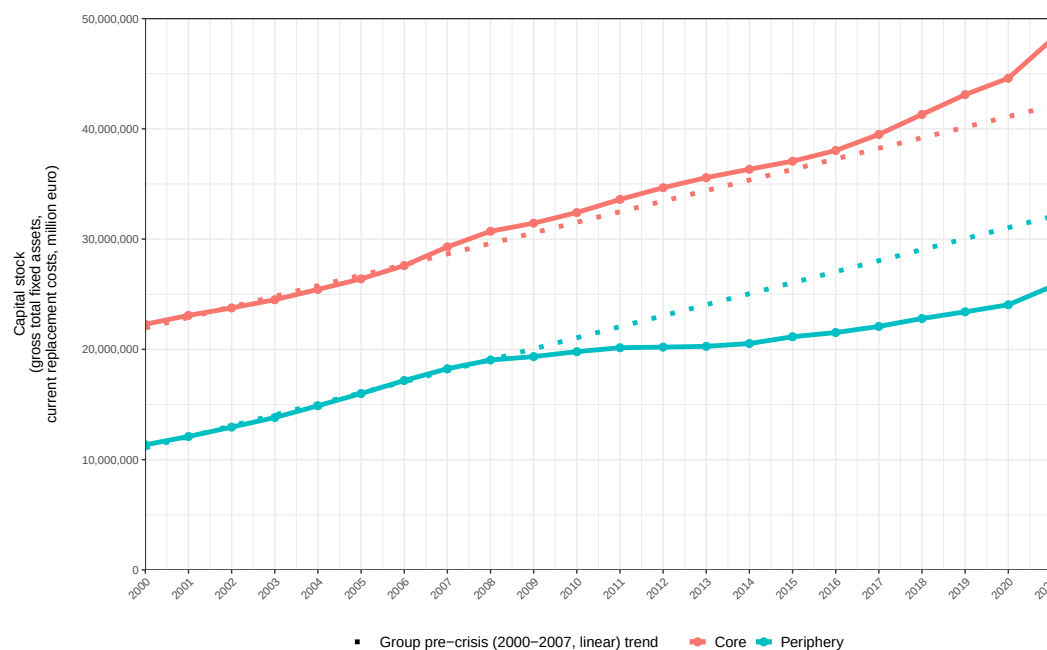


Figure 46: Gross Capital Stock and Pre-crisis Trends (2000–2021). The figure plots the evolution of the gross capital stock (total fixed assets at current replacement costs) for the Core and Periphery from 2000 to 2021. Dashed lines denote the pre-crisis linear trends (2000–2007).

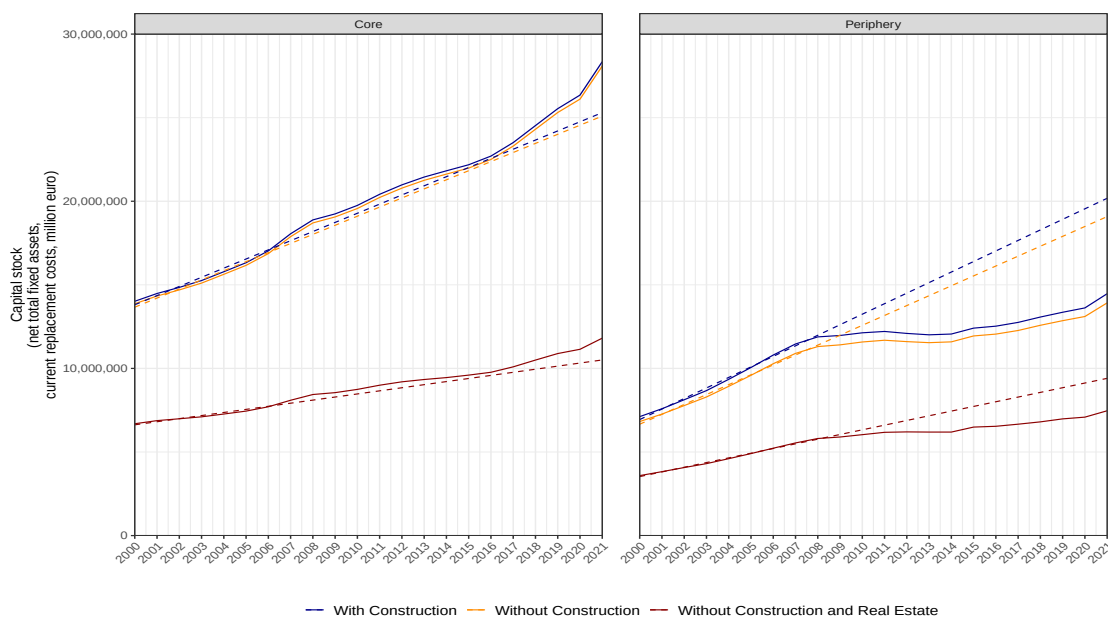


Figure 47: Net Capital Stock Excluding Construction and Real Estate (2000–2021). The figure displays the evolution of the net capital stock (current replacement costs) excluding the construction and real estate sectors.

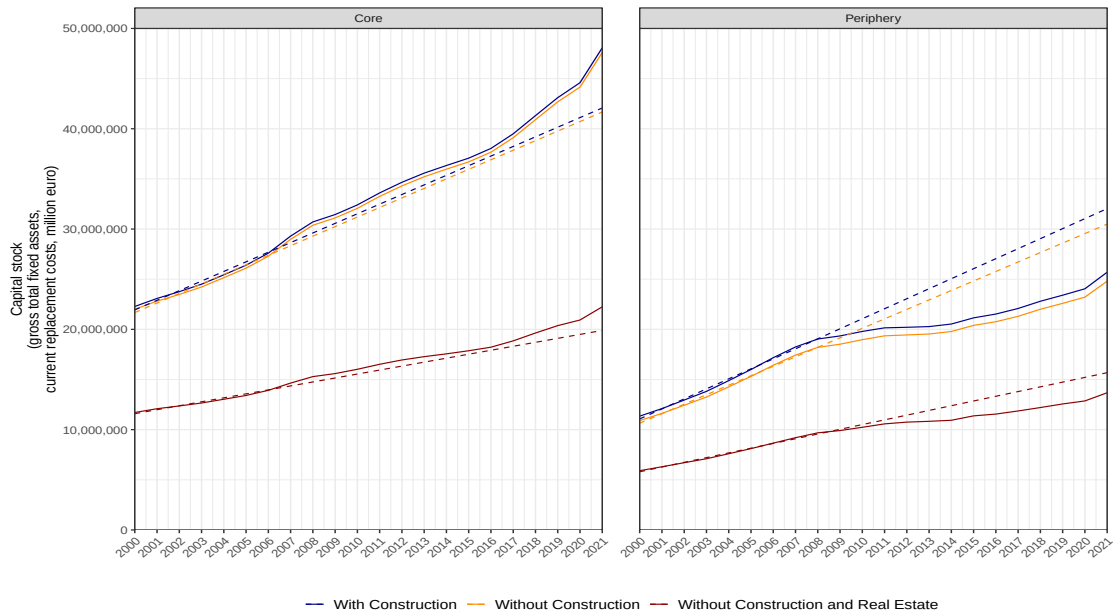


Figure 48: Gross Capital Stock Excluding Construction and Real Estate (2000–2021). The figure replicates the analysis using gross capital stock.

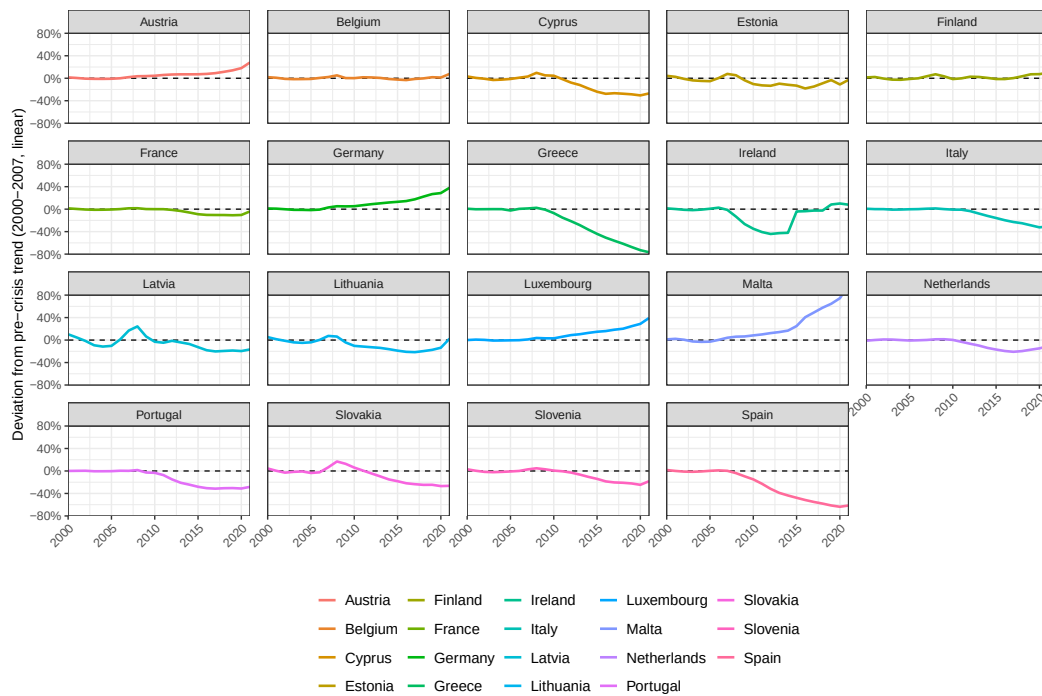


Figure 49: Deviation of National Capital Stocks from Pre-crisis Trends (2000–2021). The figure shows the deviation of each country's capital stock from its pre-crisis linear trend (2000–2007).

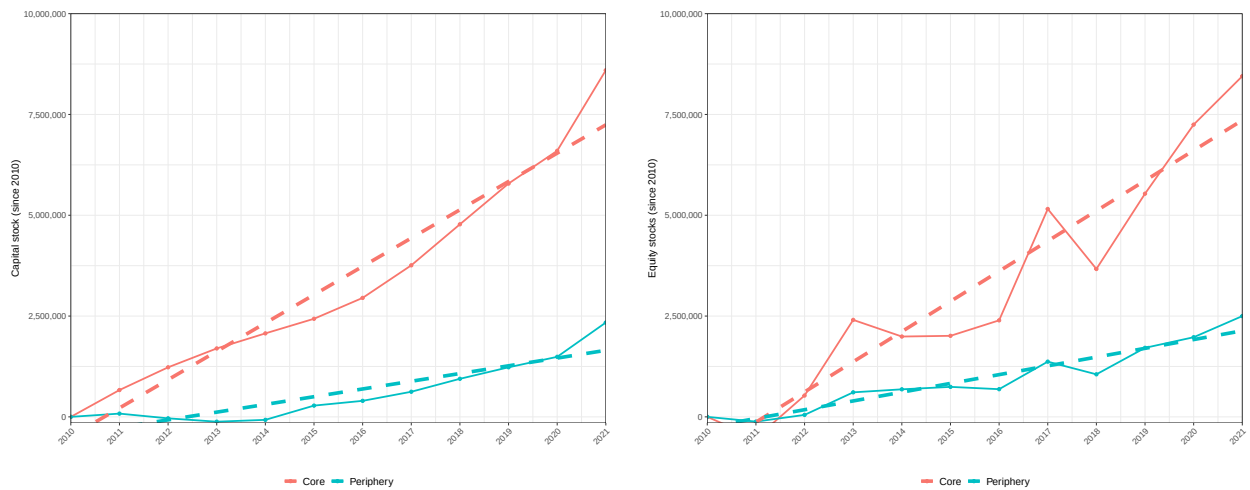


Figure 50: Capital Stock and Equity Stock Evolution (2010–2021). The figure compares the evolution of the capital stock (Eurostat) and equity stock (CPIS) for the Core and Periphery, normalized to zero in 2010.

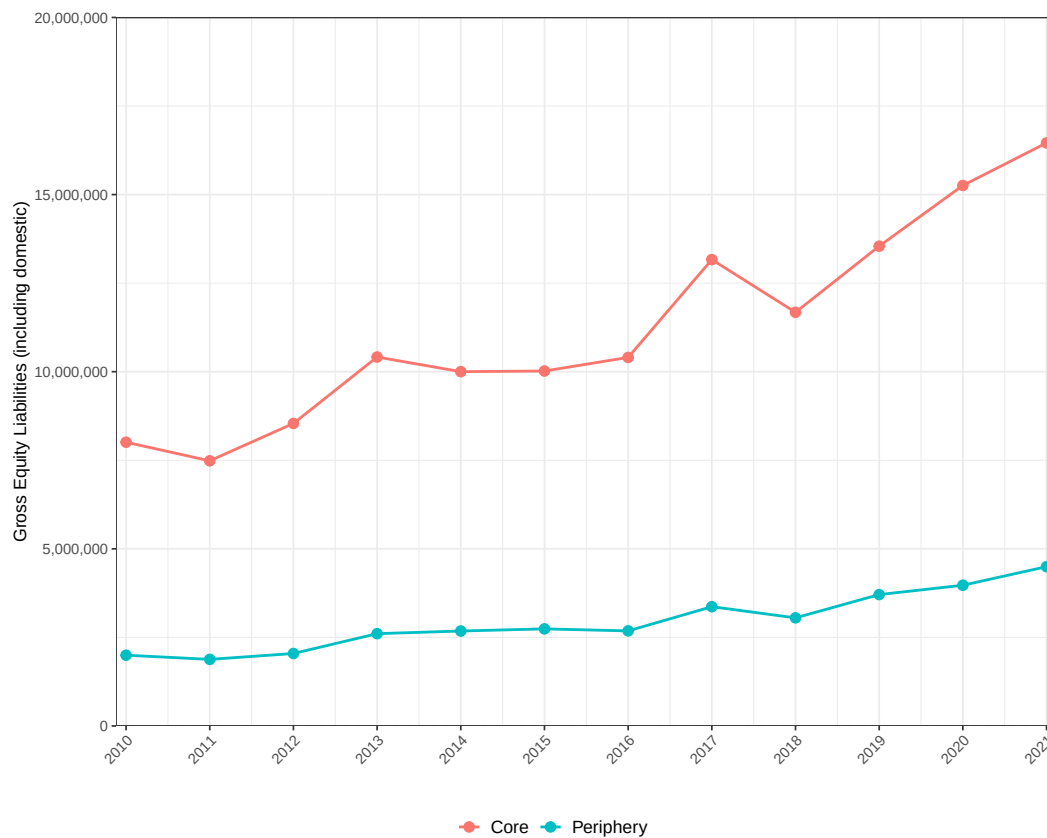


Figure 51: Gross Equity Liabilities by Group (2010–2021). The figure plots the value of gross equity liabilities as reported in the CPIS for the Core and Periphery.

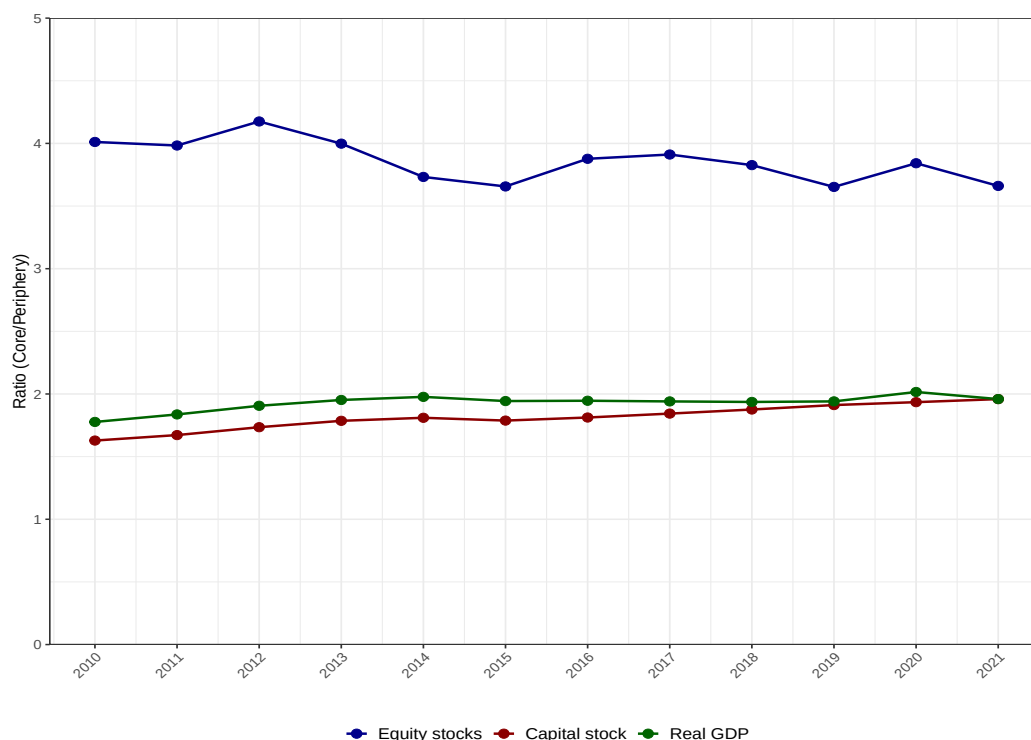


Figure 52: Ratios of Equity Stock, Capital Stock and Real GDP (2010–2021). The figure reports the ratio of equity stock, capital stock and real GDP for the Core and Periphery.

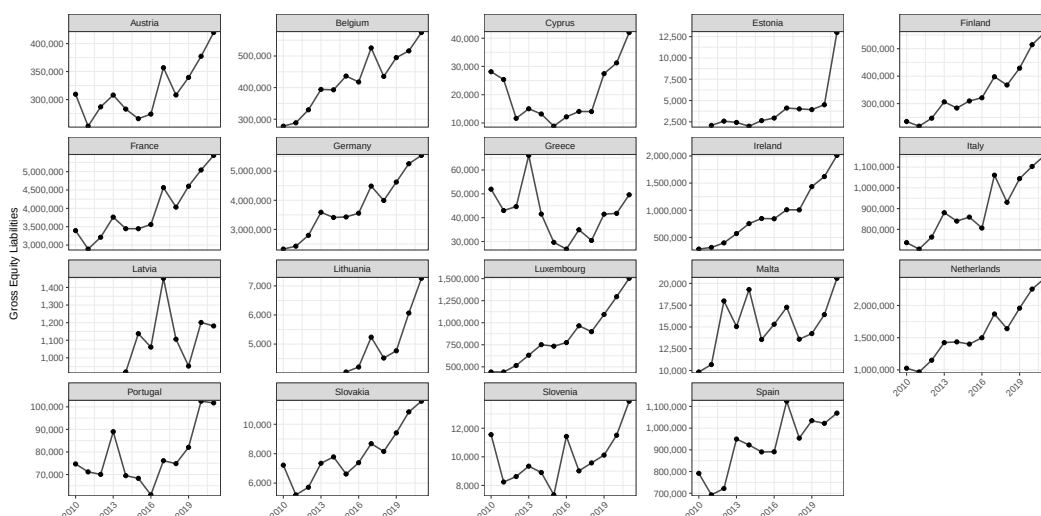


Figure 53: Gross Equity Liabilities by Country (2010–2021). The figure plots the value of gross equity liabilities as reported in the CPIS for euro area countries.

A.3 Two Dimensions of Integration

A.3.1 Financialization

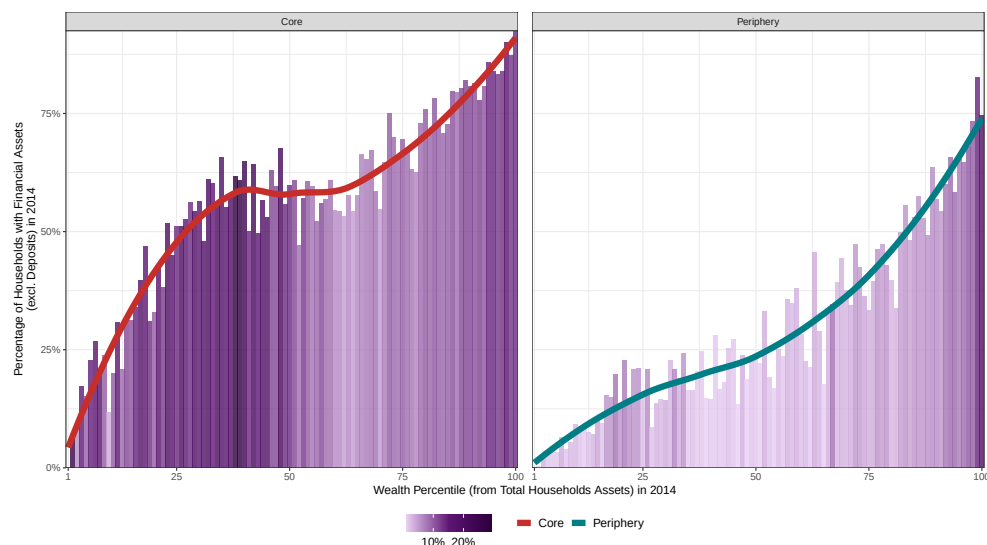


Figure 54: Financial Participation along the Wealth Distribution (2014). The figure plots, for each wealth percentile, the share of households holding financial assets (curve, extensive margin) and the share of financial assets in total household wealth (bars, intensive margin).

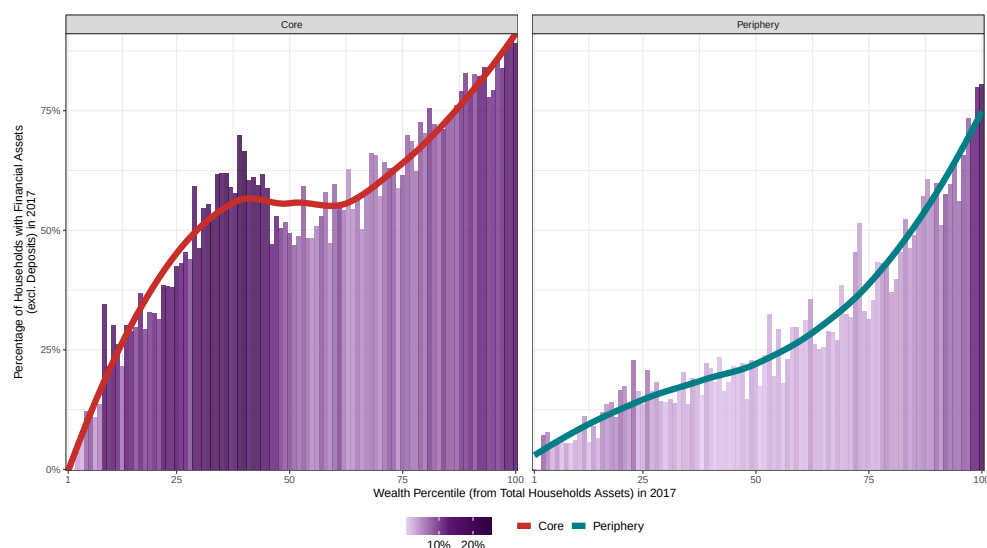


Figure 55: Financial Participation along the Wealth Distribution (2017). The figure shows the evolution of financialization relative to earlier waves.

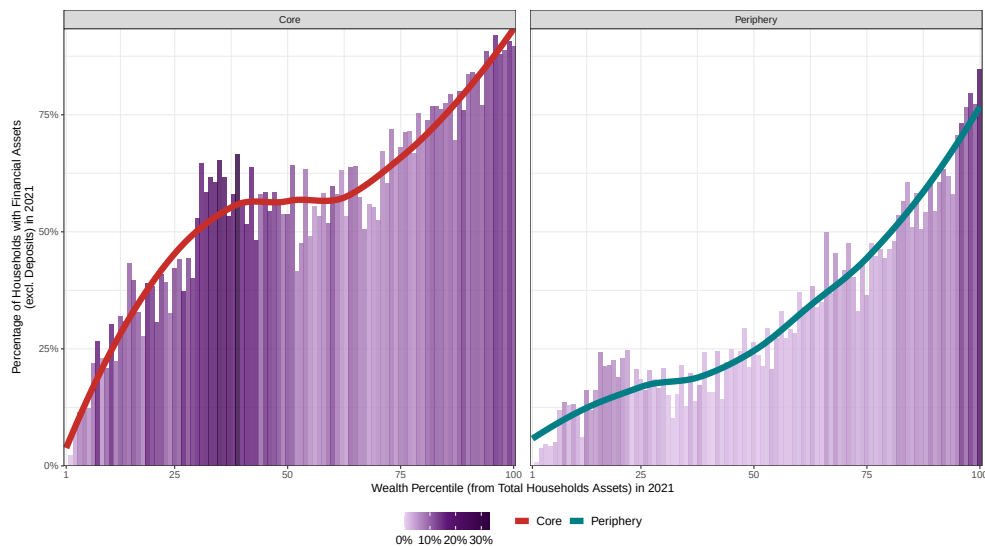


Figure 56: Financial Participation along the Wealth Distribution (2021). The figure reports financial participation and asset shares by wealth percentile, showing a slight increase in financialization since 2017 but a persistent gap between the Core and the Periphery.

A.3.2 Wealth Decomposition



Figure 57: Composition of Wealth along the Wealth Distribution (2010). The figure shows the composition of total household wealth by percentile of the wealth distribution for the Core (left) and the Periphery (right) in 2010. Each area represents the share of deposits (green), financial assets (orange), and real assets (blue) in total wealth.



Figure 58: Composition of Wealth along the Wealth Distribution (2014). The figure shows the composition of total household wealth by percentile of the wealth distribution for the Core (left) and the Periphery (right) in 2014. Each area represents the share of deposits (green), financial assets (orange), and real assets (blue) in total wealth.



Figure 59: Composition of Wealth along the Wealth Distribution (2017). The figure shows the composition of total household wealth by percentile of the wealth distribution for the Core (left) and the Periphery (right) in 2017. Each area represents the share of deposits (green), financial assets (orange), and real assets (blue) in total wealth.



Figure 60: Composition of Wealth along the Wealth Distribution (2021). The figure shows the composition of total household wealth by percentile of the wealth distribution for the Core (left) and the Periphery (right) in 2021. Each area represents the share of deposits (green), financial assets (orange), and real assets (blue) in total wealth.

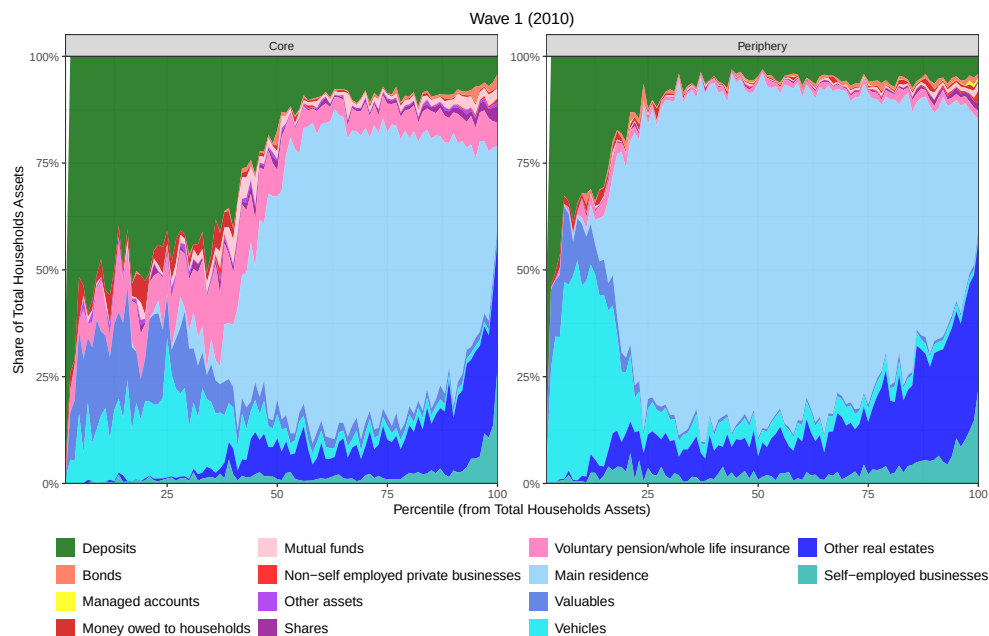


Figure 61: Detailed Composition of Wealth along the Wealth Distribution (2010). The figure displays the detailed breakdown of household wealth by percentile for the Core (left) and the Periphery (right) in 2010. Each color represents a distinct asset category: deposits, bonds, managed accounts, mutual funds, shares, voluntary pension and life insurance, money owed to households, self-employed and non-self-employed private businesses, valuables, main residences, other real estate, and vehicles.

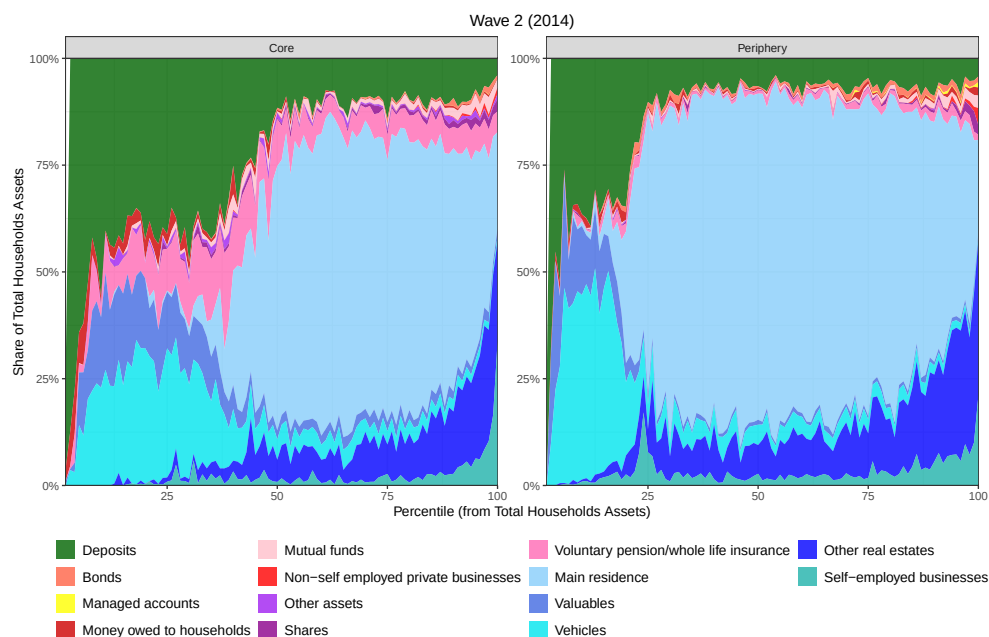


Figure 62: Detailed Composition of Wealth along the Wealth Distribution (2014). The figure displays the detailed breakdown of household wealth by percentile for the Core (left) and the Periphery (right) in 2014.

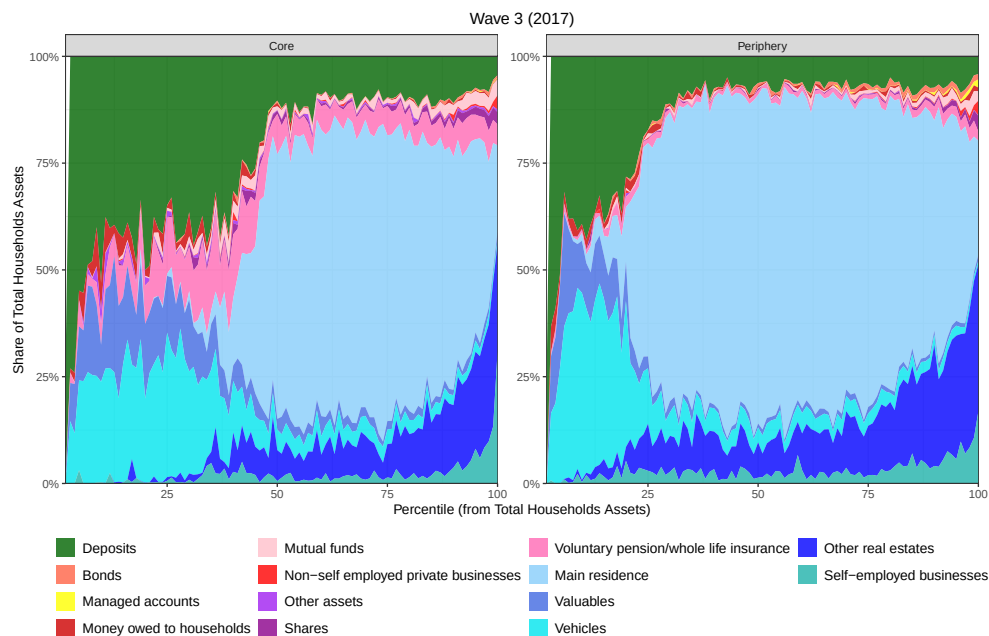


Figure 63: Detailed Composition of Wealth along the Wealth Distribution (2017). The figure displays the detailed breakdown of household wealth by percentile for the Core (left) and the Periphery (right) in 2017.

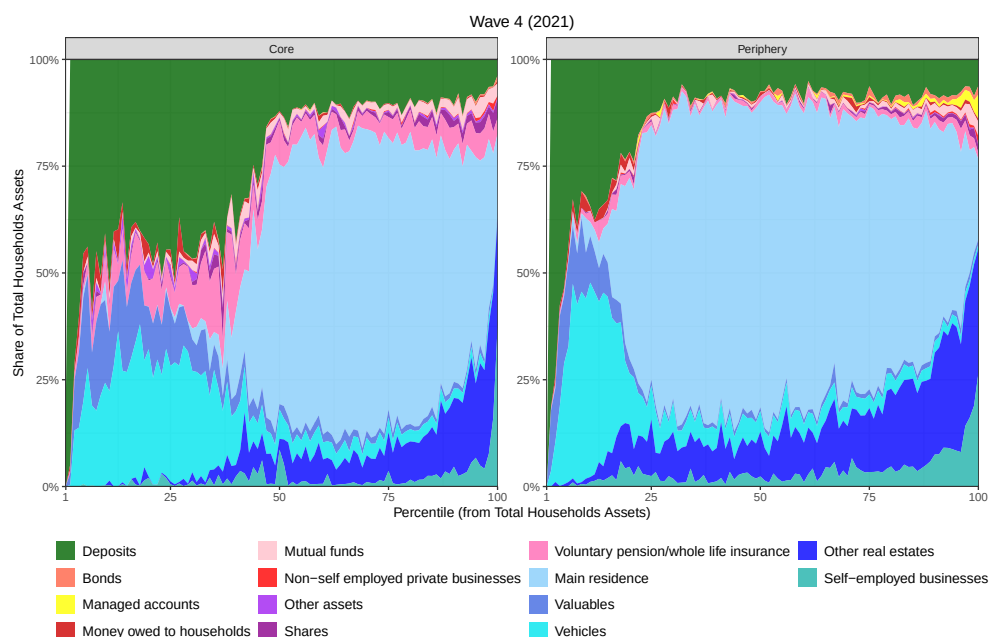


Figure 64: Detailed Composition of Wealth along the Wealth Distribution (2021). The figure displays the detailed breakdown of household wealth by percentile for the Core (left) and the Periphery (right) in 2021.

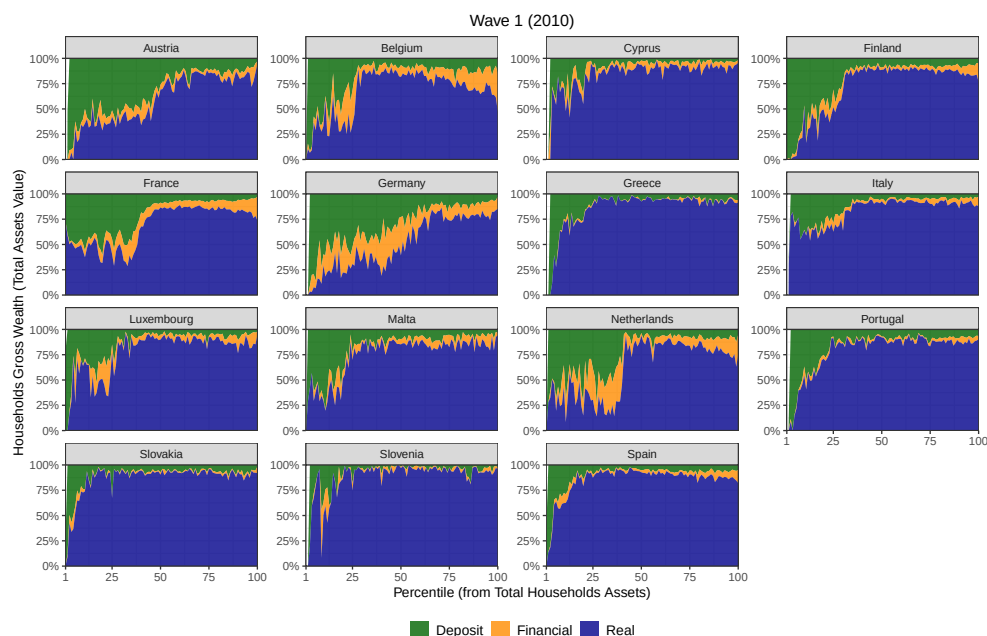


Figure 65: Composition of Wealth along the Wealth Distribution by Country (2010). The figure shows the composition of total household wealth by percentile of the wealth distribution for euro area countries in 2010. Each area represents the share of deposits (green), financial assets (orange), and real assets (blue) in total wealth.

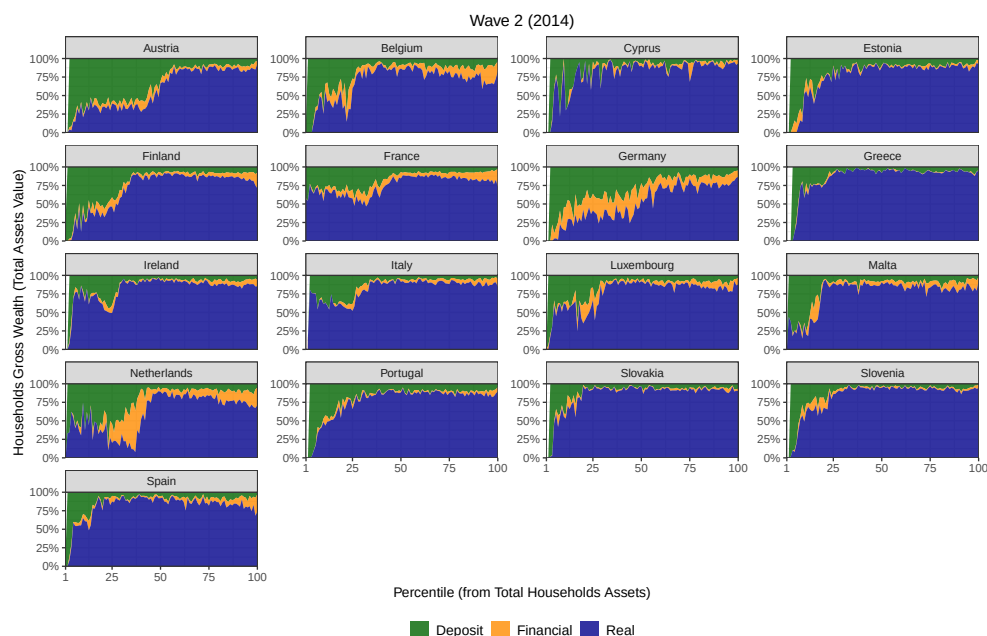


Figure 66: Composition of Wealth along the Wealth Distribution by Country (2014). The figure shows the composition of total household wealth by percentile of the wealth distribution for euro area countries in 2014. Each area represents the share of deposits (green), financial assets (orange), and real assets (blue) in total wealth.

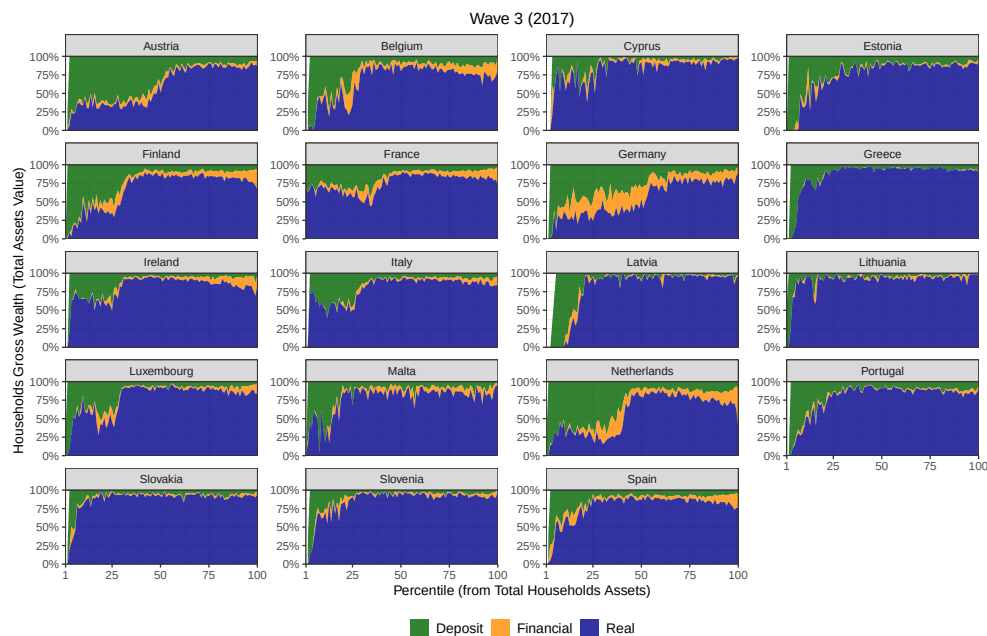


Figure 67: Composition of Wealth along the Wealth Distribution by Country (2017). The figure shows the composition of total household wealth by percentile of the wealth distribution for euro area countries in 2017. Each area represents the share of deposits (green), financial assets (orange), and real assets (blue) in total wealth.

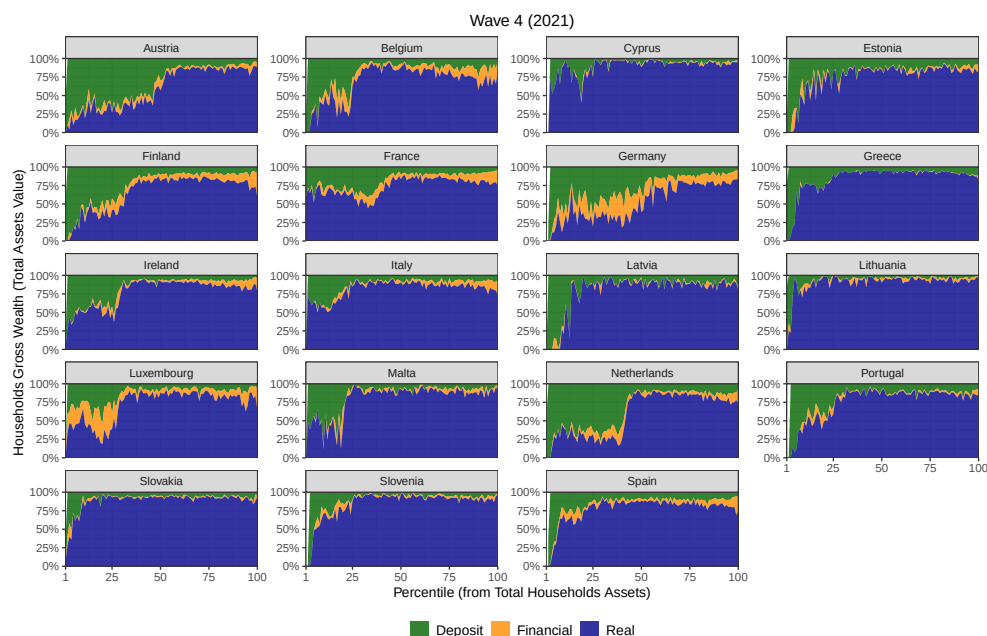


Figure 68: Composition of Wealth along the Wealth Distribution by Country (2021). The figure shows the composition of total household wealth by percentile of the wealth distribution for euro area countries in 2021. Each area represents the share of deposits (green), financial assets (orange), and real assets (blue) in total wealth.

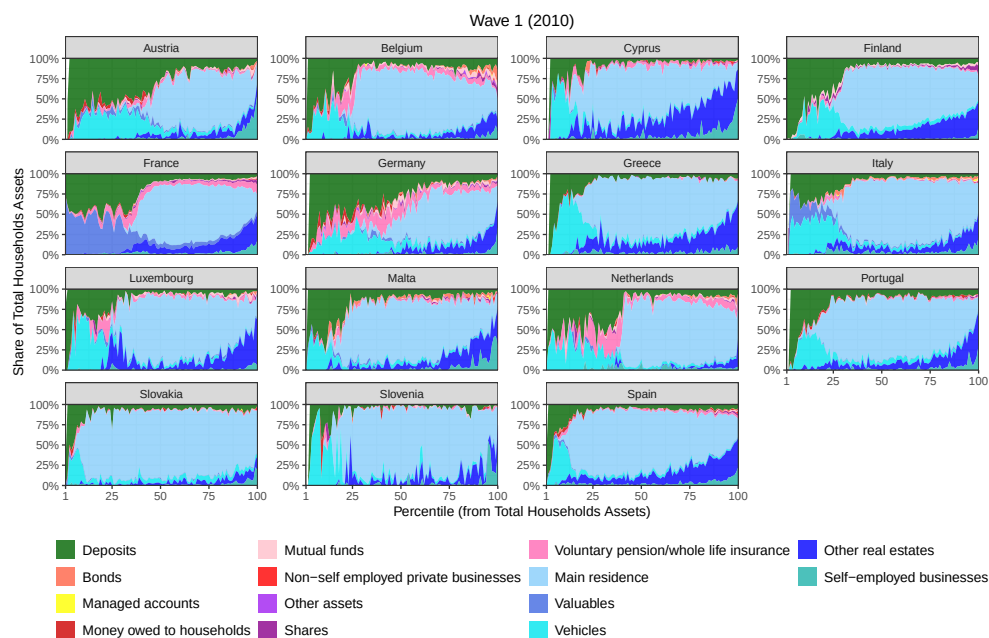


Figure 69: Detailed Composition of Wealth along the Wealth Distribution by Country (2010). The figure displays the detailed breakdown of household wealth by percentile for euro area countries in 2010. Each color represents a distinct asset category: deposits, bonds, managed accounts, mutual funds, shares, voluntary pension and life insurance, money owed to households, self-employed and non-self-employed private businesses, valuables, main residences, other real estate, and vehicles.

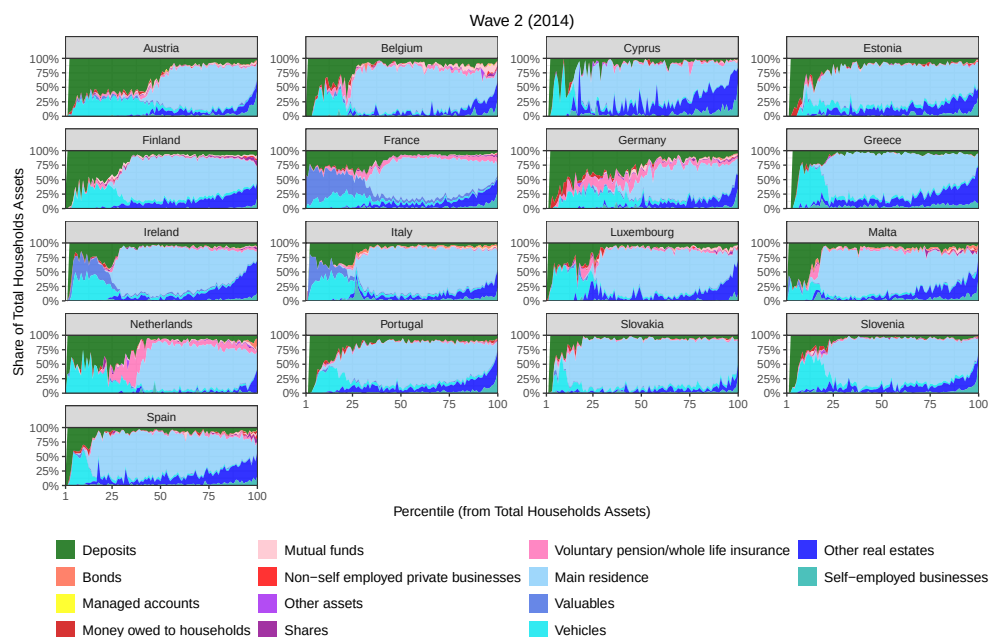


Figure 70: Detailed Composition of Wealth along the Wealth Distribution by Country (2014). The figure displays the detailed breakdown of household wealth by percentile for euro area countries in 2014.

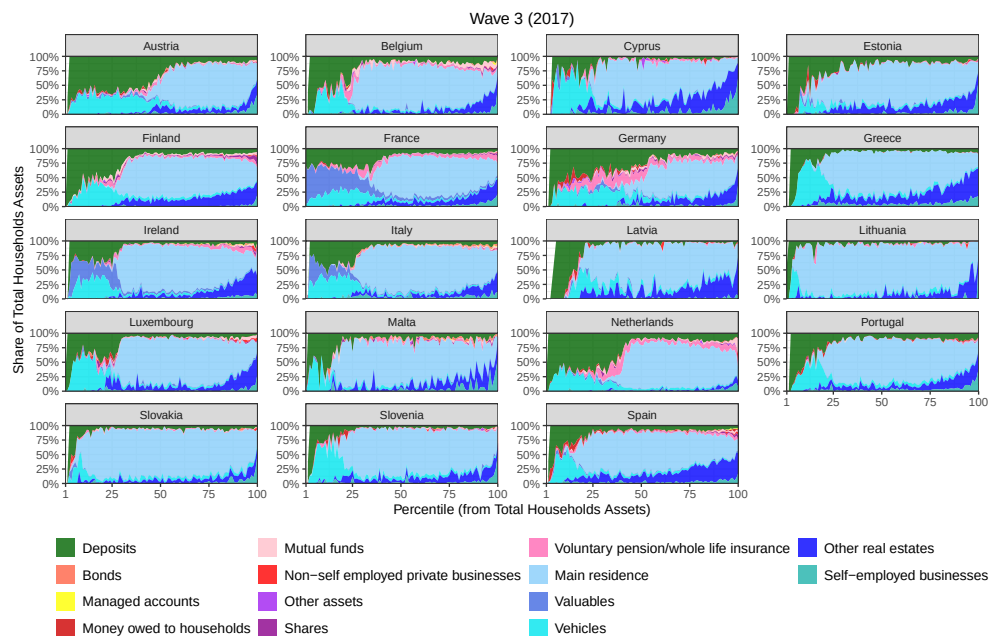


Figure 71: Detailed Composition of Wealth along the Wealth Distribution by Country (2017). The figure displays the detailed breakdown of household wealth by percentile for euro area countries in 2017.

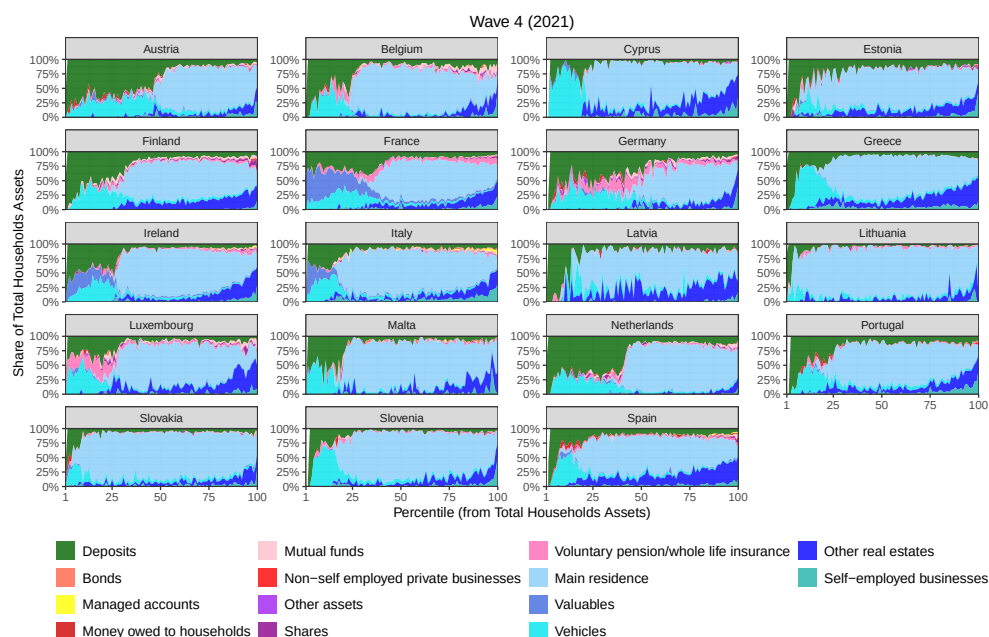


Figure 72: Detailed Composition of Wealth along the Wealth Distribution by Country (2021). The figure displays the detailed breakdown of household wealth by percentile for euro area countries in 2021.

A.3.3 Limited Evidence of Diversification at the Household Level

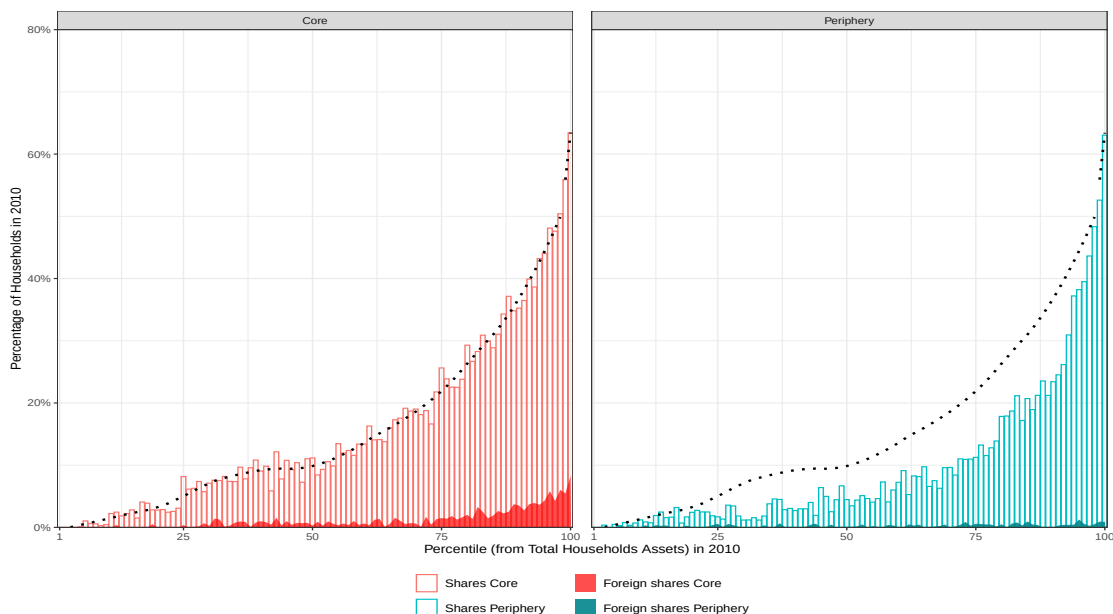


Figure 73: Share and Foreign Share Ownership Along the Wealth Distribution (2010). The figure displays the percentage of households holding shares (solid bars) and foreign shares (filled bars) by wealth percentile for the Core and the Periphery. The dotted line represents the Core distribution, allowing for a visual comparison of the shape and concentration of ownership between the two groups.

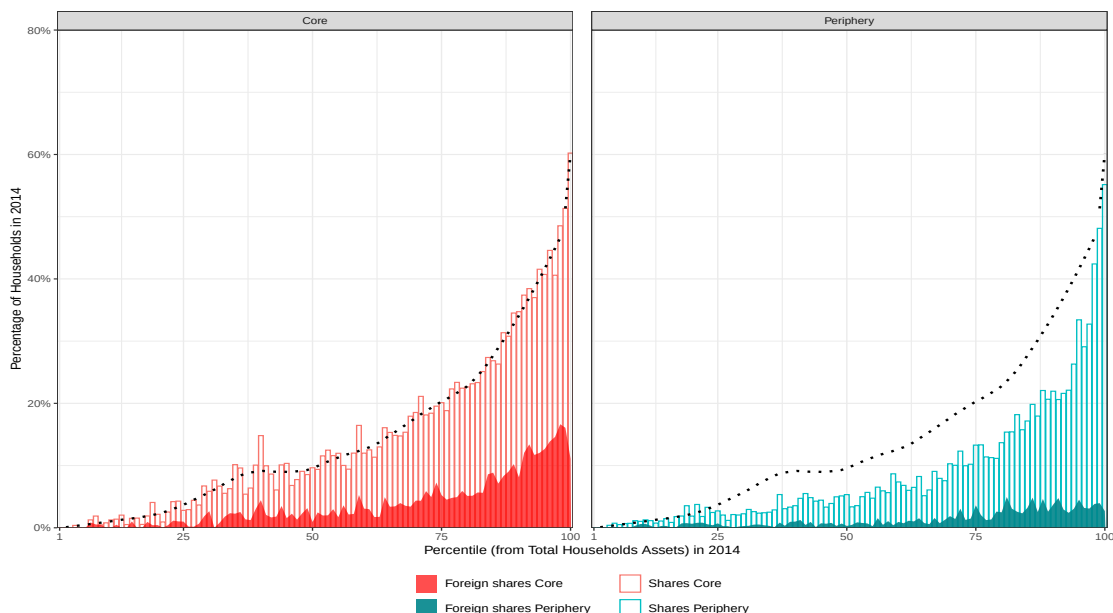


Figure 74: Share and Foreign Share Ownership Along the Wealth Distribution (2014). The figure displays the percentage of households holding shares (solid bars) and foreign shares (filled bars) by wealth percentile for the Core and the Periphery. The dotted line represents the Core distribution.

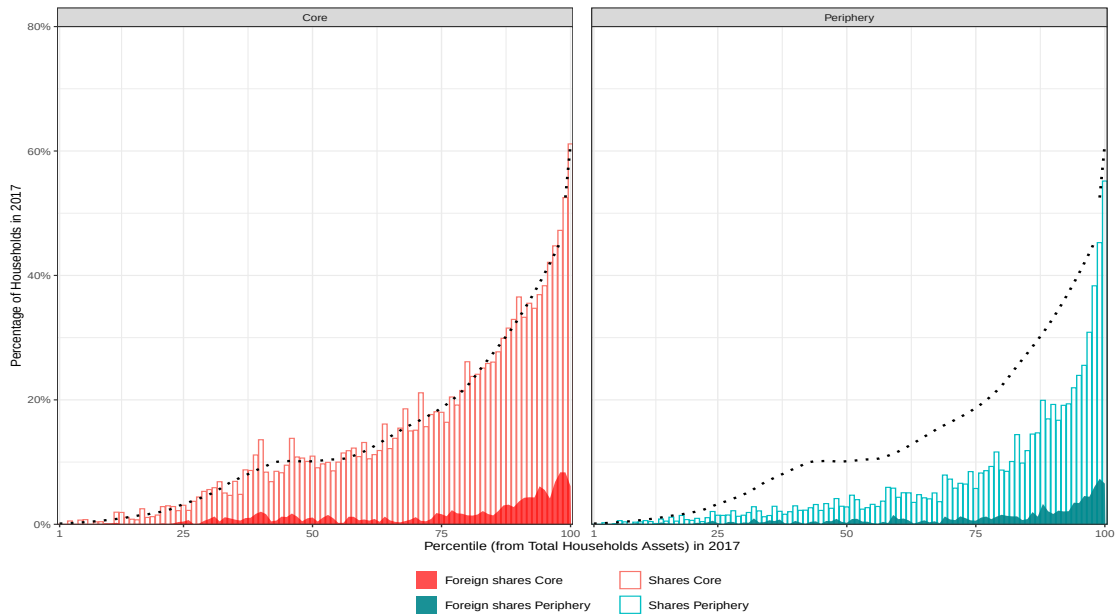


Figure 75: Share and Foreign Share Ownership Along the Wealth Distribution (2017). The figure displays the percentage of households holding shares (solid bars) and foreign shares (filled bars) by wealth percentile for the Core and the Periphery. The dotted line represents the Core distribution.

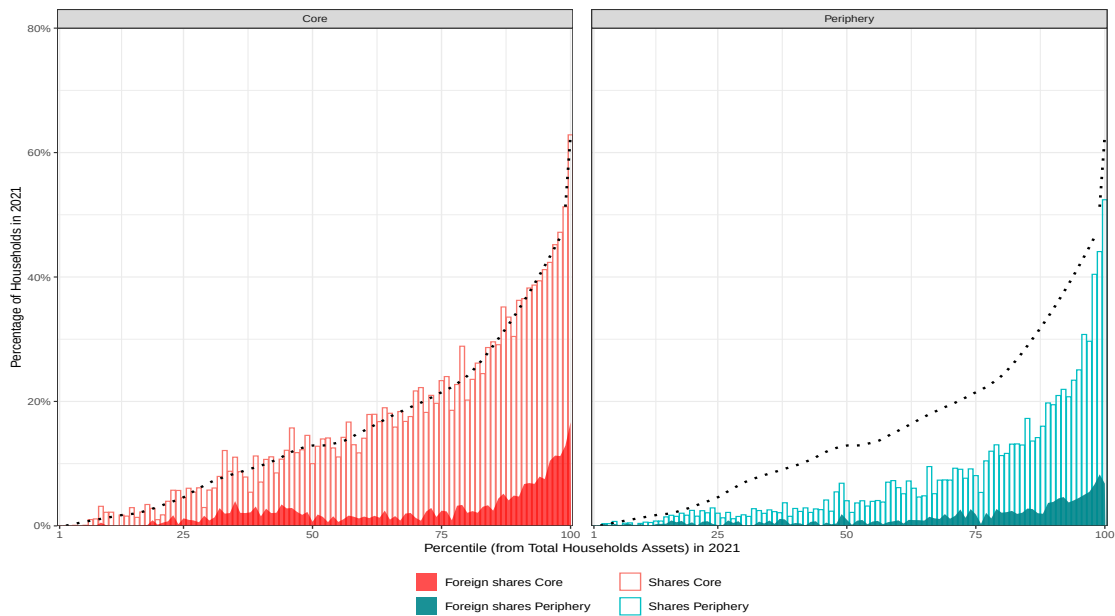


Figure 76: Share and Foreign Share Ownership Along the Wealth Distribution (2021). The figure displays the percentage of households holding shares (solid bars) and foreign shares (filled bars) by wealth percentile for the Core and the Periphery. The dotted line represents the Core distribution.

A.3.4 Home Bias

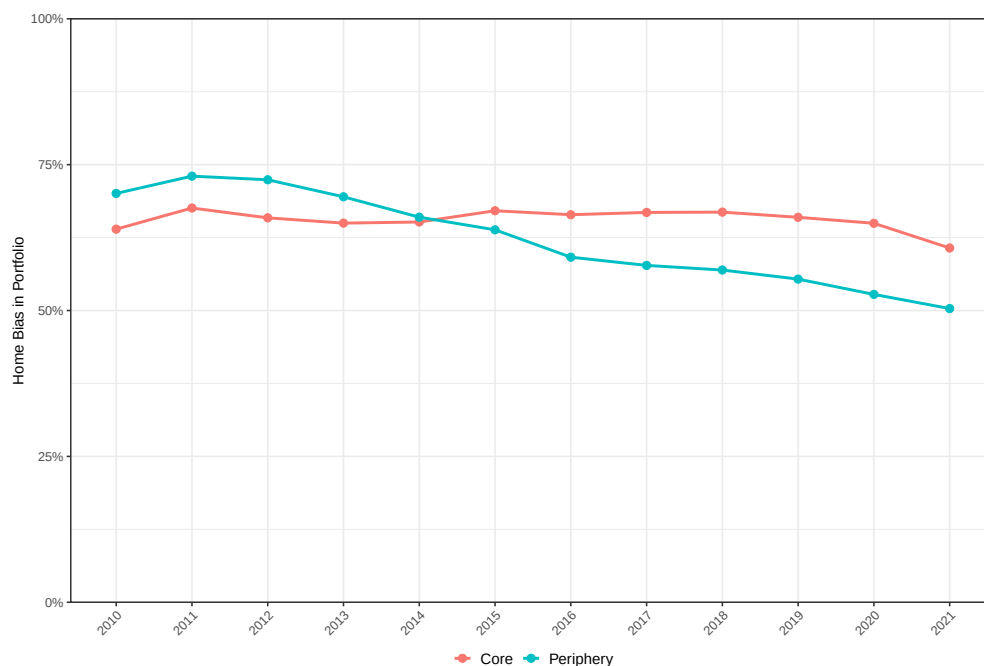


Figure 77: Home Bias in Portfolio Holdings, including the Rest of the World (2010–2021). The figure shows the home bias in equity portfolios for Core and Periphery euro area countries, including the Rest of the World.



Figure 78: Home Bias in Portfolio Holdings by Country (2010–2021). The figure shows the home bias in equity portfolios across euro area countries, restricted to euro area countries.

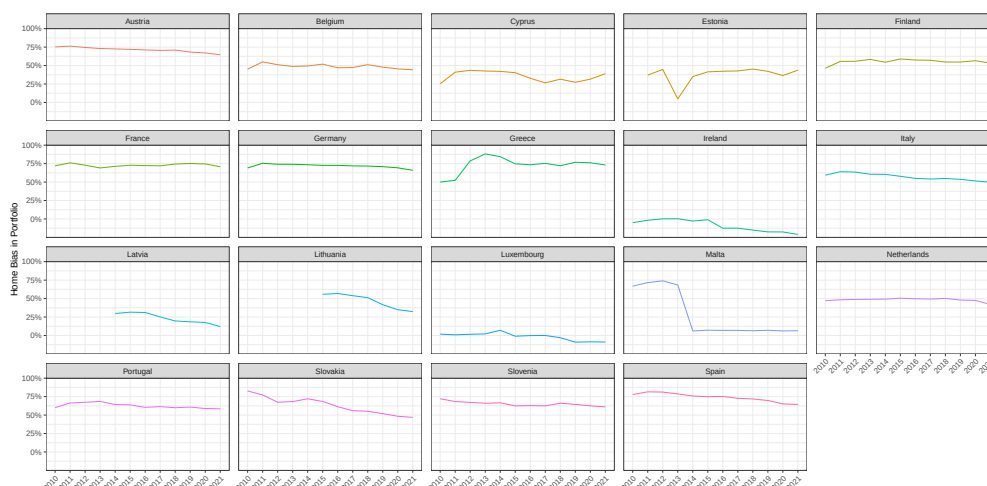


Figure 79: Home Bias in Portfolio Holdings by Country, including the Rest of the World (2010–2021). The figure shows the home bias in equity portfolios across euro area countries, including the Rest of the World.

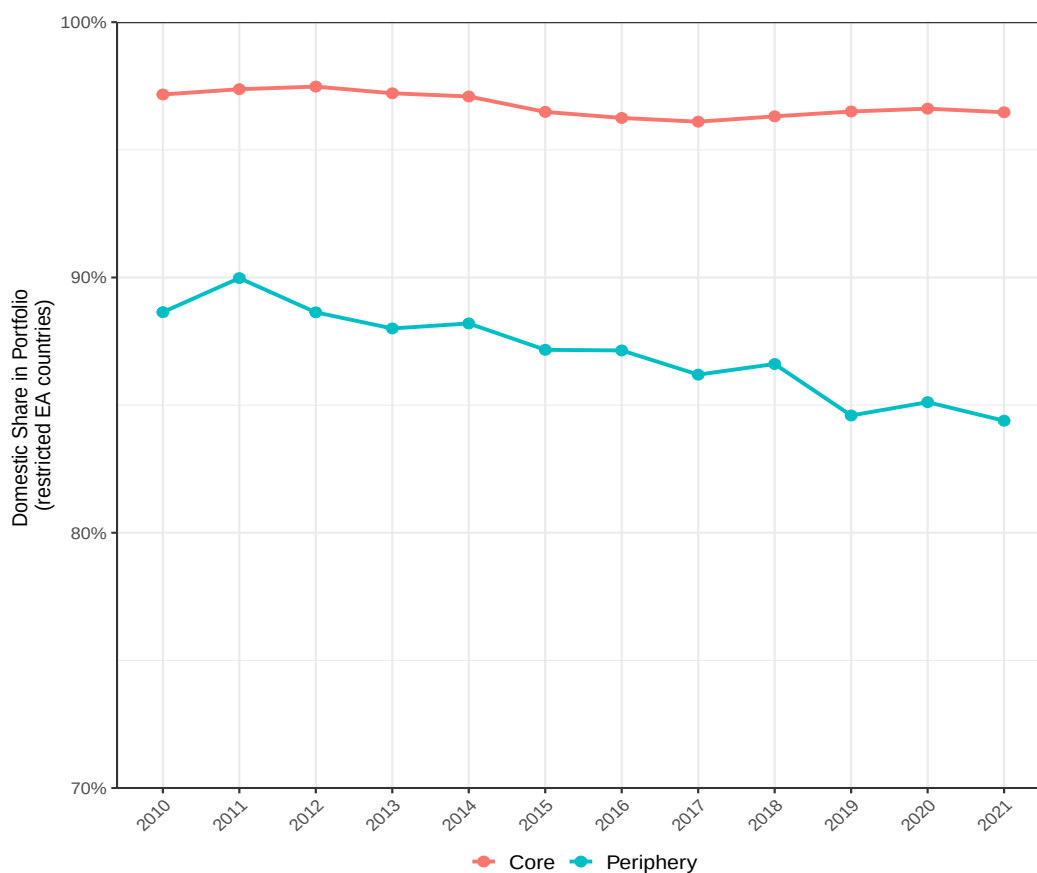


Figure 80: Domestic share in Portfolio (2010–2021). The figure shows the share of domestic assets in equity portfolios for Core and Periphery euro area countries, restricted to euro area countries.

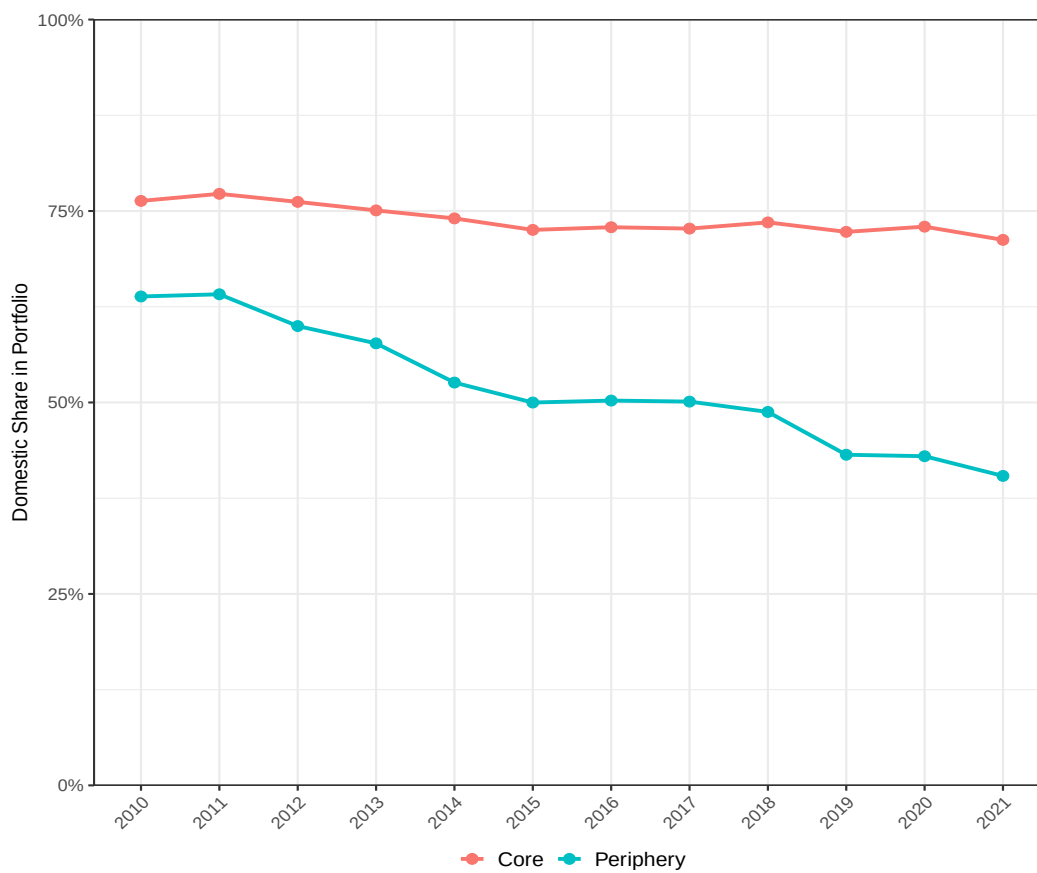


Figure 81: Domestic share in Portfolio, including the Rest of the World (2010–2021). The figure shows the share of domestic assets in equity portfolios for Core and Periphery euro area countries, including the Rest of the World.

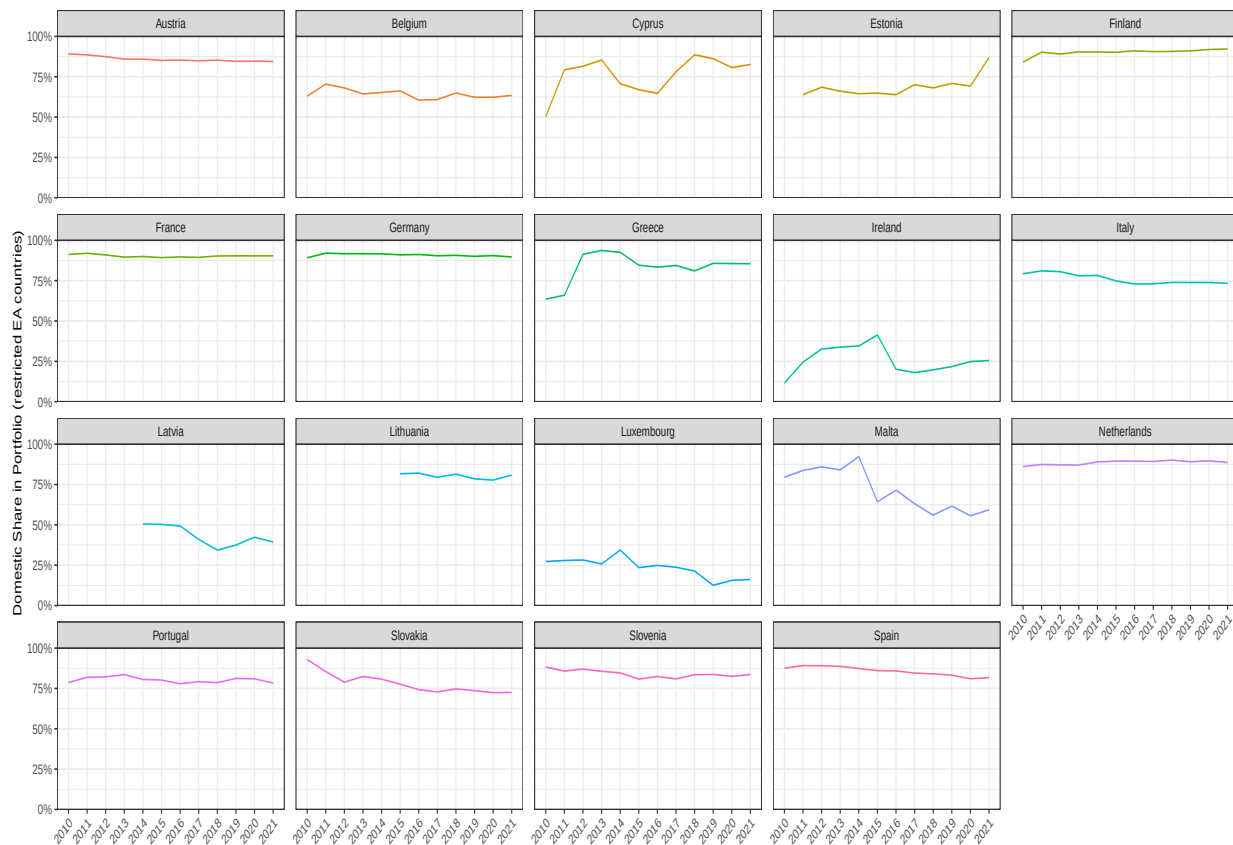


Figure 82: Domestic share in Portfolio by Country (2010–2021). The figure shows the share of domestic assets in equity portfolios across euro area countries.

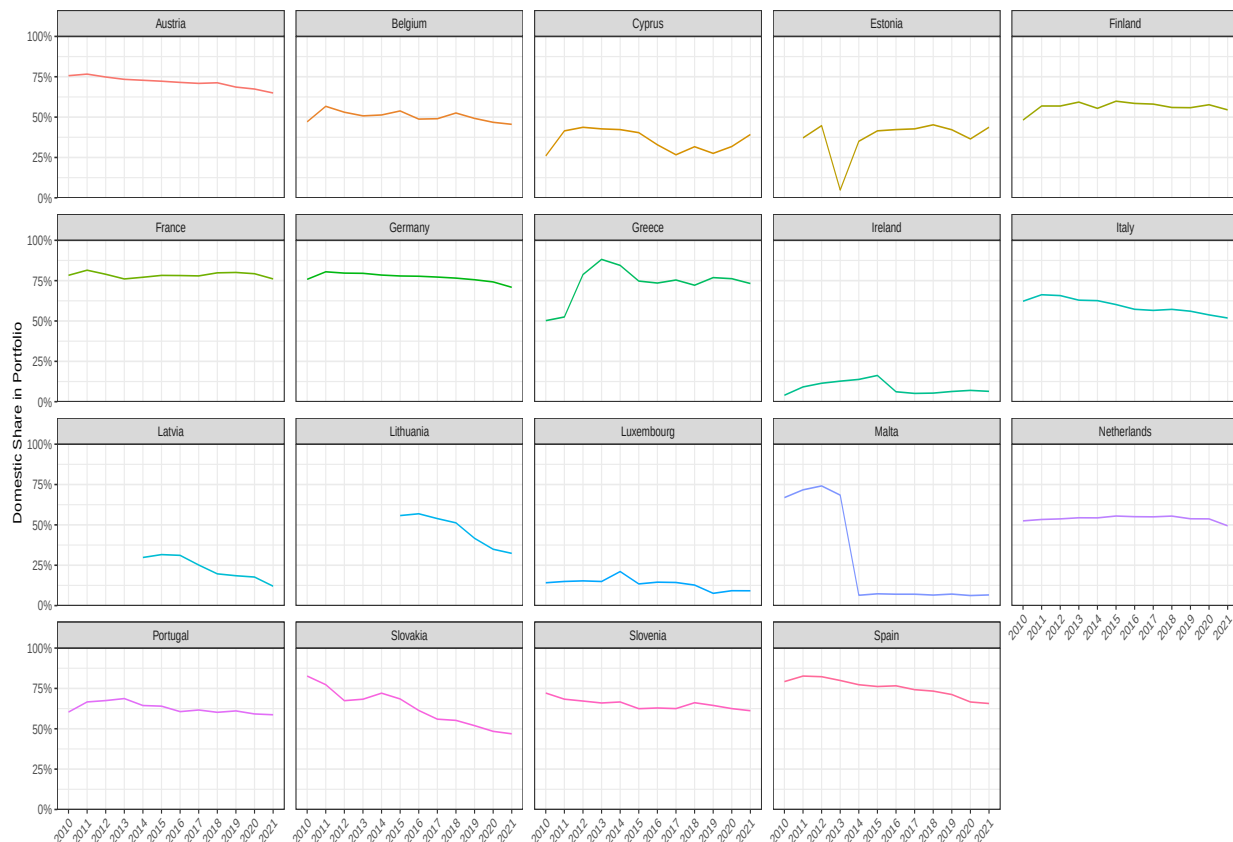


Figure 83: Domestic share in Portfolio by Country, including the Rest of the World (2010–2021). The figure shows the share of domestic assets in equity portfolios across euro area countries, including the Rest of the World.

B Two-Agent New Keynesian (TANK) Model

B.1 One-asset equivalent

In the baseline model, Saver households invest separately in two assets: equity claims issued by the domestic mutual fund, E_t^i , and one-period nominal union bonds, b_t^i . This section presents an equivalent formulation in which households instead hold a single composite asset issued by the mutual fund, denoted A_t^i , which internally reallocates between equity and bonds. This alternative representation is analytically equivalent to the baseline model but simplifies the household problem by consolidating all financial decisions into one aggregate asset.

B.1.1 Household

Households now purchase only the composite mutual fund claim A_t^i , instead of holding E_t^i and b_t^i separately. Their first-order condition for A_t^i becomes:

$$C_{S,t}^i{}^{-\sigma} \left(1 + p_t^i \left[\frac{E_t^i}{A_t^i} \frac{\kappa_E}{(1-\lambda^i)} (E_t^i - \bar{E}^i) + \frac{b_t^i}{A_t^i} \frac{\kappa_D}{(1-\lambda^i)} b_t^i \right] \right) = \beta \mathbb{E}_t \left[C_{S,t+1}^i{}^{-\sigma} r_{t+1}^A \right],$$

where r_{t+1}^A denotes the gross real return on the aggregate fund. The term in parentheses captures the adjustment costs from changing equity and bond positions within the fund, weighted by their respective portfolio shares in A_t^i .

B.1.2 Risk-Neutral Mutual Fund

The mutual fund in each country i issues a composite claim A_t^i , whose value equals the sum of the fund's equity and bond holdings:

$$A_t^i = E_t^i + b_t^i.$$

Internally, the fund allocates its portfolio between equity and bonds according to optimal returns. The equity component E_t^i consists of both domestic and foreign capital holdings:

$$E_t^i = K_{i,t}^i + S_t K_{i,t}^j,$$

where S_t is the relative price of the foreign good in terms of the domestic consumption basket.

The fund's realized return is the weighted average of returns on equity and bonds:

$$r_t^A A_{t-1}^i = r_t^i E_{t-1}^i + \frac{r_{t-1}^U}{\pi_t^{CPI,i}} b_{t-1}^i,$$

where r_t^i is the real return on equity, and $\frac{r_{t-1}^U}{\pi_t^{CPI,i}}$ is the real return on the nominal union bond.

The mutual fund's no-arbitrage condition links the expected return on equity to that on

bonds, taking into account portfolio adjustment costs:

$$\mathbb{E}_t(r_{t+1}^i) + \frac{\kappa_D}{(1 - \lambda^i)} p_t^i b_t^i = \frac{r_t^U}{\pi_{t+1}^{CPI,i}} + \frac{\kappa_E}{(1 - \lambda^i)} p_t^i (E_t^i - \bar{E}^i).$$

This condition states that expected returns must equalize across assets once adjustment costs are accounted for. A higher κ_D or κ_E increases the effective cost of reallocating wealth between bonds and equity, thereby reducing the responsiveness of portfolios to changes in relative returns.

The equity component of the fund obeys the same no-profit condition as in the baseline model:

$$r_t^i E_{t-1}^i = (r_t^{K,i} + 1 - \delta^i) K_{i,t-1}^i + (r_t^{K,j} + 1 - \delta^j) S_t K_{i,t-1}^j - \frac{\kappa^i}{2} \left(\frac{K_{i,t-1}^i}{E_{t-1}^i} - \frac{\bar{K}_i^i}{\bar{E}^i} \right)^2 E_{t-1}^i.$$

The corresponding no-arbitrage condition between domestic and foreign capital holdings is:

$$\mathbb{E}_t(r_{t+1}^{K,i} + 1 - \delta^i) = \mathbb{E}_t \left[(r_{t+1}^{K,j} + 1 - \delta^j) \frac{S_{t+1}}{S_t} \right] + \kappa^i \left(\frac{K_{i,t}^i}{E_t^i} - \frac{\bar{K}_i^i}{\bar{E}^i} \right).$$

Thus, the internal portfolio of the mutual fund, composed of both equity and bonds, satisfies the same optimality and no-arbitrage conditions as in the baseline model. The only difference is that the household now interacts with the mutual fund through a single composite claim, while the fund reallocates internally across assets to ensure optimal portfolio returns.

This one-asset equivalent formulation provides a convenient way to represent the fund as the unique financial intermediary in the economy. It unifies the household's asset choice, while preserving the model's two key mechanisms: (i) portfolio composition across domestic and foreign capital (ζ^i), and (ii) portfolio reallocation dynamics governed by adjustment costs (κ^i). The simplification is particularly useful when extending the framework to heterogeneous-agent settings, where computational efficiency and aggregation are essential.

C Results - Two-Agent Model

C.1 Steady-state Portfolios Determination

In the steady state, portfolio allocations are determined by the domestic ownership shares of capital in each country. Let θ^i denote the share of country i 's total capital that is domestically owned, such that:

$$K_i^i = \theta^i K^i, \quad K_j^i = (1 - \theta^i) K^i.$$

Total equity held by residents of country i is therefore given by:

$$E^i = K_i^i + S K_i^j = \theta^i K^i + (1 - \theta^i) S K^j,$$

where S is the steady-state relative price (terms of trade) between the two countries.

The steady-state domestic portfolio share ζ^i , that is, the fraction of total equity owned domestically, is defined as:

$$\zeta^i \equiv \frac{\bar{K}_i^i}{\bar{E}^i} = \frac{\theta^i \bar{K}^i}{\theta^i \bar{K}^i + (1 - \theta^i) S \bar{K}^j}.$$

This relationship links the observed steady-state portfolio composition to the underlying domestic ownership shares (θ^C, θ^P) . Not all combinations of (θ^C, θ^P) yield interior solutions for ζ^C and ζ^P : some parameter pairs lead to corner solutions where portfolios are fully domestic or fully foreign.

Figure 84 illustrate this mapping under a symmetric calibration, showing how the steady-state domestic portfolio shares ζ^C and ζ^P vary with the domestic ownership parameters θ^C and θ^P . The yellow regions correspond to corner solutions, where full domestic ownership ($\zeta = 1$) prevails, while the darker regions represent interior cases with partial diversification.

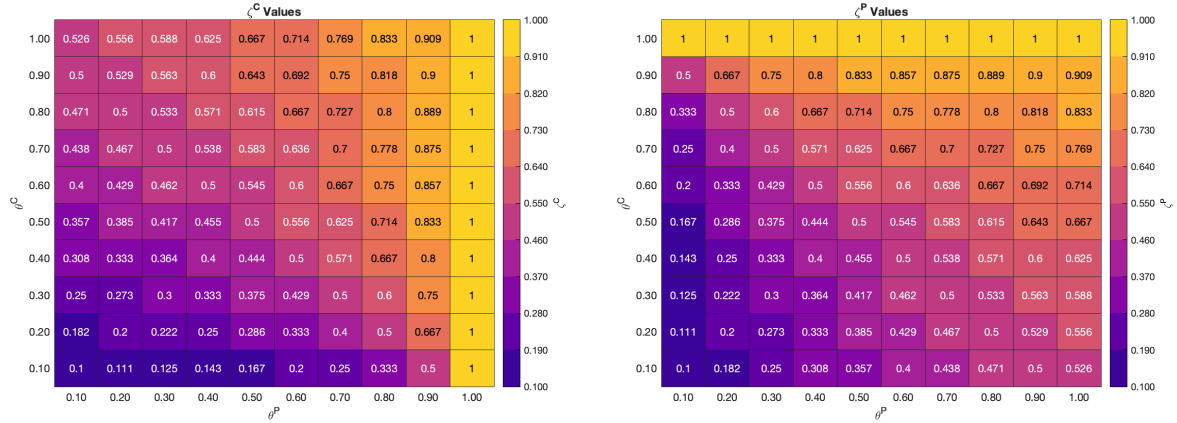


Figure 84: Steady-State Portfolio Shares ζ^C and ζ^P under Symmetric Calibration. The figures show the mapping between the domestic ownership shares (θ^C, θ^P) and the corresponding steady-state domestic portfolio shares (ζ^C, ζ^P).

To match the empirical values of ζ^C and ζ^P , I select the triplet $(\theta^C, \theta^P, \psi)$, where ψ denotes the ratio of the Periphery's to the Core's total GDP, calibrated in order to match the relative economic size and the ratio of total equity between the two blocs in 2010. Together, (θ^C, θ^P) determine the domestic ownership structure within each country, while ψ ensures that the relative scale of capital and equity holdings is consistent with observed macroeconomic aggregates. This approach guarantees internal consistency between the steady-state portfolio composition, the relative capital stocks, and the empirical Core–Periphery asymmetries in financial integration. Table 11 reports the calibrated parameter values for 2010.

Table 11: Calibration of Portfolio Determination Parameters (2010)

Parameter (Symbol and Description)	Value	Target
Domestic ownership share in the Core, θ^C	0.974	Domestic portfolio share in the Periphery, $\zeta^C = 0.972$
Domestic ownership share in the Periphery, θ^P	0.877	Domestic portfolio share in the Periphery, $\zeta^P = 0.886$
GDP ratio (Periphery/Core), ψ	0.231	Ratio of total equity holdings (Periphery/Core)

C.2 Calibrated case to 2010 with Standard Asymmetric Shock

In this section, I replicate the results from Section 4.2 under a temporary 1% decline in productivity in the Periphery only (a MIT shock), which is a more standard specification. The calibration remains unchanged, except that $\sigma_z^P = 0.01$ while $\sigma_z^C = 0$.

Figure 85 compares the impulse responses of capital and consumption for Savers and Non-Savers under both configurations. The dotted lines correspond to equity autarky, while the solid lines represent the calibrated 2010 economy. It provides similar qualitative results than under a large estimation of the shock.

Relative to autarky, capital market integration amplifies the cross-country divergence in investment and consumption dynamics. Following a negative technology shock in the Periphery, capital falls more sharply there (−3 percentage points) and rises in the Core (+0.7 percentage points), as savings are reallocated toward the higher-return region.

For Savers, the effects reflect the interaction of diversification and reallocation. In the Periphery, Savers experience a smaller decline in consumption (+3 percentage points relative to autarky) owing to income insurance from Core asset holdings. Core Savers, by contrast, face a modestly larger decline (−0.4 percentage points) due to exposure to the Periphery downturn. Yet, because the Periphery is more diversified than the Core, the gains for Periphery Savers outweigh the losses for Core Savers.

For Non-Savers, the transmission operates through wages. The stronger contraction in Periphery capital depresses labor demand and real wages (−0.78 percentage points), reducing Non-Saver consumption. In the Core, higher capital inflows increase labor demand, moderating the wage decline and slightly raising Non-Saver consumption (+0.21 percentage points).

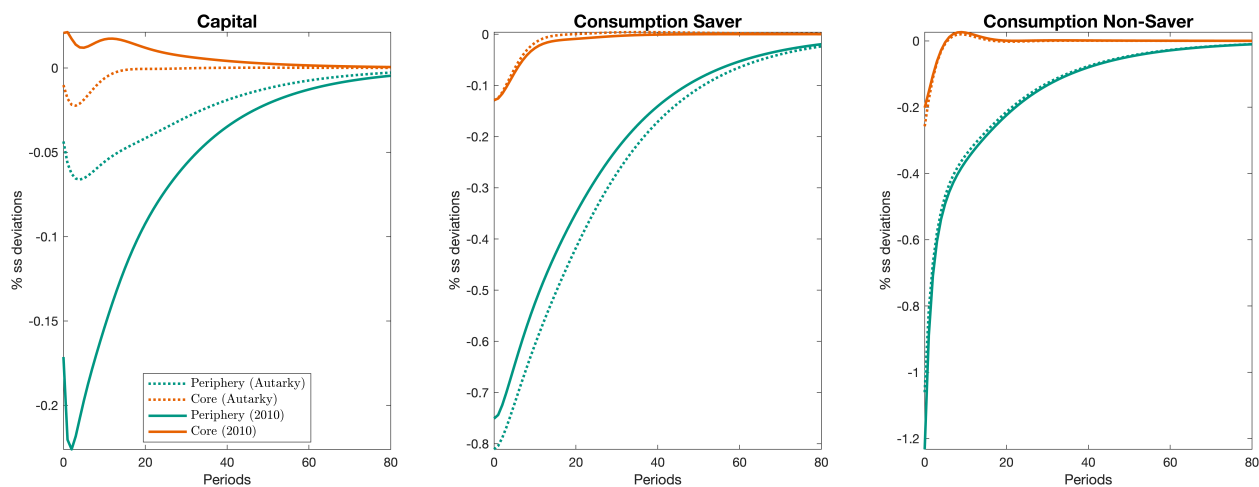


Figure 85: Impulse Responses of Capital and Consumption for Savers and Non-Savers in the Core and Periphery. Comparison between equity autarky (dotted) and calibrated 2010 (solid) configurations.

The welfare decomposition in Table 4 confirms these patterns: capital market integration improves welfare for Savers in the Periphery and for Non-Savers in the Core, while

Non-Savers in the Periphery experience losses driven by labor market effects.

Table 12: Welfare Decomposition Relative to Autarky (Discounted Utilities)

Household	ΔU^C	ΔU^L	ΔU^{Total}
Diversification only			
Core Saver	+0.046	+0.013	+0.059
Core Non-Saver	+0.002	+0.013	+0.015
Periphery Saver	-1.286	-0.188	-1.474
Periphery Non-Saver	+0.048	-0.188	-0.139
Capital market integration			
Core Saver	+0.006	-0.003	+0.004
Core Non-Saver	+0.064	-0.003	+0.061
Periphery Saver	+0.831	+0.045	+0.876
Periphery Non-Saver	-1.526	+0.045	-1.480

Interestingly, introducing realistic euro area asymmetries substantially attenuates the redistributive impact of integration. In the symmetric benchmark, integration benefits Savers in the Periphery and harms those in the Core, while Non-Savers gain in the Core and lose in the Periphery. Under the 2010 calibration, these effects are much weaker. The small relative size of the Periphery, the pronounced home bias of Core investors, and limited portfolio reallocation restrict risk-sharing. Consequently, Core returns and Periphery wages fall less, and welfare gains and losses are smaller overall. Capital market integration remains stabilizing in aggregate but continues to generate unequal gains, favoring asset holders in the Periphery and wage earners in the Core, while leaving liquidity-constrained households most exposed.

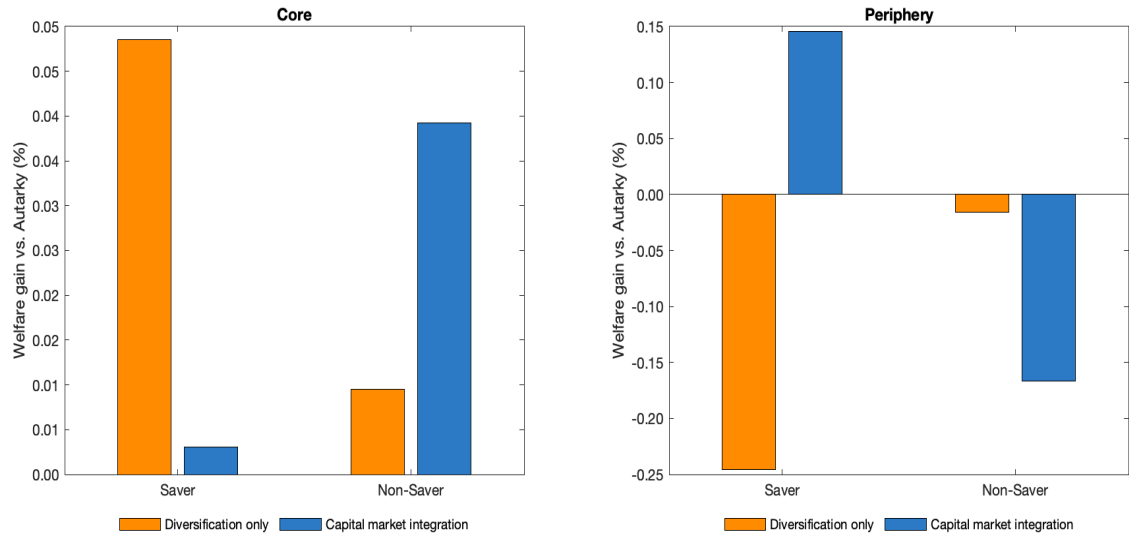


Figure 86: Welfare Gains Relative to Equity Autarky

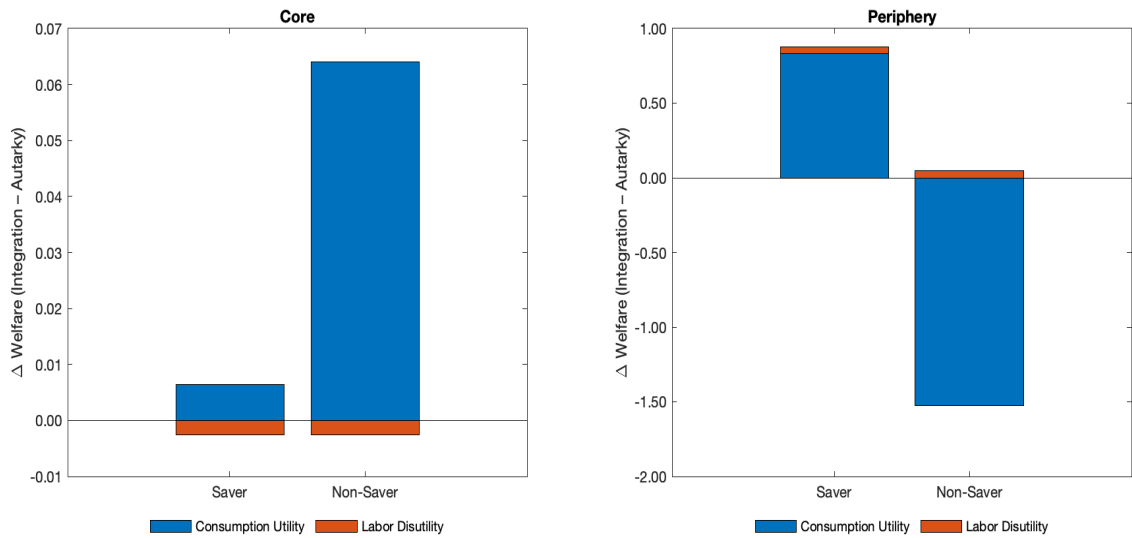


Figure 87: Welfare Decomposition between Consumption and Labor

Consistent with the empirical evidence in the paper, the calibrated model predicts that integration reduces cross-country inequality among Savers but widens within-country inequality, as the welfare gap between Savers and Non-Savers expands in the Periphery and narrows in the Core.

Inequality. Figure 88 compares the impulse responses of capital and consumption for Savers and Non-Savers under both configurations. The dotted lines correspond to the representative-agent benchmark, while the solid lines represent the calibrated 2010 economy. It provides similar qualitative results as under a larger estimation of the shock, though magnitudes differ due to the presence of household heterogeneity.

Relative to the representative-agent benchmark, introducing inequality amplifies the asymmetry of the response to shocks across countries and household types. In the Periphery, capital declines less sharply (+1.23 percentage points relative to the representative-agent case), reflecting weaker aggregate demand when a large share of households cannot smooth consumption. In contrast, capital rises more in the Core (+0.42 percentage points), as excess savings from unconstrained households are redirected toward the higher-return region. This reallocation of savings mitigates the output decline in the Core but exacerbates the downturn in the Periphery.

For Savers, the effects reflect the interaction between diversification and reallocation. In the Periphery, Savers' consumption falls slightly more (-0.6 percentage points relative to the representative-agent case) due to lower returns and depressed local demand. In the Core, Savers also face a larger decline (-0.52 percentage points) owing to their exposure to the Periphery's downturn. However, these effects are moderate compared to the large losses experienced by liquidity-constrained households.

For Non-Savers, the main transmission operates through wages. The contraction in Periphery capital depresses labor demand and real wages, leading to a substantial fall in Non-Saver consumption (-14.1 percentage points). In the Core, by contrast, stronger capital inflows support labor demand, slightly reducing the fall in Non-Saver consumption (-0.02 percentage points). Overall, inequality amplifies within-country heterogeneity, as liquidity-constrained households bear the brunt of the adjustment in the Periphery.

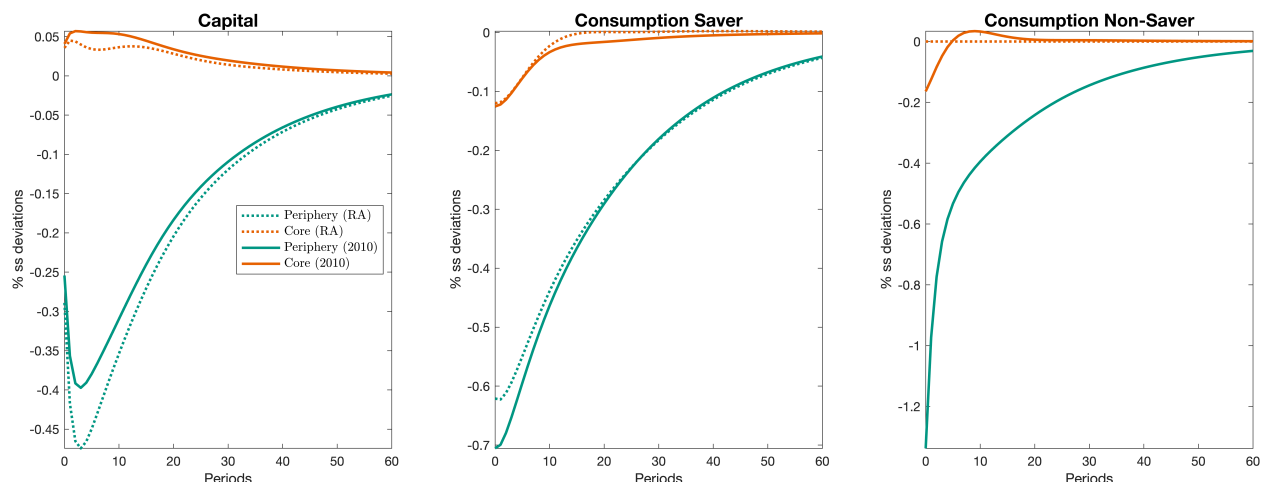


Figure 88: Impulse Responses of Capital and Consumption for Savers and Non-Savers in the Core and Periphery. Comparison between representative-agent (dotted) and calibrated 2010 (solid) configurations.

The welfare decomposition in Table 13 confirms these patterns. Introducing heterogeneity generates highly unequal welfare effects across countries and households. When only the Periphery is heterogeneous, welfare improves for Periphery Savers (+143.0) but falls sharply for Periphery Non-Savers (-1116.8), reflecting adverse labor market dynamics and limited access to capital income. When only the Core is heterogeneous, welfare gains are concentrated among Core Savers (+11.3), while Core Non-Savers face losses (-194.9) due to increased inequality in labor income. Under full inequality, the pattern of heterogeneous effects persists but is somewhat attenuated, as partial risk-sharing across countries moderates the extreme welfare losses among constrained households.

Table 13: Welfare Decomposition Relative to Representative Agent Baseline (Discounted Utilities)

Household	ΔU^C	ΔU^L	ΔU^{Total}
Periphery heterogeneity only			
Core Saver	+0.000	+0.001	+0.001
Core Non-Saver	-156.335	-36.812	-193.148
Periphery Saver	+163.978	-20.976	+143.002
Periphery Non-Saver	-886.777	-230.000	-1116.777
Core heterogeneity only			
Core Saver	+13.044	-1.780	+11.265
Core Non-Saver	-156.336	-38.593	-194.929
Periphery Saver	+0.005	-0.002	+0.004
Periphery Non-Saver	-887.006	-209.026	-1096.032
Full inequality			
Core Saver	+13.010	-1.802	+11.208
Core Non-Saver	-156.260	-38.616	-194.875
Periphery Saver	+166.132	-20.760	+145.372
Periphery Non-Saver	-888.317	-229.784	-1118.102

Consistent with the macroeconomic adjustment described above, heterogeneity increases within-country inequality in the Periphery, as constrained households suffer disproportionately from the shock. In the Core, heterogeneity primarily redistributes income among households but leaves aggregate outcomes largely unaffected. Overall, inequality amplifies welfare dispersion within countries while slightly improving risk-sharing across them, underscoring the asymmetric role of household heterogeneity in shaping the transmission of shocks in the euro area.

C.3 Symmetric Case with Asymmetric Shocks

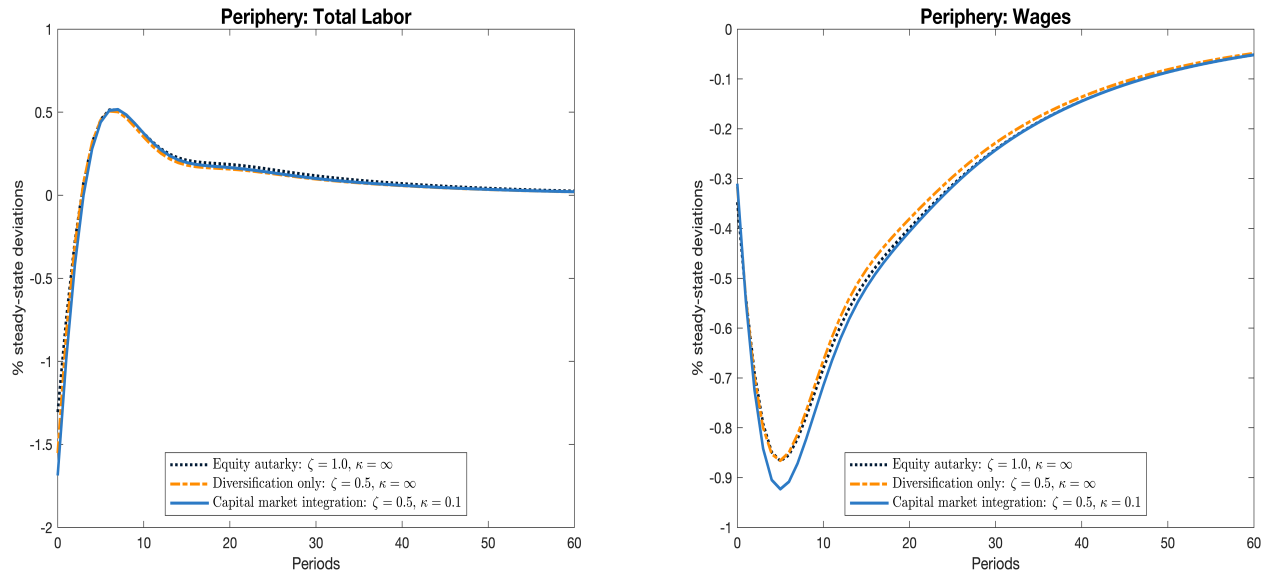


Figure 89: Impulse Response of Wages and Total Labor in the Periphery

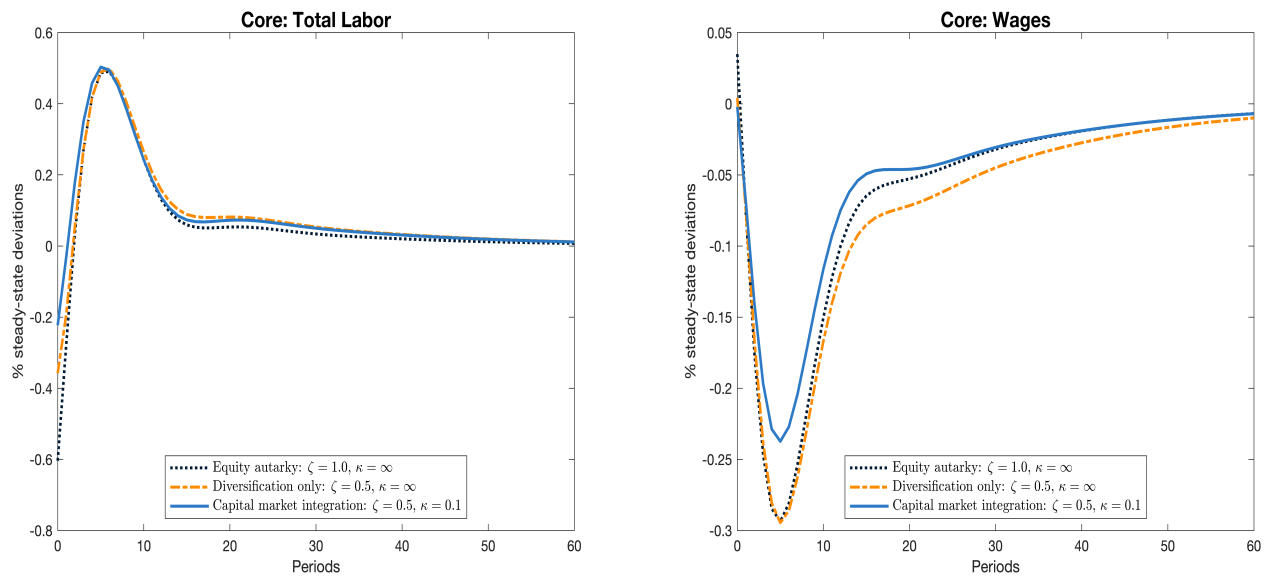


Figure 90: Impulse Response of Wages and Total Labor in the Core

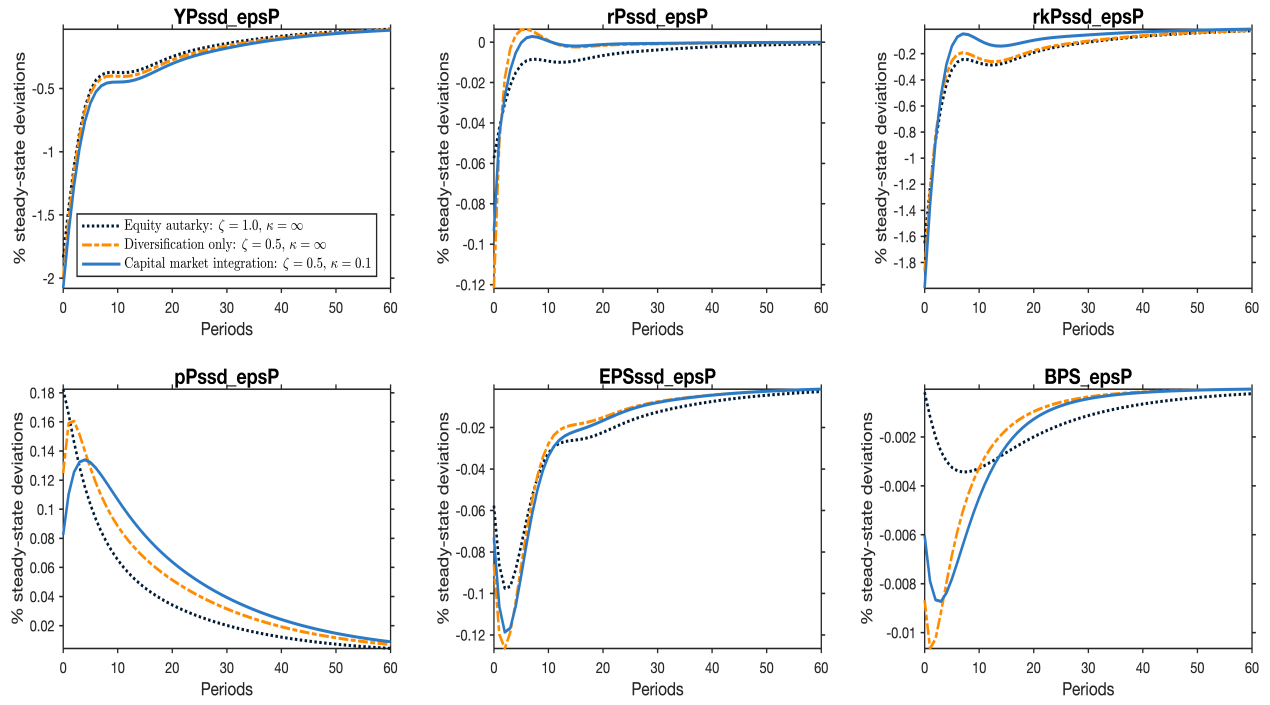


Figure 91: Impulse Response of Selected Variable in the Periphery

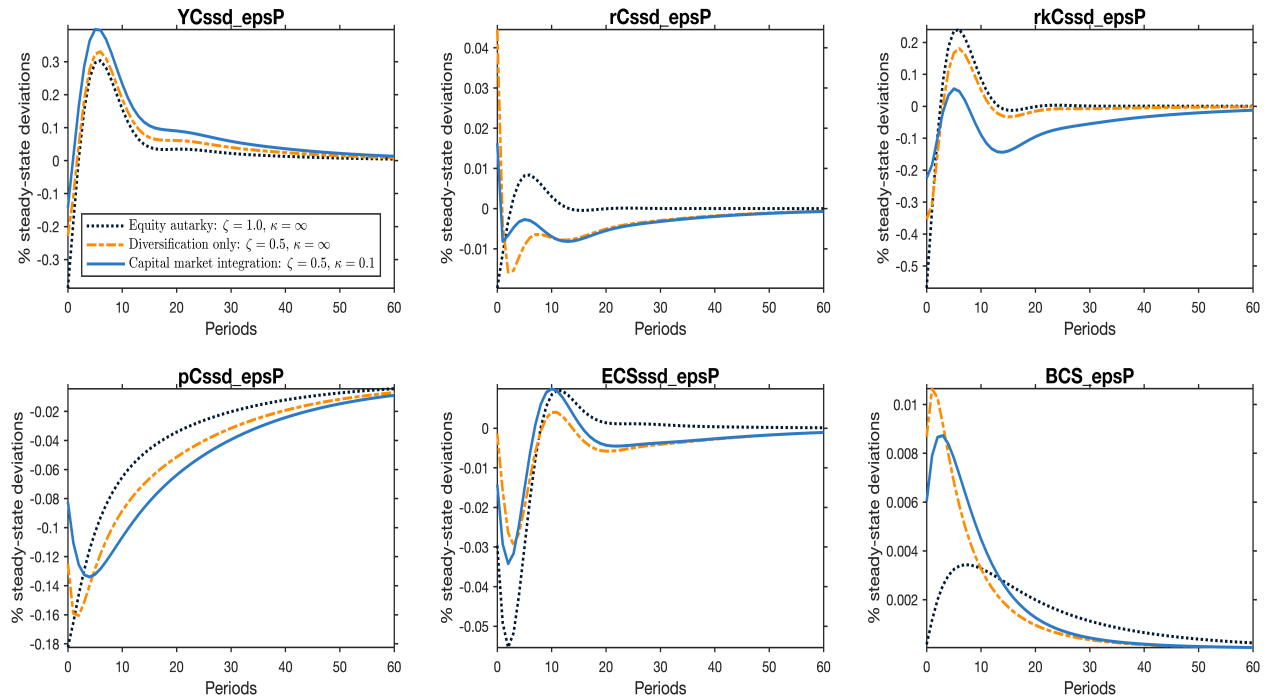


Figure 92: Impulse Response of Selected Variables in the Core

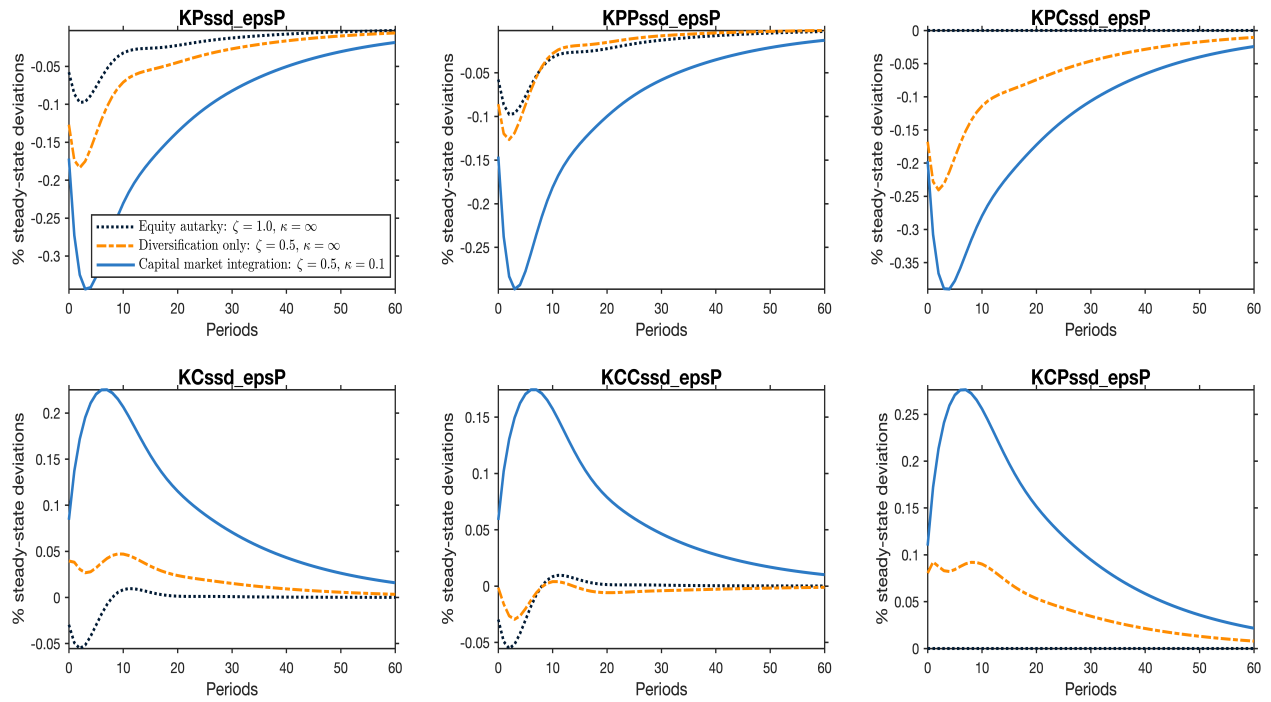


Figure 93: Impulse Response of Selected Variables for Capital

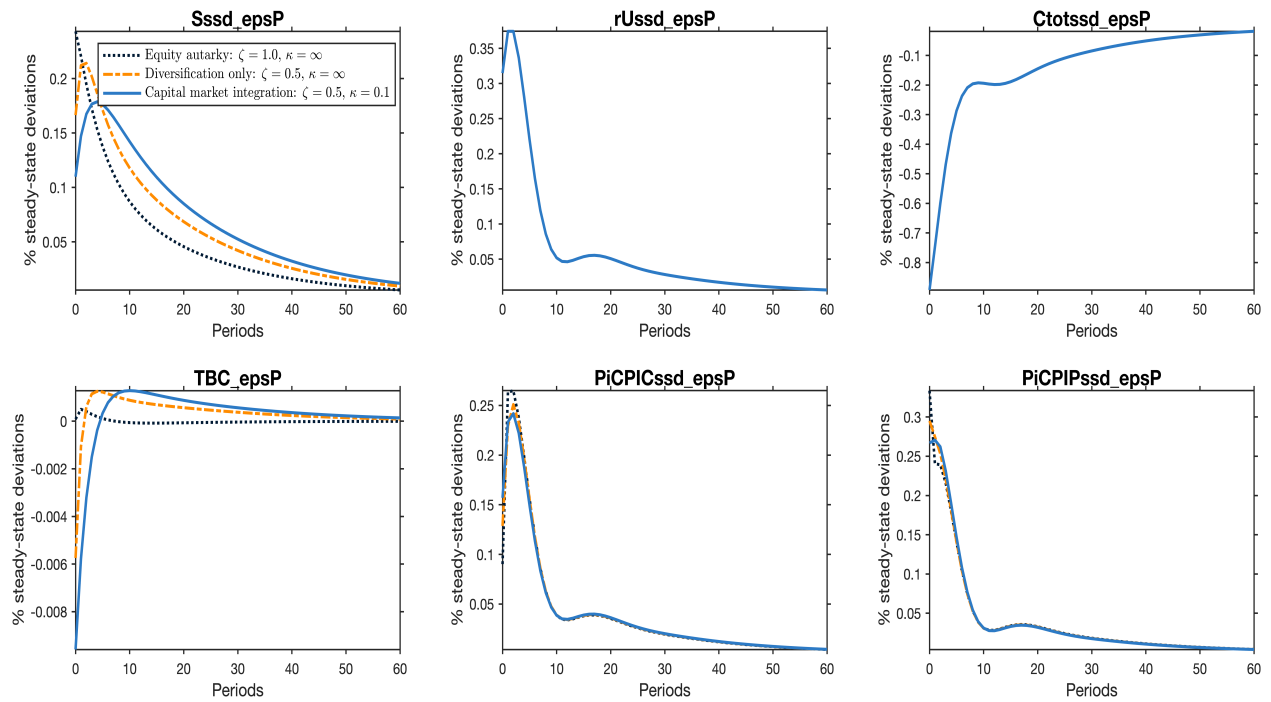


Figure 94: Impulse Response of Selected Variables for Prices

C.4 Welfare under Symmetric Case with Asymmetric Shocks

Figure 95 plots the welfare gains relative to equity autarky. The combined effect of diversification and reallocation mitigates the amplitude of welfare changes observed in the Core. Reallocation allows Core households to benefit through higher labor income and to reallocate assets domestically, whereas diversification alone reduces their welfare through lower financial returns. For Non-Savers in the Core, welfare remains nearly unchanged under integration, as the decline in domestic demand, driven by lower Saver income, offsets the gains from a milder wage decline. In the Periphery, the joint operation of diversification and reallocation also dampens the positive welfare effects observed under diversification alone. While diversification provides Savers with insurance through exposure to foreign assets, reallocation amplifies the fall in domestic capital and real wages, reducing labor income and partially offsetting the benefits from financial income. As a result, overall welfare gains for Periphery Savers decline once reallocation is introduced, and Non-Savers experience welfare losses due to stronger wage effects.

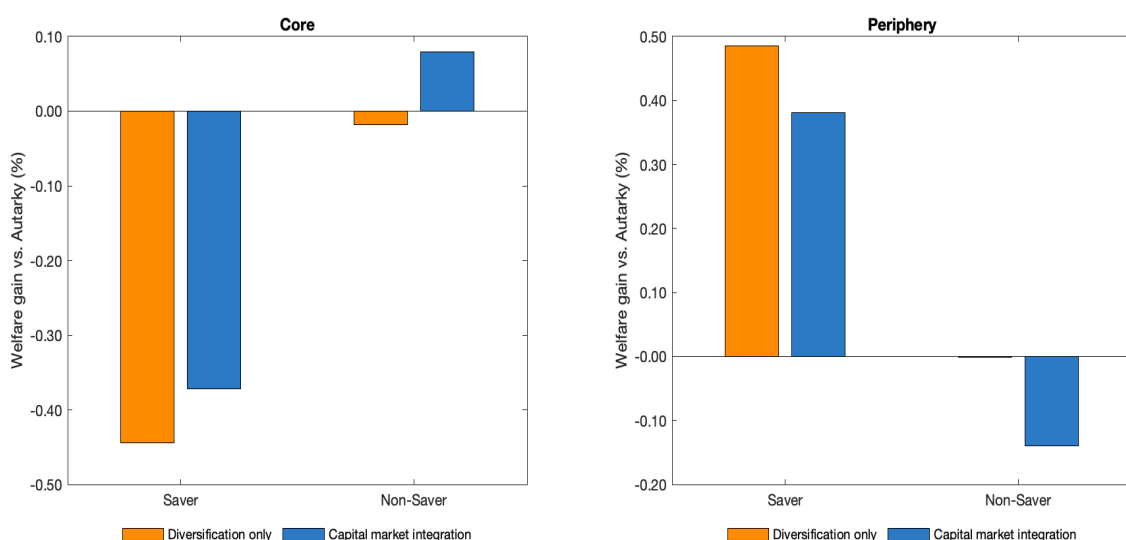


Figure 95: Welfare Gains Relative to Equity Autarky

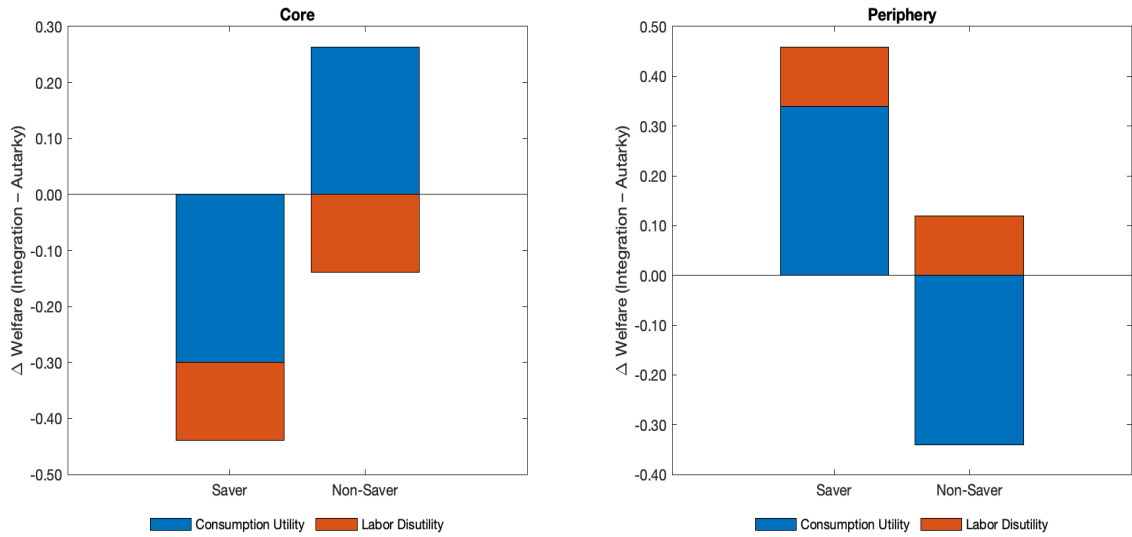


Figure 96: Welfare Decomposition between Consumption and Labor

C.5 Variance under Symmetric Case with Uncorrelated Asymmetric Shocks

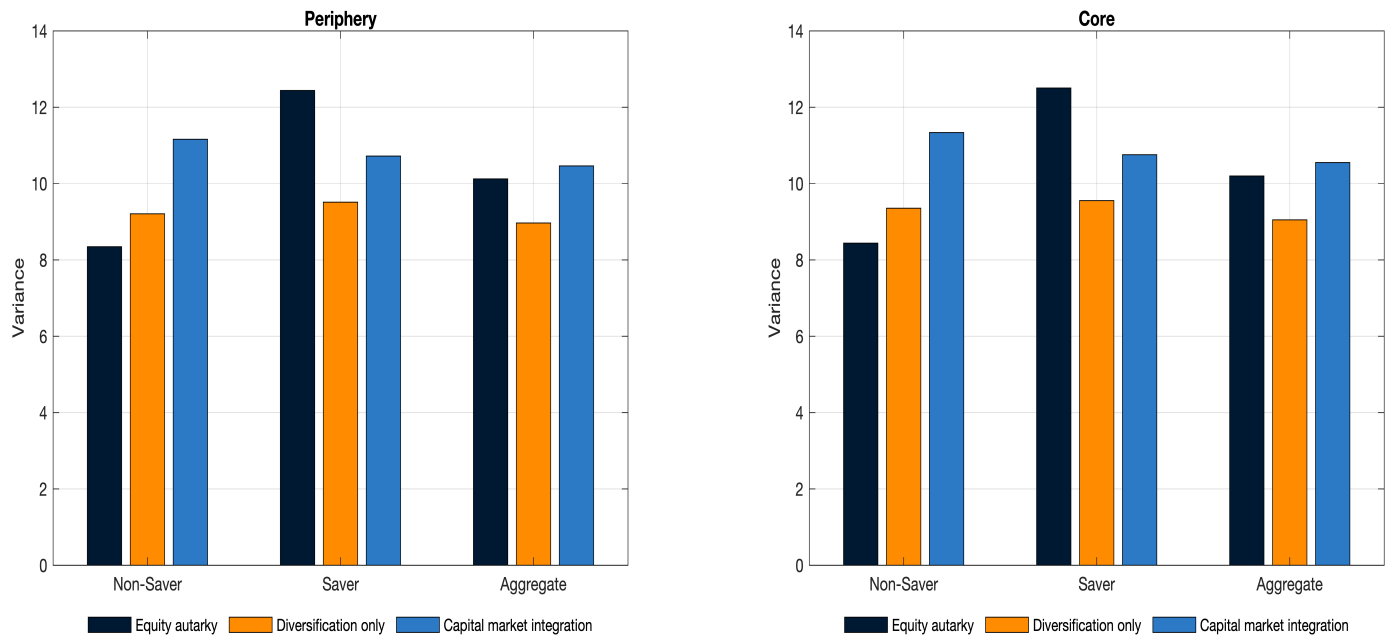


Figure 97: Variance of Consumption across regimes

C.6 Sensitivity to Heterogeneity in Financial Participation and Portfolio Composition

This appendix reports the full results underlying Section 4.3.3. The analysis explores two dimensions of heterogeneity that shape the distributional effects of capital market integration: (i) the share of Non-Savers, capturing differences in financial participation across countries; and (ii) the domestic portfolio share, capturing differences in home bias and diversification.

C.6.1 Varying the Share of Non-Savers

Figure 98 presents simulations under alternative configurations of the share of Non-Savers. The benchmark assumes a symmetric 50

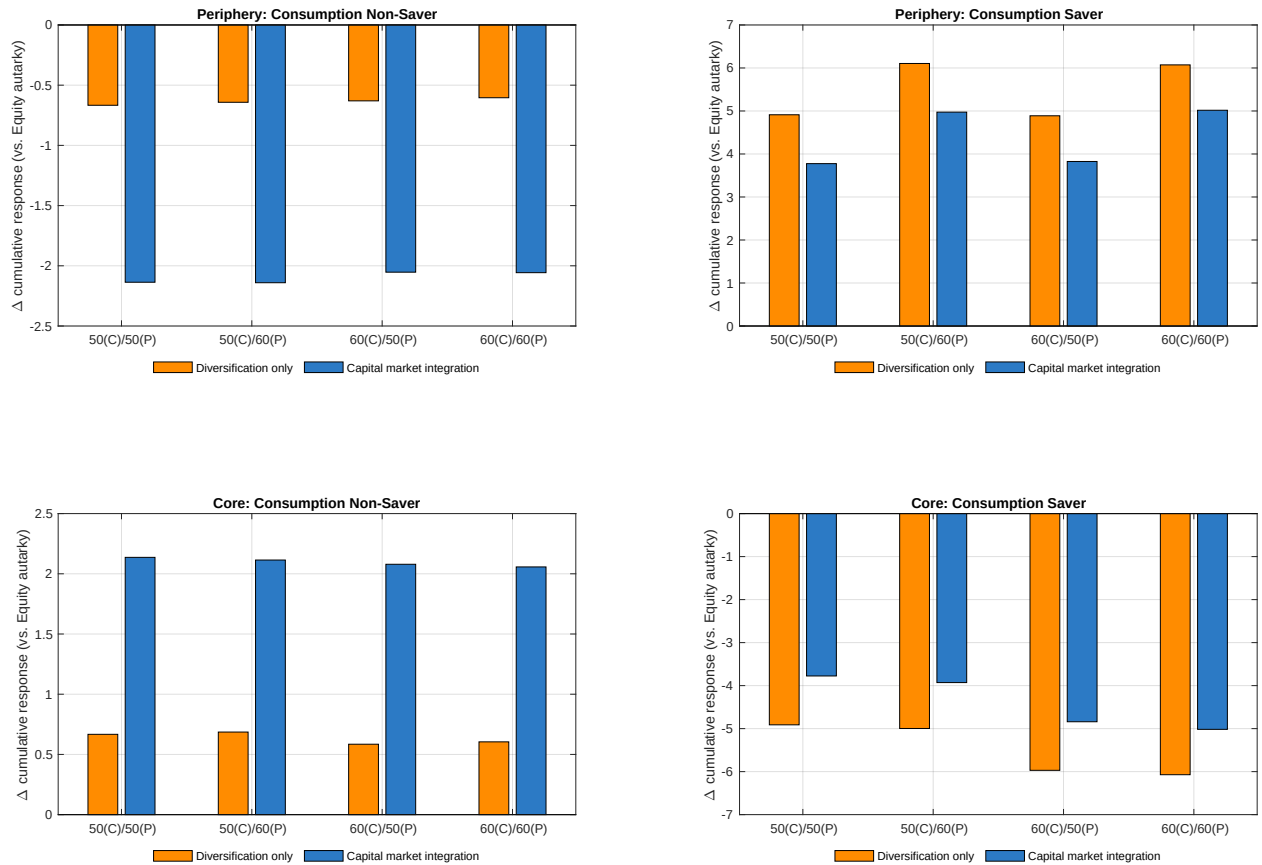


Figure 98: Sensitivity of Distributional Effects to the Share of Non-Saver Households (Deviations from Benchmark). Bars show cumulative effects of each channel, measured as the sum of differences in IRFs over 100 periods relative to the benchmark. The x-axis reports the share of Non-Savers (with the associated country in parentheses).

Results show that the benefits of integration accrue primarily to Savers and become more concentrated as the share of Non-Savers rises. Since only Savers hold financial assets, a smaller investor base implies that aggregate gains from diversification are distributed among fewer households. Non-Savers, who rely solely on labor income, experience little direct change, as wage and capital dynamics remain broadly stable across configurations. The overall distributional outcome depends mainly on the aggregate share of Non-Savers rather than its cross-country distribution.

C.6.2 Varying the Degree of Home Bias

Figure 99 reports simulations where countries differ in their domestic portfolio share. The benchmark assumes symmetric portfolios with 90

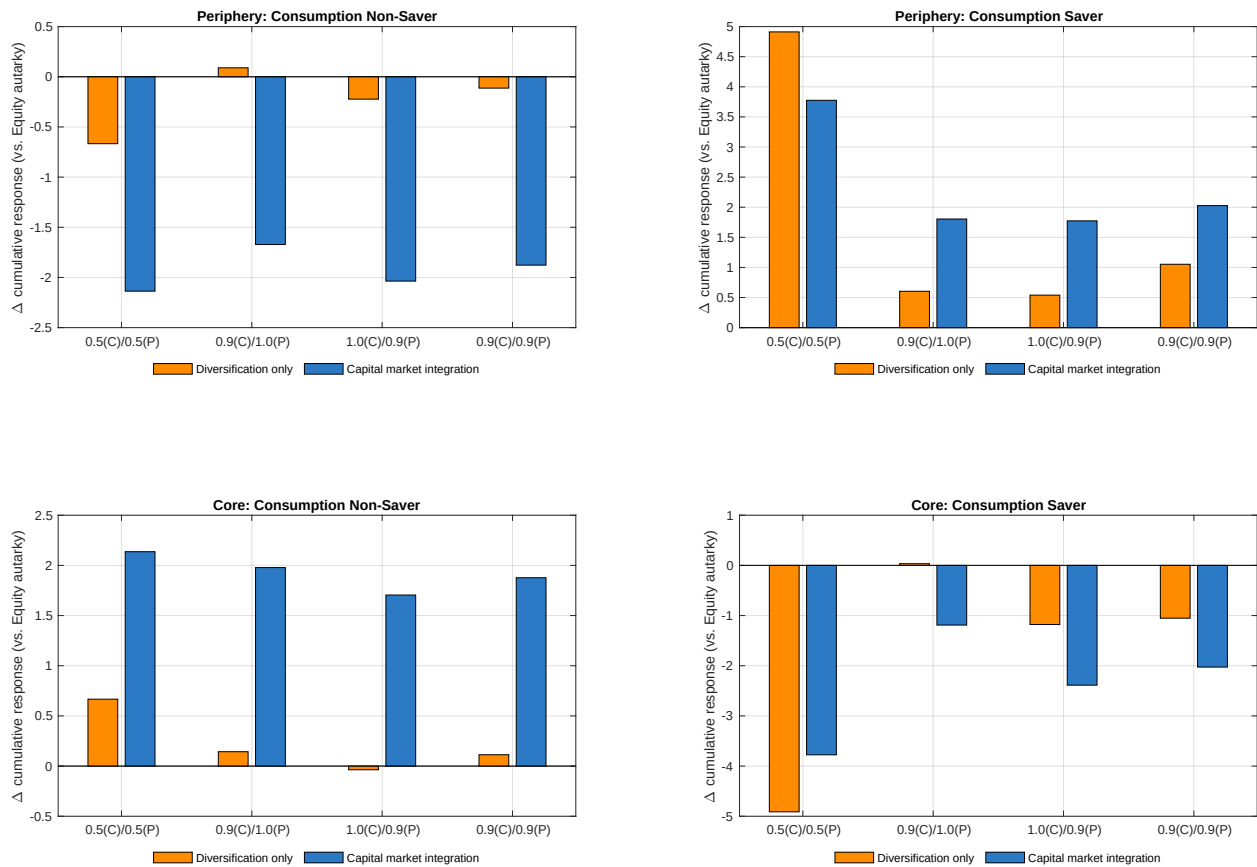


Figure 99: Sensitivity of Distributional Effects to Asymmetric Domestic Portfolio Shares (Deviations from Benchmark). Bars show cumulative effects of each channel, measured as the sum of differences in IRFs over 100 periods relative to the benchmark. The x-axis reports the degree of domestic portfolio share (with the associated country in parentheses).

When the country hit by the shock is more diversified, Savers benefit from foreign

income that stabilizes domestic demand and supports wages, moderating consumption losses for both household types. When the shocked country is more home-biased, the absence of foreign income insurance magnifies domestic contractions and transmits the downturn abroad, worsening outcomes for Non-Savers in both regions.

These results demonstrate that asymmetric diversification amplifies cross-country spillovers and has important implications even for households without financial assets. The structure of portfolio holdings shapes how shocks are absorbed and transmitted through income and labor channels, influencing both within- and between-country inequality.